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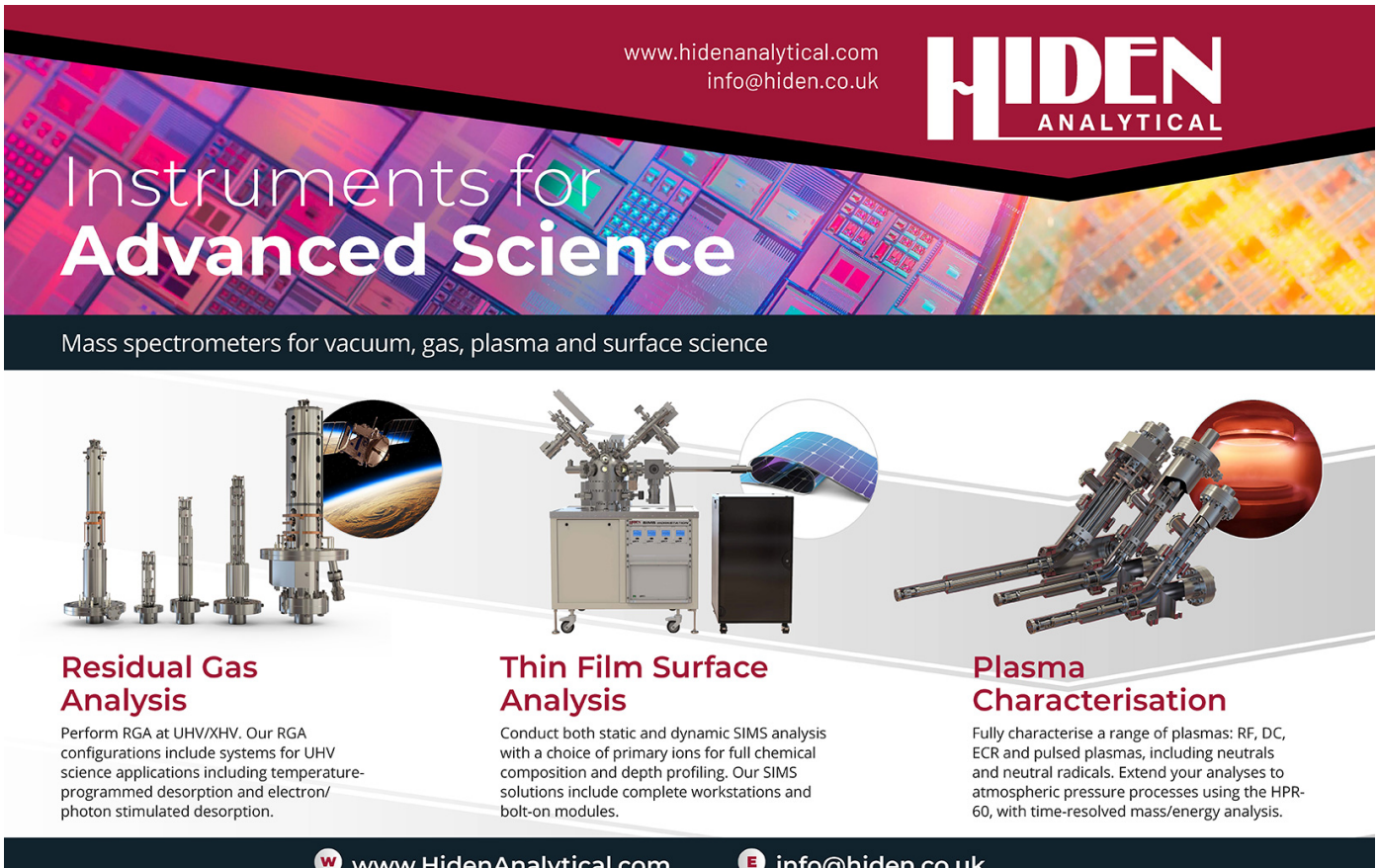
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Applications of B-splines in atomic and molecular physics

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Abstract

One of the most significant developments in computational atomic and molecular physics in recent years has been the introduction of B-spline basis sets in calculations of atomic and molecular structure and dynamics. B-splines were introduced in applied mathematics more than 50 years ago, but it has been in the 1990s, with the advent of powerful computers, that the number of applications has grown exponentially. In this review we present the main properties of B-splines and discuss why they are useful to solve different problems in atomic and molecular physics. We provide an extensive reference list of theoretical works that have made use of B-spline basis sets up to 2000. Among these, we have focused on those applications that have led to the discovery of new interesting phenomena and pointed out the reasons behind the success of the approach.

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1. Introduction

The use of basis sets to solve the Schrödinger equation has a long history in physics. It transforms the solution of a differential equation into an algebraic eigenvalue problem, and, particularly in conjunction with the introduction of electronic computers, this approach has become popular since linear algebra is one of the best developed branches of numerical computation. Today large matrices can be routinely diagonalized with high accuracy in reasonably short times. Many types of basis set have been tried in the past but we focus in this review on the use of B-splines which is a relatively new development in atomic and molecular physics. In molecular physics, the use of basis sets of different types has been standard practice for a long time, but in atomic physics their use has been much more restricted. The reason is that the finite-difference methods, introduced by Hartree, for atoms with spherical symmetry have been found to be more accurate than the use of a limited basis set. This approach, in which a numerical wavefunction is determined at a limited number of mesh points in the self-consistent field (SCF) approach, using for example the Numerov algorithm, is well suited to atomic problems because of their spherical symmetry, which allows the separation of angular and radial variables. In particular, the multi-configuration Hartree–Fock (HF) and Dirac–Fock (DF) programs due to Froese Fischer and co-workers [337] and Grant and co-workers [338], respectively, have been found to be able to provide very accurate wavefunctions for atomic states. Generally speaking the reason for the superiority of the finite-difference methods for atoms is that the basis sets which have been tried need to be large in order to be 'complete enough', but linear dependences usually effectively restrict the size of basis that can be used before the accuracy attainable with the finite-difference methods has been reached.

B-splines have the property of being 'complete enough' with a relatively small number of basis functions and, since linear dependences are negligible, even for a large basis, it is possible for such a basis set to compete with the finite-difference methods. This accounts for the recent swing towards the use of B-splines in atomic calculations, while also for molecular problems B-splines have considerable advantages over earlier basis sets.

The finite-difference and the basis set methods can be characterized as 'local' and 'global' methods, respectively. In local methods, the solution at one point depends on the solution in nearby points only. As a consequence some initial estimate is required to get the algorithm started and normalization must be performed afterwards, while in the global methods the solution at one point is linked to the solutions at all other points so that an initial estimate is unnecessary and normalization is implicit. These are important advantages but there are compensating disadvantages with the usual basis sets. For example, for Slater type orbital (STO) basis sets, which have been the most popular basis for atomic calculations, nonlinear optimization is required. This is a difficult process to control, which means, for example, that the STO basis sets of HF quality established a long time ago by Clementi and Roetti [336] for the ground states of a number of atoms and ions still are routinely used as initial estimates for excited states of atoms because of the large amount of optimization which would be required to start the procedure from scratch. A consequence is that a good working knowledge of which basis set to use under particular circumstances is required. The use of a B-spline basis would not require nonlinear optimization, which is an important difference that still has to be exploited.

Splines of various types have been used, particularly for fitting purposes, in numerical analysis for a long time, and they are a standard part of fitting routines in commercial program packages. The name is derived from a draftsman's spline, a flexible piece of rubber, which can be used to draw smooth curves. The spline is pinned down at known points (called knot points

in the following) on the otherwise perhaps unknown or perhaps irregular curve. The flexibility of the spline now allows us to draw a continuous curve between the points with continuous first and second derivatives at all points. The use of splines in numerical analysis is designed to obtain the same properties in numerical fitting to known or partially known functions. Spline basis functions are compact L^2 integrable functions defined in a restricted space usually referred to as a box, in contrast to such L^2 basis sets as Gaussians or Slater orbitals that extend to infinity. In the box, a knot sequence is defined and the freedom to define the knots to suit the particular problem is, as we shall see, another important advantage of the method. The knot sequence defines the extent of the individual splines. Cubic splines, for example, are piecewise third-degree polynomials. They are used in many fitting programs and usually require a decision from the user concerning the boundary conditions to be used in each individual case. B-spline basis sets of order four are designed to provide the most compact basis for cubic splines and each spline is different from zero only between four adjacent knot points, except possibly the first and last splines. (Note that the order of the B-spline corresponds to the degree of the polynomials involved plus one. Both notations will be used throughout this review.) Thus B-splines are in principle defined as a set (B stands for basis) but for convenience we shall also refer to the individual splines in the set as B-splines. B-splines are positive definite functions, which leads to matrices that are easier to diagonalize. This is in contrast to cardinal splines, for example, that are different from zero over the whole box and not positive definite functions. Compactness has another advantage in matrix calculations, since integrals over two B-splines for example will be zero if the two splines are different from zero in disjoint intervals. The resulting matrices are sparse and often, but not always, banded, which is advantageous not only in the diagonalization procedure but also when calculating the matrix elements.

B-splines were introduced by Schoenberg in 1946 [1] and he also coined the name in 1967 [43], but it was only with the publication of the monograph by de Boor [5] that their application to atomic physics started in earnest. De Boor published FORTRAN routines that makes it possible to define and manipulate B-splines of arbitrary order and knot point distribution and this meant that the extension to B-splines of order higher than four became straightforward. This is essential for obtaining a competitive degree of accuracy in the solution of the Schrödinger equation. In the 1970s Shore published a series of papers [8–12] which nicely illustrates the development. In his first papers, Shore used cubic splines to solve a single-particle Schrödinger equation while in later work Shore apparently became the first to use B-splines for atomic structure calculations using mainly local potentials, and he introduced for example the exponential knot sequence to solve bound-state problems that we shall describe later. In his final paper Shore used B-splines of order higher than four but the main focus of his papers was to compare different basis sets, and he concluded that for comparable sizes B-splines provide the most accurate approximation to the true function, particularly for higher eigenvalues. Shore discussed exclusively different types of single-particle problem and did not consider SCF procedures.

Following Shore, Gilbert and co-workers [13, 14, 42] published work on the use of several types of spline as basis sets for SCF calculations. The first studies used cardinal splines but as mentioned above these have disadvantages in numerical calculations and therefore Altenberger-Siczec and Gilbert [42] switched to B-splines. The origin of their interest was molecular calculations, but the first HF application in 1976 was to He, and they concluded that the calculation of the HF operator in this basis using the so-called supermatrix approach required too much computer power. Alternatively the SCF procedure was proposed, in which the two-electron integrals should be recalculated for each iteration. Nevertheless, this was also deemed to be too time consuming to make a spline basis feasible. However, the advances in computer power have reduced this problem to manageable proportions. In addition, Altenberger-Siczec

and Gilbert based their conclusion on the use of cubic splines, and the introduction of higher-order splines means that basis sets smaller than was foreseen by them are required. The latter was in fact established shortly after the Altenberger-Siczes–Gilbert paper by Silverstone and collaborators [15–17]. Silverstone *et al* [16] also proposed that the use of piece-wise polynomials as basis would be an advantage for many-body perturbation theory (MBPT).

Nevertheless, it took another ten years with the attendant increase in computer power before the next attempt was made to make use of the properties of B-splines . First in 1986, on the instigation of Bottcher, Johnson and Sapirstein [104] introduced a B-spline basis set for the calculation of second- and higher-order corrections to properties of heavy atoms using MBPT. In their work, Johnson, Sapirstein and co-workers [99, 105] constructed orbitals in numerical form from the finite number of eigenfunctions to the DF equation for a valence electron in an alkaline atom in a B-spline basis. The set of numerical orbitals was used in the subsequent MBPT calculations. Thus the only property of the B-spline basis which is used is the 'completeness' that is retained when the wavefunctions are transformed to numerical form. This approach has the advantage that existing numerical computer programs can be used in the MBPT calculations with perhaps minor changes.

Other applications of B-splines take advantage of different properties of the basis. An approach that requires relatively simple computations is the collocation approach. However, this has been shown already by Shore to be much less accurate than the Raleigh–Ritz–Galerkin procedures which form the main subject of this review although the method has been revived by Bottcher and co-workers [28,29,32]. For Hermitian (symmetric) matrices the Raleigh–Ritz variational approach is well known, but for nonsymmetric matrices that appear, for example, when the boundary conditions are different from zero on the outer side of the box, the Galerkin approach is very useful, while for symmetric matrices it is equivalent to the Raleigh–Ritz procedure [339]. The various ways of implementing the Galerkin conditions is a prominent aspect of the technical part of this review.

Since the early work by Johnson and Sapirstein, B-splines have been used in many other types of calculation of various properties for both bound and continuum states and this is the main subject of the physics part of this review. In particular, B-splines are able to provide a very accurate representation of continuum states, which makes them superior to more conventional L^2 basis sets. Although the first applications were to static properties, it has turned out that B-splines perhaps are even more important for the calculation of dynamic properties and we shall discuss subjects such as multiphoton excitation, above-threshold ionization (ATI) and high-order harmonic generation (HOHG) in atoms and dissociation of molecules for which the use of B-splines have meant that effects could be taken into account which theoretically were beyond reach earlier.

The structure of the review is as follows. In the next two sections we first describe briefly the mathematical properties of B-splines and then how to use them to solve the single-particle equation. Section 4 describes the basic techniques for more electrons and applications. Readers who have little previous experience with B-splines would want to start with section 3 while readers with some knowledge of splines and interested in applications might go directly to section 4 (atomic structure), 5 (atoms in strong fields), 6 (diatomic molecules) or 7 (polyatomic molecules and clusters), depending on their particular interest. Sections 5–7 are fairly independent of each other. They are also independent of section 4.3, which is exclusively devoted to applications in atomic structure calculations. To reduce the size of the review we shall largely restrict ourselves to nonrelativistic basis sets and applications, except for relativistic configuration interaction (CI), which is close to other nonrelativistic applications.

We end with a bibliography of all publications we have found concerning the use of B-splines in atomic and molecular physics. The bibliography is very large and not all the

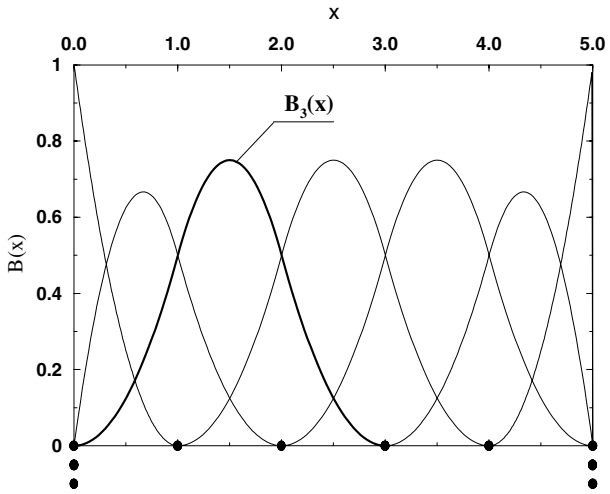


Figure 1. The full set of B-splines of order $k = 3$ relative to the knot sequence $\{0, 0, 0, 1, 2, 3, 4, 5, 5, 5\}$. Knots are represented by full circles.

references are discussed in the review. To help the reader we have divided the papers into sections although some papers in practice cover more than one category. The general references have been given separately for each section to reduce the time needed to find a particular reference. If a reference is mentioned in several chapters it is included only once (the first time).

Throughout, we use atomic units (au) unless specifically indicated.

2. Description of B-splines

B-splines are functions designed to generalize polynomials for the purpose of approximating arbitrary functions. One can view them as new elementary functions such as $\sin x$ or $j_l(x)$. One becomes familiar with them by understanding their qualitative behaviour and how to use them, that is how to obtain values of the functions, their derivatives or integrals. This is the topic of this section. A complete description of B-splines and their properties can be found in de Boor's book [5]. Just like $\sin x$, efficient numerical evaluation can be accomplished by black-box routines, and it is not generally important to know the algorithm for proficient use.

Let us introduce a few definitions.

- The polynomials of order k (maximum degree $k - 1$) are
$$p(x) = a_0 + a_1 x + \cdots + a_{k-1} x^{k-1}.$$
- A function which is continuous (on a given interval) together with its derivatives up to order n, that is $f(x)$, $Df, \ldots, D^n f$, is said to be of class C^n. Then C^0 means that only f is continuous and C^{-1} that f is discontinuous.
- Consider an interval $I = [a, b]$ divided into l subintervals $I_j = [\xi_j, \xi_{j+1}]$ by a sequence of $l + 1$ points $\{\xi_j\}$ in strict ascending order
$$a = \xi_1 < \xi_2 < \cdots < \xi_{l+1} = b.$$
The ξ_j will be called *breakpoints* (bps).

A full set of B-splines (to be defined later) over the interval $[0, 5]$ is shown in figure 1, which we shall repeatedly use for illustration. In this example, the interval is divided into five

subintervals by the bps $\{0, 1, 2, 3, 4, 5\}$. Each B-spline (consider B_3 for example) is a function made up of different polynomial pieces on adjacent subintervals, of a fixed order k ($k = 3$ in the present case), joined with a certain degree of continuity at the bps. In this case each B-spline is continuous together with its first derivative (class C^1) at the interior bps, while at $x = 0$ B_3 is still C^1, but B_2 has discontinuous first derivative, and B_1 is discontinuous (they are C^0 and C^{-1} respectively). The same is true for B_6 and B_7 at $x = 5$, and it will be typical of the B-spline bases commonly employed. An explicit representation of $B_3(x)$ in the example is

$$B_3(x) = \begin{cases} \frac{1}{2}x^2 & 0 \leqslant x < 1 \\ x^2 + 3x - \frac{3}{2} & 1 \leqslant x < 2 \\ \frac{1}{2}x^2 - 3x + \frac{9}{2} & 2 \leqslant x < 3 \\ 0 & \text{otherwise} \end{cases}$$

although this is rarely needed.

- Let us associate with interior bps ξ_j, $j = 2, \ldots, l$, a second sequence of non-negative integers ν_j, $j = 2, \ldots, l$, $\nu_j \geqslant 0$, which define continuity condition $C^{\nu_j - 1}$ at the associated bps ξ_j. With the end bps ξ_1 and ξ_{l+1} we associate $\nu_1 = \nu_{l+1} = 0$, that is we do not require any continuity. This is natural since we are only interested in the interval $[a, b]$. Further restrictions may be dictated by boundary conditions at the endpoints, and are easily implemented as we shall see. In the example we therefore have $\{\nu_j\} = \{0, 2, 2, 2, 2, 0\}$.
- Finally let us call *knots* another sequence of points $\{t_i\}$ in ascending order, not necessarily distinct,

  $$\{t_i\}_{i=1,\ldots,m}, \qquad t_1 \leqslant t_2 \leqslant \cdots \leqslant t_m$$

  associated with ξ_j and ν_j as follows:

  $$\begin{aligned} &t_1 = t_2 = \cdots = t_{\mu_1} = \xi_1; && \mu_1 = k \\ &t_{\mu_1+1} = \cdots = t_{\mu_1+\mu_2} = \xi_2 \\ &\cdots \\ &t_{p+1} = \cdots = t_{p+\mu_i} = \xi_i; && p = \mu_1 + \mu_2 + \cdots + \mu_{i-1} \\ &\cdots \\ &t_{n+1} = \cdots = t_{n+k} = \xi_{l+1}; && \mu_{l+1} = k; \qquad n = \mu_1 + \cdots + \mu_l \end{aligned}$$

  where μ_j is the multiplicity of the knots t_i at ξ_j and is given by $\mu_j = k - \nu_j$. In the example $\{t_i\} = \{0, 0, 0, 1, 2, 3, 4, 5, 5, 5\}$ and $\{\mu_j\} = \{3, 1, 1, 1, 1, 3\}$. Actually only multiplicity at the inner bps is important, and we shall always choose maximum multiplicity $\mu_1 = \mu_{l+1} = k$ at the endpoints.

The most common choice for knot multiplicity at inner bps is unity, corresponding to maximum continuity, that is C^{k-2}. This choice will be employed throughout this present paper, unless otherwise stated. With this choice the number of B-spline functions n ($n = 7$ in the example) is given by

$$n = l + k - 1.$$

Continuity may be relaxed at the inner bps by increasing the associated knot multiplicity there.

Any function expressed as a linear combination of the B_i will be called a *piecewise polynomial function* (pp-function) over $[a, b]$.

$$f = \sum_{i=1}^{n} c_i B_i$$

and it can be shown [5] that f is the most general function made of l polynomial pieces of order k, one for each subinterval I_j joined at the interior bps with continuity C^{ν_j-1}. Note that there is no point in asking for more than C^{k-2} continuity, since otherwise f reduces to a single polynomial over all I, so some discontinuity is the essence of pp-functions. On the other hand it is rare to employ discontinuity in the lowest derivatives or the function itself. If this is the case, for instance with no continuity constraints, $f \in C^{-1}$, one should be careful to consider that $f(\xi_j^-) \neq f(\xi_j^+)$; the function has different left and right limits at the bps. To make f and $D^r f$ uniquely defined over all I, de Boor [5] arbitrarily chooses right continuity, $f(\xi_j) = f(\xi_j^+)$. Careful consideration of discontinuities is, however, not necessary if they are confined to derivatives higher than those actually needed, which is always the case in applications. We define the rth derivative of a pp-function, $D^r f$, as the pp-function of order $k - r$ which has the same sequence ξ and is written in terms of $D^r p_j$ polynomials.

2.1. The basis of B-splines

A single B-spline $B(x)$ is defined by the order $k > 0$, and a set of $k + 1$ knots, $\{t_i, \ldots, t_{i+k}\}$, such that $t_i < t_{i+k}$.

Important properties, apparent in figure 1, are the following.

- $B(x)$ is a pp-function of order k over $[t_i, t_{i+k}]$.
- $B(x) > 0$ for $x \in]t_i, t_{i+k}[$.
- $B(x) = 0$ for $x \notin [t_i, t_{i+k}]$.
- For $x = \xi_j$ $B(x) \in C^{k-1-\mu_j}$, where μ_j is the multiplicity at ξ_j. The maximum multiplicity is k, giving discontinuous functions, the minimum is one, giving $B(x) \in C^{k-2}$.
- The knots need not be equidistant and the shape of $B(x)$ changes smoothly with the change of the knots.

In general a family of B-spline functions, $B_i(x)$, $i = 1, \ldots, n$ is completely defined given $k > 0$, $n > 0$ and a sequence of knots $\boldsymbol{t} = \{t_i\}_{i=1,\ldots,n+k}$ (actually $\{t_i\}$ may be infinite at one or both ends). Since k and t are usually fixed, one writes simply $B_{t,i}^k = B_i$. Each B_i is defined over an interval $[t_i, t_{i+k}]$, which contains $k + 1$ consecutive knots, and is indexed by the knot where it starts, so exactly one B_i starts at each knot t_i, $i = 1, \ldots, n$, and ends k knots later (see figure 1). In the example B_3 starts at $t_3 = 0$ and ends at $t_6 = 3$. $B_i > 0$ over the interval $]t_i, t_{i+k}[$, which is called the support of B_i, and $B_i = 0$ outside it. In fact, of all pp-functions, B-splines are designed to have minimal support. Some general properties of B-splines are the following.

- Over each interval $]t_i, t_{i+1}[$, $t_i < t_{i+1}$, exactly k B-splines are nonzero,

  $$B_j(x) \neq 0 \qquad \text{for} \quad j = i - k + 1, \ldots, i$$

  the first being B_{i-k+1}, which ends at t_{i+1}, and the last is B_i, which starts at t_i. Therefore we have identically

  $$B_i(x) \cdot B_j(x) = 0 \qquad \text{for} \quad |i - j| \geqslant k.$$

- In the expansion of an arbitrary function

  $$f(x) = \sum_{j=1}^{n} c_j B_j(x) = \sum_{j=i-k+1}^{i} c_j B_j(x) \qquad \text{for} \quad x \in [t_i, t_{i+1}]$$

  one always has only k terms contributing, so a minimal number of operations is needed.

- Because the B-splines are non-negative with minimal support, the expansion coefficients of an arbitrary function f are close to the function values at the knots. This means that wild oscillations in the coefficients are avoided, cancellation errors are minimal and numerical stability maximal (see for instance figure 4 in section 3.2).
- Each interval $I_j = [\xi_j, \xi_{j+1}] = [t_i, t_{i+1}]$ is characterized by a pair of consecutive knots $t_i < t_{i+1}$. t_i is called the left knot of interval I_j, and determines the indices of the B_i contributing over I_j, which are $B_{i-k+1}, \ldots, B_i$.
- They are normalized as $\sum_i B_i(x) = 1$ over the whole $[t_k, t_n]$.
- For simple equidistant knots each B_i is just a translation by one interval of the previous one. If the knots are not equidistant there is a smooth change in shape.
- B-splines [5, p 130], satisfy the recursion relation

$$B_i^k(x) = \frac{x - t_i}{t_{i+k-1} - t_i} B_i^{k-1}(x) + \frac{t_{i+k} - x}{t_{i+k} - t_{i+1}} B_{i+1}^{k-1}(x). \tag{1}$$

  Together with the definition of B-splines of order $k = 1$

$$B_i^1(x) = 1 \qquad t_i \leqslant x < t_{i+1} \qquad \text{and} \qquad B_i^1(x) = 0 \quad \text{otherwise}$$

  this gives rise to the algorithm employed for the practical evaluation of B-splines: given a point x, one generates by recursion the values of all the k B-splines which are nonzero at x, as illustrated in figure 2. The derivative of a B-spline of order k, being a pp-function of order $k - 1$, can be also expressed as a linear combination of B-splines of the same order [5, p 138],

$$DB_i^k(x) = \frac{k-1}{t_{i+k-1} - t_i} B_i^{k-1}(x) - \frac{k-1}{t_{i+k} - t_{i+1}} B_{i+1}^{k-1}(x).$$

Actually there is complete freedom in the choice of the first and last k knots, the only requirement being

$$t_1 \leqslant \cdots \leqslant t_k \leqslant \xi_1 \qquad \xi_{l+1} \leqslant t_{n+1} \leqslant \cdots \leqslant t_{n+k}.$$

Their choice affects the behaviour of the first and last k B-splines, but is of no consequence on their basis set property for pp-functions on $[\xi_1, \xi_{l+1}]$. The standard choice $t_1 = \cdots = t_k = \xi_1$ and $t_{n+1} = \cdots = t_{n+k} = \xi_{l+1}$ is particularly convenient. In this case (see figure 1) all $B_i = 0$ outside $[a, b]$. Moreover only the first B-spline is discontinuous at $x = a$, $B_1(x) \neq 0$, the same for DB_1 and DB_2, and likewise for the higher derivatives; the same at $x = b$ for B_n, DB_n and DB_{n-1}, and so on. This makes it very easy to implement boundary conditions at the endpoints. For instance $f(a) = 0$ is satisfied by deleting B_1 from the set, $Df(b) = 0$ by combining B_{n-1} and B_n as $\tilde{B}_{n-1} = c_1 B_{n-1} + c_2 B_n$ so that $D\tilde{B}_{n-1}(b) = 0$, and so on.

The most common choice is therefore to employ knots of multiplicity k at the endpoints and simple knots at the interior points, unless there is a known singularity at some point, where continuity may be reduced by suitably increasing the knot multiplicity. In general, when approximating analytic functions, it is best to employ high-order splines, compatible with the numerical stability and roundoff errors, typically in the range $k = 7$–10. Note that the error will be bound by

$$\varepsilon \sim \frac{h_j^k}{k!} |D^k f(\eta_j)|$$

where h_j is the width of the interval I_j and $\eta_j \in I_j$. This is the main advantage of B-splines over global bases, in that the error can be controlled by step size, in an analogous manner to finite-difference approaches, but retaining all the advantages of basis set expansions. The completeness property of the basis may be characterized by the fact that for a given threshold

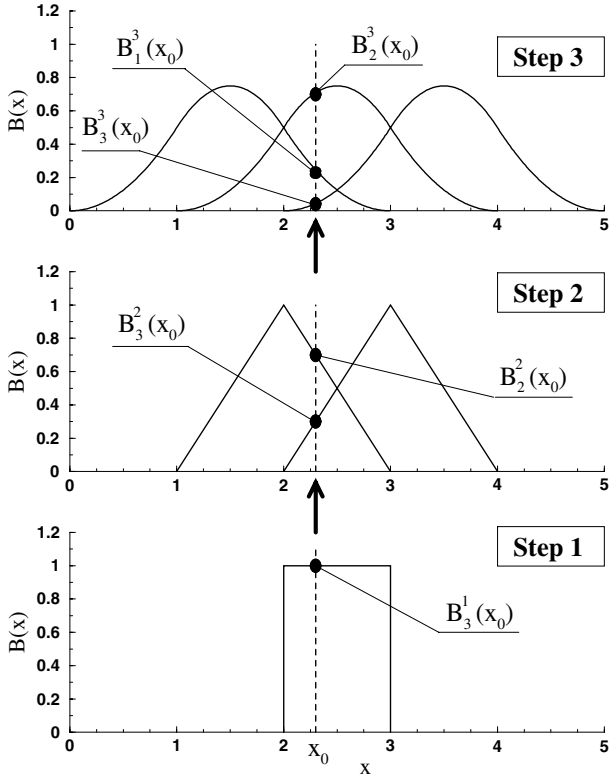


Figure 2. Recursive evaluation of B-splines up to order $k = 3$, relative to the knot sequence $\{0, 1, 2, 3, 4, 5\}$, which are not zero at $2 < x < 3$. Each set is obtained from the previous one by applying the recursion formula (1). Notice that at each x one obtains k nonzero B-spline values, which add up to unity.

ε all sufficiently smooth functions will be approximated within ε, provided that the derivatives are conveniently limited, that is up to a cut-off on the maximum local value of the derivatives. We shall see in section 3.3 that, in the quantum mechanical case, this translates into a limit on the maximum kinetic energy of the states considered, which can be tuned by the choice of the interval step h.

From a practical point of view, stable evaluation of splines is accomplished by the recursion algorithm (1) (see chapter X in [5]) and is efficiently implemented in subroutine BSPLVP (ibid., p 134), which requires as input values the spline order, the knot sequence, the value of the abscissa and the index of the 'left knot' t_i. The latter can be located given the abscissa and the knot sequence, by subroutine INTERV (ibid., p 92). The routine returns the values of the k B-splines that are nonzero at x, that is $B_{i-k+1}(x), \ldots, B_i(x)$. If derivatives are also needed they can be evaluated as combination of B-splines of lower order by subroutine BSPLVD (ibid., p 288).

In quantum mechanical applications mostly matrix elements, i.e. definite integrals involving B-splines and their derivatives, are involved. These are best computed to machine accuracy employing Gauss–Legendre (GL) integration [6, 7] of appropriate order over each subinterval I_j. Recall that GL of order n (n points) is exact for a polynomial integrand of order $2n$. So k GL points integrate exactly the product $B_i B_j$, and a little more is usually completely

accurate even for nonpolynomial integrands, provided no singularity is present in the interior of I_i (see appendix A.3).

3. Solving the Schrödinger equation with B-splines

This section introduces the basic techniques in the description of atomic structure with B-spline basis set. Before extending those techniques to multi-electron atoms and even more complicated systems such as molecules, we focus on the simplest atomic systems: those with one electron bound in a central potential. This class of systems obviously includes the hydrogen atom and any hydrogen-like ions characterized by the Coulomb potential, but also true one-valence electron systems such as alkaline atoms (Na, K, Rb, Cs, ...) or alkaline-like ions (Be^+, Mg^+, ...) provided a sufficiently accurate model or pseudo-potential is available to describe the frozen ion core potential. Less intuitive is the modelling of closed-shell systems such as rare gases in the single-active-electron (SAE) approximation [340, 341]. For example, a model potential recently appeared in the literature [342] that is able to describe the argon atomic structure and fairly well reproduce the dynamics of ATI by intense laser fields.

As demonstrated below, the flexibility of the present approach allows us to calculate both the energies and the wavefunctions in a given central potential with an arbitrary accuracy that can be as high as the machine precision on table top workstations taking only a few minutes. The limitations of the potential will determine the overall accuracy.

3.1. One-electron atomic systems

An eigenstate $\psi(\boldsymbol{r})$ with energy E describing a single-electron atom is a solution of the stationary Schrödinger equation:

$$[-\tfrac{1}{2}\boldsymbol{\nabla}^2 + V(\boldsymbol{r})]\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r}) \tag{2}$$

where $V(\boldsymbol{r}) = V(r)$ is the atomic central potential. Accounting for the symmetry, it is more convenient to separate the spherical variables and to look for solutions of the form

$$\psi_{nlm}(\boldsymbol{r}) = \frac{\mathcal{U}_{nl}(r)}{r} Y_l^m(\theta, \phi) \tag{3}$$

where $Y_l^m(\theta, \phi)$ is a spherical harmonic function. The radial function $\mathcal{U}_{nl}(r)$ is thus a solution for the reduced Schrödinger equation:

$$\left[-\frac{\mathrm{d}^2}{2\,\mathrm{d}r^2} + \frac{l(l+1)}{2r^2} + V(r)\right]\mathcal{U}_{n,l}(r) = E_{n,l}\mathcal{U}_{n,l}(r) \tag{4}$$

with the boundary condition $\mathcal{U}_{n,l}(r = 0) = 0$.

The radial equation (4) is solved numerically in a subspace by assuming that any solution $\mathcal{U}_{nl}(r)$ can be approximated, in a given cavity, by spline functions. As explained in section 2.1, such a subspace can be generated by the basis of B-spline functions. It is thus natural to expand the solutions belonging to that space onto the B-spline basis set as

$$\mathcal{U}_{nl}(r) = \sum_{i=1}^{N} c_i^{nl} B_i(r) \tag{5}$$

where $B_i(r)$ is the ith B-spline of order k as defined in section 2.1. Building the basis requires a knot sequence (see section 2.1 for a description of the full procedure) which depends on the following parameters: a set of mesh points called the breakpoint sequence defined in $[0, r_{\max}]$ (see appendix A.1 for a list of possible arrangements), the order k of the piecewise polynomials

and the continuity conditions at each breakpoint. All these parameters are chosen the properties of the physical problem in mind.

In the present case, the cavity (often called the box), in which we solve the problem, is defined by letting the radial coordinate r vary within the interval $[0, r_{\max}]$. This interval is now divided into segments whose endpoints form the breakpoint sequence. There is a total freedom in choosing the mesh of bps. An optimal choice will depend on the type of result we are particularly interested in. For instance, to obtain an accurate bound-state wavefunction, it is more appropriate if the bps are concentrated close to the nucleus (exponential, polynomial or power breakpoint sequence) because the solution vanishes at some finite distance. In contrast, a continuum wavefunction, which oscillates to infinity, is therefore uniformly represented if the bps are equally spaced (linear sequence). Alternatively, one may need accuracy for the initial ground state (GS) and, at the same time, a fairly good description of the continuum as is the case in ionization cross-section calculations or in the study of ATI for example (see section 5.2). These applications use a mixed sequence which is polynomial up to some distances from the nucleus and linear afterwards. There is no physical reason for not imposing the maximum-continuity conditions ($\nu_i = k - 1$) at each breakpoint in $]0, r_{\max}[$. However, to satisfy the boundary conditions, we require the solution to vanish at the endpoints: $\mathcal{U}_{n,l}(r = 0) = \mathcal{U}_{n,l}(r = r_{\max}) = 0$ (see section 3.3 for a different choice of boundary conditions at $r_{\max}$). These conditions are fulfilled either by removing the first and last B-splines from the basis set or by setting the continuity conditions $\nu_i = 1$ at the boundaries.

For a given fixed angular momentum l, the atomic states (energy and wavefunction) satisfying (4) are calculated by solving the system of N linear equations obtained when substituting (5) in (4) and projecting on $B_j(r)$. Written in matrix form, this procedure is equivalent to solving the following generalized eigensystem:

$$\boldsymbol{H}_l \cdot \boldsymbol{c} = E\ \boldsymbol{S} \cdot \boldsymbol{c} \tag{6}$$

for E_{nl} and $\boldsymbol{c}_{nl} = \{c_i^{nl}\}_1^N$, where

$$\begin{aligned}(\boldsymbol{H}_l)_{ij} = &-\frac{1}{2}\int_0^{r_{\max}} B_i(r)\frac{\mathrm{d}^2}{\mathrm{d}r^2}B_j(r)\,\mathrm{d}r + \frac{l(l+1)}{2}\int_0^{r_{\max}}\frac{B_i(r)B_j(r)}{r^2}\,\mathrm{d}r \\ &+ \int_0^{r_{\max}} B_i(r)V(r)B_j(r)\,\mathrm{d}r \\ (\boldsymbol{S})_{ij} = &\int_0^{r_{\max}} B_i(r)B_j(r)\,\mathrm{d}r.\end{aligned} \tag{7}$$

The overlap matrix $\boldsymbol{S}$ originates from the fact that B-splines do not form an orthonormal set of basis functions. All matrix elements are computed exactly using a Gauss quadrature. A GL quadrature with n GL points calculates exactly the integral of a polynomial of degree $2n + 1$ over a closed interval. Since a B-spline of order k is a particular polynomial of degree $k - 1$ on each segment, one easily sees that the integrands of the element $(\boldsymbol{S})_{ij}$ and the kinetic term of the element $(\boldsymbol{H}_l)_{ij}$ are polynomials with maximum degree $2k - 2$. Applying the GL procedure *on each segment* leads to an exact numerical evaluation of these two integrals. The case of the centrifugal and the potential term of the element $(\boldsymbol{H}_l)_{ij}$ is slightly different. Here the integrand is no longer a polynomial of some order but rather a rational fraction and the above statement does not hold. However, it can be shown that a small increment in the number of GL points results in an immediate convergence to machine accuracy. All matrix elements are therefore implicitly computed exactly.

Another key property directly related to the use of a B-spline basis follows from the fact that B-splines are compact functions: a B-spline of order k differs from zero only on k successive segments. Therefore, any local operator expressed in the B-spline basis appears

Table 1. Energies (in au) of all eigenstates of hydrogen (s states) computed with B-splines and compared with the exact values ($k = 9$, $r_{\max} = 100$ au, $N = 100$, exponential sequence with $\gamma = 5$).

n	E_n	$E_n^{\text{exact}} = -1/2n^2$	ΔE
1	−0.500 000 0000	−0.500 000 0000	9.0×10^{-13}
2	−0.125 000 0000	−0.125 000 0000	1.0×10^{-13}
3	−0.055 555 5555	−0.055 555 5555	4.4×10^{-14}
4	−0.031 249 9999	−0.031 250 0000	3.9×10^{-13}
5	−0.019 999 9715	−0.020 000 0000	2.8×10^{-08}
6	−0.013 868 4777	−0.013 888 8889	2.0×10^{-05}
7	−0.009 596 3749	−0.010 204 0816	6.1×10^{-04}
8	−0.004 662 8369	−0.007 812 5000	3.1×10^{-03}

as a matrix having nonvanishing values on a diagonal band of width $2k - 1$. Moreover, the Hamiltonian as well as the overlap matrix are symmetric matrices. It thus turns out that, for a given l, out of the N^2 matrix elements, only Nk need to be computed and stored in memory.

Solving the eigensystem (6) is now straightforward if one accounts for the fact that the overlap matrix $\boldsymbol{S}$ is positive definite. The symmetric band generalized eigensystem (6) can then be transformed into a regular eigensystem with the same band structure. A direct diagonalization of the new system provides all eigenenergies but this procedure is not able to recover the corresponding eigenfunctions since the transform matrix (full N^2 matrix) has been dropped. Having all eigenvalues, the eigenvectors are computed one by one in a second step by inverse iteration when necessary. This procedure never requires more than three iterations for a relative error of the order of the machine accuracy.

3.2. Bound states

The diagonalization procedure provides a set of eigenenergies that are associated with two different kinds of state. The energies lying below the ionization threshold (negative energies in the case of a central potential vanishing at infinity) correspond to the bound atomic states while the energies above that threshold are associated with continuum states. This subsection and the next one are devoted to explaining the influence of the basis parameters on the atomic bound states as well as to giving an interpretation of the so-called discretized continuum states. In what follows, the discussions are illustrated in the context of atomic hydrogen.

An example of a bound-state calculation is given in table 1, where we show all bound states (with angular momentum $l = 0$) obtained with the following B-spline basis: $k = 9$, $r_{\max} = 100$ au, $N = 100$, exponential breakpoint sequence with $\gamma = 5$ (see appendix A.1 for definitions of breakpoint sequences). Although the basis size is rather small, the agreement is excellent (machine accuracy) for the first four states. This is partly due to the choice of an exponential breakpoint sequence. Indeed, bound states vary close to the nucleus before vanishing exponentially at larger distances. Such a sequence defining a finer mesh (small segments) at small distances and larger segments away from the nucleus is therefore better adapted to the computation of bound states [9, 322]. As an example, the GS energy of hydrogen computed with the same basis but a linear sequence gives a deviation from the exact value of $\Delta E = 2.4 \times 10^{-4}$ au. Alternatively, one can use other types of sequence that concentrate bps near the nucleus such as power, sinelike, polynomial etc sequences (see appendix A.1).

Table 2. The different sets of parameters (k, N) achieving a particular level of accuracy ΔE on the GS of hydrogen. The optimal set (depicted with bold-face type) is chosen as being the least CPU time consuming (linear sequence with $r_{\max} = 1000$ au).

	$\Delta E < 10^{-3}$		$\Delta E < 10^{-6}$		$\Delta E < 10^{-12}$	
k	N	Time	N	Time	N	Time
4	860	17	3000	174.5		
5	530	10	1370	49.1		
6	**370**	**9.7**	860	31.7	4000	403.5
7	280	12.1	**620**	**31.6**	2370	243
8	220	16.1	490	39.5	1590	192
9	190	23.4	400	52.1	**1160**	**175**
10			340	70	890	201
11					710	242
12					620	316

In our context, the B-spline approach requires that the problem is confined in a sphere, which is in contradiction with the fact that, for a Coulomb potential, the spatial extension of the wavefunction tends to infinity as the principal quantum number increases. In fact, in hydrogen, one has $\langle r\rangle_{nl} = \frac{1}{2}\left[3n^2 - l(l+1)\right]$. Obviously, the box size $r_{\max}$ will limit the number of bound states obtained by this technique. As an example, although $\langle r\rangle_{1s} = 1.5$ au, it requires a box of $r_{\max} = 30$ au to obtain the proper energy of E_{1s}. Therefore, $r_{\max}$ must be set large enough for the box to contain all the atomic bound states of interest. The increasing deviation ΔE, shown in table 1, as n gets bigger is not a consequence of inaccuracy but is due to the fact that high-n states are eigenstates of the atomic Hamiltonian within the box (often referred to as pseudo-states). Indeed, the boundary conditions imposed here are equivalent to adding, to the atomic potential, a potential step of infinite height located at $r = r_{\max}$.

Given a box size $r_{\max}$ and a breakpoint sequence, higher accuracy is achieved by either increasing the order k or the density of bps, that is, since $r_{\max}$ is fixed, the basis dimension N. The rate of convergence is different for the two parameters as well as the increase in computational cost. Thus, depending on the degree of accuracy required, there exist optimal sets of parameters (k, N) in terms of CPU time as shown in table 2. This table clearly shows why high-order splines are much more efficient than simple cubic splines ($k = 4$). In practice, the order k is chosen between six and nine (the range commonly used in the literature being $k \in [7, 11]$) and the basis dimension adjusted to meet the desired accuracy.

The accuracy and computational stability of the B-spline approach compared with other basis expansions is clearly illustrated by comparing figures 3 and 4. Both represent a plot of the 4s wavefunction in H but one is built with STOs while the other uses B-splines. Figure 3 shows that the value of the wavefunction results from large cancellations in the linear combination of STOs (things worsen as n is increased). Cancellations, on the other hand, are dramatically minimized when B-splines are used, as seen in figure 4, due to the property that B-splines have minimal support. Indeed, stability in the wavefunction evaluation is enforced by the fact that there exist only k nonvanishing B-splines of order k at any point. The result is that the sum (see equation (5)) involved in the evaluation of the wavefunction at a particular position reduces to

$$\mathcal{U}_{nl}(r) = \sum_{j=i-k+1}^{i} c_j^{nl} B_j(r) \tag{8}$$

where $r \in [t_i, t_{i+1}]$. The evaluation thus always involves the sum of k terms whatever the basis dimension N is.

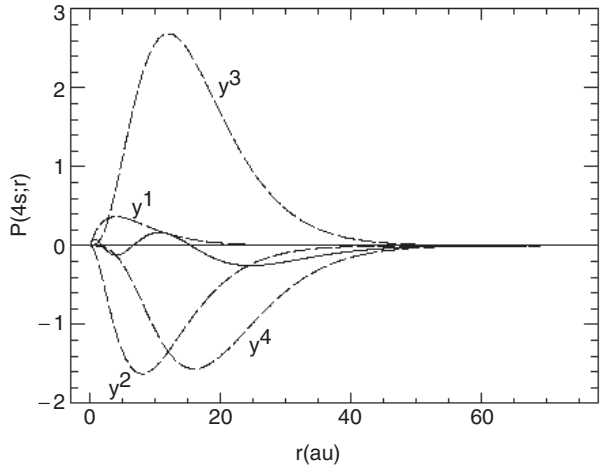


Figure 3. Expansion of the 4s wavefunction in H (solid curve) in terms of four STOs. The dashed curves represent the contribution of each orbital ($y^n = A_n r^n \mathrm{e}^{-\alpha r}$) to the wavefunction. Note the large cancellations between the various components (from [321]).

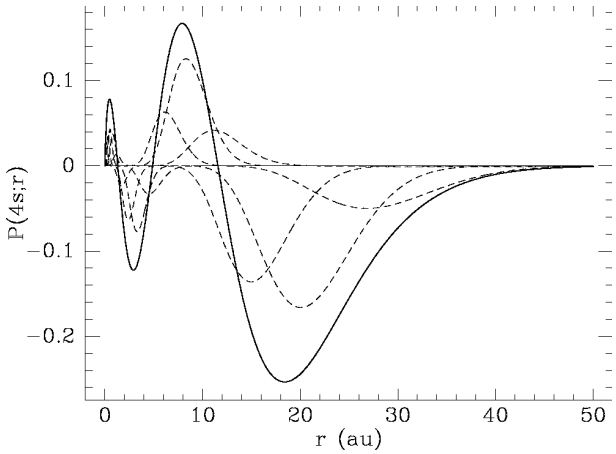


Figure 4. Expansion of the 4s wavefunction in H in terms of B-splines. The dashed curves represent the contribution of each B-spline i of order $k = 5$ (the terms $c_i B_i(r)$ in equation (5)). Note here the quasi-absence of cancellations (from [321]).

3.3. Continuum states

Formally, the solutions of the Schrödinger equation for positive energies appear as a family of functions depending on the continuous parameter E. In our approach however, the diagonalization procedure only provides a discrete set of electronic wavefunctions. These are the particular stationary solutions satisfying $\mathcal{U}_{nl}(r = r_{\max}) = 0$ due to the fact that we have restricted the problem to a finite part of the space. Nevertheless, these functions can be interpreted as a representation of the true electronic continuum with a different normalization (see below). One of the great advantages of making use of B-spline functions is that the density of discretized continuum states can be adjusted with a total degree of freedom simply by modifying $r_{\max}$ as shown in figure 5.

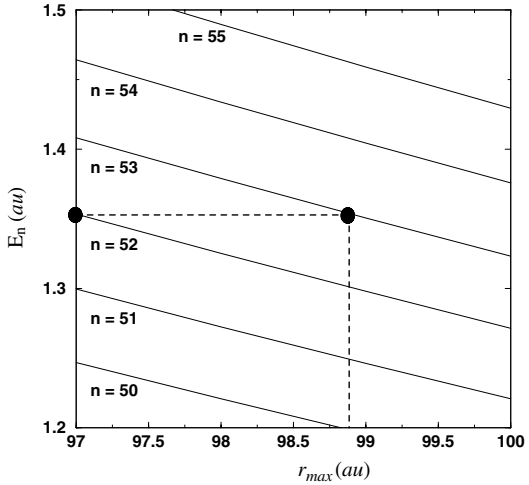


Figure 5. Plot of a selection of discretized continuum energies as a function of $r_{\max}$ ($N = 100$). Note that by varying $r_{\max}$ from 97 to 98.9 au, state number 53 will scan the empty energy gap that separates it from state number 52. Note also that the curves are not straight lines but vary as $1/r_{\max}^2$.

The box size $r_{\max}$ is a very important and useful parameter for manipulating discretized continuum states. The value of $r_{\max}$ uniquely sets the energy of each discretized continuum state and, consequently, the density of states. This is better understood if one considers the asymptotic behaviour of a Coulomb partial continuum wavefunction, which reads, when $kr \to \infty$,

$$\mathcal{U}_{E,l}(r) = \sqrt{\frac{2}{\pi k}} \sin\left(kr + \frac{Z}{k}\ln(2kr) - \frac{l\pi}{2} + \arg\,\Gamma\left(l + 1 - \frac{\mathrm{i}Z}{k}\right) + \sigma_l\right) \quad (9)$$

where $k = \sqrt{2E}$, l is the angular momentum, Z the nuclear charge, $\arg\,\Gamma(l + 1 - \mathrm{i}Z/k)$ the Coulomb phase and σ_l the non-coulombic phase ($\sigma_l = 0$ for hydrogen). Since our method implies that all solutions of the Schrödinger equation vanish at $r = r_{\max}$ ($\mathcal{U}_{E,l}(r_{\max}) = 0$), a discretized wavefunction at energy $E = k^2/2$ satisfies

$$kr_{\max} + \frac{Z}{k}\ln(2kr_{\max}) - \frac{l\pi}{2} + \arg\,\Gamma\left(l + 1 - \frac{\mathrm{i}Z}{k}\right) + \sigma_l = n\pi \quad (10)$$

where n is an integer. Equation (10) defines a relation between the energy and a discrete parameter n, which can be used as a wavefunction index. In a first approximation (when $kr_{\max}$ is bigger than all the other terms) this relation leads to the energy of a particle in a box [330]:

$$E_n = \frac{n^2\pi^2}{2r_{\max}^2}. \quad (11)$$

This latter property, which is illustrated in figure 5, makes it possible to compute continuum wavefunctions at any energy E simply by choosing the proper value of $r_{\max}$. This is achieved either by solving equation (10) for $r_{\max}$ with $k = \sqrt{2E}$, or through an inverse iteration procedure [343] which varies the box size until $E_n = E$. Some applications, like the computation of multiphoton ionization (MPI) cross-sections, may require a resolution in the final-state energy sometimes smaller than the energy spacing obtained through the discretization procedure. There are two ways to tackle this problem. One of them is to repeat

the computation with slightly different box sizes. For example, as shown in figure 5, varying $r_{\max}$ between 97 and 98.9 au shifts the energy of state $n = 53$ from 1.41 down to 1.35 au. Notice that this trick barely affects the energy spacing. The other way is to increase the box size (keeping the breakpoint spacing constant), thus reducing the energy spacing. As a matter of fact, the quality of the continuum discretization improves as the density of states increases (i.e. the energy spacing decreases). The density, defined as $\rho(E_i) = \frac{\partial n}{\partial E_n}|_{n=i}$ (see below), can be evaluated from equation (11) as

$$\rho(E) = \frac{1}{\pi\sqrt{2}} \frac{r_{\max}}{\sqrt{E}}. \tag{12}$$

Thus, the density of states can be adjusted to meet any requirement since it varies linearly with the box size $r_{\max}$.

An alternative approach, that can yield the continuum wavefunction at any selected energy E employing a fixed basis set, without the need to change box size, is the so-called Galerkin [18], or least-squares, approach [26,274], or free boundary condition approach [245]. It is based on the idea of minimizing the residual vector $R_E = (\mathcal{H} - E)\mathcal{U}$ for a given value of E. Actually, defining

$$b_i = \langle B_i, (\mathcal{H} - E)\mathcal{U}\rangle = \sum_j (\boldsymbol{A}(E))_{ij} c_j \tag{13}$$

where $\mathcal{H}$ is the reduced Hamiltonian as defined in (4) and where

$$(\boldsymbol{A}(E))_{ij} = (\mathcal{H})_{ij} - E(\boldsymbol{S})_{ij}, \qquad \mathcal{U} = \sum_i c_i B_i \tag{14}$$

one minimizes $\sum_i b_i^* b_i$, under the constraint $\sum_i c_i^* c_i = 1$ to avoid the trivial solution $\boldsymbol{c} = 0$. This is equivalent to finding the minimum of

$$\boldsymbol{c}^\dagger \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{c} - \lambda \boldsymbol{c}^\dagger \boldsymbol{c}. \tag{15}$$

The minimum corresponds to the eigenvector of the matrix $\boldsymbol{A}^\dagger \boldsymbol{A}$ whose eigenvalue is closest to zero. Recall that $\boldsymbol{A}^\dagger \boldsymbol{A}$ is Hermitian and has non-negative eigenvalues. Since with a good basis, like B-splines, the lowest eigenvalue is extremely close to zero, and separated from the higher ones by several orders of magnitude, the corresponding eigenvector can be obtained accurately and efficiently employing inverse iteration on the $\boldsymbol{A}^\dagger \boldsymbol{A}$ matrix. Of course this has to be performed for all required values of the energy E. A nice feature of the approach is that it works equally well in the multichannel case. In the latter case, when N_c channels are open, N_c independent solutions are required at each energy E (see section 4.2.1). These solutions are correctly given as N_c independent eigenvectors corresponding to the N_c lowest eigenvalues of $\boldsymbol{A}^\dagger \boldsymbol{A}$. Again, given the large separation from the following eigenvalues, the lowest ones are obtained efficiently by inverse iteration applied to a subspace of dimension N_c. It is important to note two further points.

- Since at a given energy the solution will not be zero at $r_{\max}$, it is necessary to keep the last B-spline, which is nonzero at $r_{\max}$, in the expansion, at variance with the diagonalization approach. It is precisely because of this last function that the B-spline expansion can smoothly follow the solution at each energy, between the energies corresponding to two consecutive knots of the wavefunction at $r_{\max}$, which are given by the diagonalization. The approach is like the solution of the vibrating string problem with only one boundary condition, at the origin, and a free boundary condition at the other end. This allows a solution for any frequency, at variance with the two-point boundary condition, which gives only the eigenfrequencies, and corresponds to the diagonalization process.

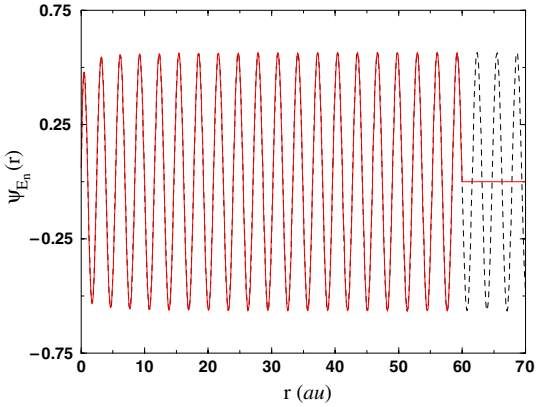


Figure 6. Comparison of a discretized continuum wavefunction (solid curve) with the exact Coulomb wavefunction (dashed curve) in H ($E_k = 54$ eV, $r_{\max} = 60$ au).

(This figure is in colour only in the electronic version)

- Because of the large separation between the eigenvalues corresponding to the required solutions and the rest, the corresponding eigenvectors are very close to those of the $\boldsymbol{A}$ matrix alone, and the same inverse iteration approach can be successfully employed with $\boldsymbol{A}$ [18]. Note that $\boldsymbol{A}$ is non-Hermitian (because of the nonzero spline at $r_{\max}$, in the kinetic energy term), and its eigenvalues are in general complex, so that solutions correspond in this case to minimum-modulus eigenvalues. In practice employing $\boldsymbol{A}$ saves a matrix multiplication, which is expensive; the $\boldsymbol{A}^\dagger\boldsymbol{A}$ formulation is Hermitian, however, and more accurate, and both are currently employed.

Unlike bound states, continuum states are ever-oscillating functions and thus vary up to infinity. Therefore, there are no reasons for not choosing a linear breakpoint sequence which divides the radial coordinates into equally spaced grid points. Although this tends to be the more appropriate sequence for the continuum wavefunctions, in many cases, if not all, one also needs accurate bound states. The idea then is to mix the two types of sequence: one concentrating points near the nucleus, which is better adapted to bound states, and the linear sequence for larger values of r necessary for continuum states (see e.g. expression (206) in appendix A.1).

The maximum energy $E_{\max}$ of the discretized continuum states will be set by both the B-spline order k and the breakpoint spacing, which can be roughly defined as $\Delta r = r_{\max}/N$ (see also the formal discussion of errors at the end of section 2.1). At large distances, a continuum wavefunction oscillates with a wavelength $\lambda = 2\pi/\sqrt{2E}$. The number of B-splines contained in one wavelength has to be large enough to fully reproduce the two sign switches. In practice, one has to achieve $\Delta r = \lambda/3$ for $k = 9$ or $\Delta r = \lambda/5$ for $k = 4$ (these numbers lead to an error on the wavefunction phase that never exceeds 0.1 rad). With such parameters, the typical accuracy for the phase shift is of the order of 10^{-8} rad. This is illustrated in figure 6, where we compare a highly oscillating discretized continuum wavefunction with the exact wavefunction. Note that there is absolutely no difference between the two wavefunctions inside the box.

We now address a final major problem known as the *normalization problem* [343] appearing in any discretization technique. On the one hand, discrete eigenstates are orthonormalized with respect to their (discrete) index as

$$\langle \psi_i | \psi_j \rangle = \delta_{ij}, \tag{16}$$

whereas continuum states, normalized on the (continuous) energy scale, satisfy

$$\langle \psi_E | \psi_{E'} \rangle = \delta(E - E'). \tag{17}$$

On the other hand, the computed discretized continuum states are normalized on the same level as the bound states, i.e. with respect to their index, because they originate from a diagonalization procedure. Any attempt to compute measurable quantities (such as cross-section, ionization rate, ...) involving a continuum state as the final state has to reestablish the correct normalization, that is find the coefficient $\mathcal{A}_n$ such that

$$\psi_{E_n} = \mathcal{A}_n \psi_n \tag{18}$$

where ψ_n is a discretized wavefunction and ψ_{E_n} the corresponding wavefunction normalized on the energy scale. We review here the two methods commonly used with B-splines. The first one compares the discretized state to the exact solution in the asymptotic region ($kr \gg 1$) where the analytical expression is often available (see e.g. equation (9) for the partial waves of a Coulomb potential). A standard fit will provide the normalization factor $\mathcal{A}_{n,l}$ as well as the non-coulombic phase σ_l if the potential is non-coulombic at the origin. This method is actually used in applications involving highly interfering nonperturbative processes [252,253,272] to check the accuracy of the discretized wavefunctions since, apart from the amplitude, it provides the most stringent test on the phase of the wavefunction. Also, it is generally used in the least-squares algorithm when solutions at any energy are obtained [19, 26]. Note that, as shown below, the square of the normalization factor is nothing other than the density of discretized continuum states: $\mathcal{A}_n^2 = \rho(E_n)$. The other technique, which is self-consistent with the discretization procedure, extracts the density of states from the set of eigenenergies without the need for external information. It can be shown [343, 344] that switching from a normalization with respect to the index n to a normalization on the energy scale is achieved by means of the relation

$$\psi_{E_n} = \left| \frac{\partial E(n')}{\partial n'} \right|_{n'=n}^{-\frac{1}{2}} \psi_n, \tag{19}$$

provided $E(n) = E_n$ is a monotonic function of n, which is satisfied in view of the approximate expression (11). Through a Taylor expansion up to third order, one obtains the formal expression

$$\left. \frac{\partial E(n')}{\partial n'} \right|_{n'=n} = \frac{E_{n+1} - E_{n-1}}{2} + \frac{1}{6} \left. \frac{\partial^3 E(n')}{\partial n'^3} \right|_{n'=n}. \tag{20}$$

The absence of the second-order term together with the weak curvature of the function $E(n)$ (see equation (11)) leads to the rather accurate approximation [343]

$$\rho(E_n) = \frac{2}{E_{n+1} - E_{n-1}}. \tag{21}$$

More accurate expressions for the density of states are obtained by polynomial interpolation of E_n and are listed in appendix A.2. Figure 7 clearly shows that the relative error produced by using the approximate expression (21) for evaluating the density of states is already very small. Making use of the five-, seven- or nine-point formula further lowers the error to a level close to machine accuracy with no extra cost, thus reinforcing the use of this self-consistent discretization/normalization procedure.

4. Atomic structure

After the early work by Shore [8–12] and Silverstone and collaborators [15–17] mentioned in the introduction, the first application of B-splines to atomic structure calculations was reported

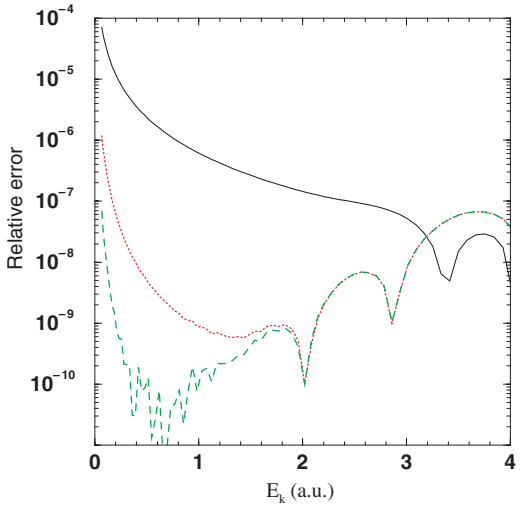


Figure 7. Relative error obtained when approximating the density of discretized continuum states with the three-point formula (equation (21)), solid curve, the five-point formula (see equations (208) of appendix A.2), dotted curve, and the seven-point formula (see equations (208)), dashed curve (linear–parabolic sequence, $r_{max} = 100$ au, $N = 500$ B-splines and $k = 7$).

(This figure is in colour only in the electronic version)

by Johnson and Sapirstein [104] in the framework of (relativistic) many-body perturbation theory ((R)MBPT) for GS properties of heavy alkali atoms using basis sets determined from a frozen-core DF approximation. Following this publication a number of applications have been reported primarily for lighter atoms using mainly CI methods but with the basis sets determined by a variety of methods, such as hydrogenic, model potential, frozen-core HF and density functional theories (DFT). However, more direct approaches to HF and DF calculations have also been tried as well as an implementation of the close-coupling (CC) approximation expressed in B-splines, called spline–Galerkin, as an approach to the calculation of bound and resonance levels. For two-electron systems Bottcher *et al* [32] have introduced a direct method based on collocation on a discrete lattice in space.

A common feature in all these methods (as well as in specific methods developed for the evaluation of continuum states) is the evaluation of two-electron integrals. This is a bottleneck in most computations of atomic structure properties. For this reason, the construction of efficient algorithms for the evaluation of two-electron integrals is a very important issue. This topic is discussed in the appendix. Here we present a brief summary of the most common methods used in atomic structure calculations and how they are implemented with B-spline basis sets. Later we shall present the methods developed for the evaluation of continuum states in atoms. New features of many-electron atoms, not discussed in section 3.3, are multichannel continua and Feshbach resonances. For the description of these types of continuum problem, one must use special methods that will be discussed in section 4.2. A selection of applications in atomic structure, autoionization and photoionization of atoms is presented at the end of this section.

4.1. Methods for bound and quasi-bound states

In this section we shall distinguish between those methods that explicitly provide the many-electron wavefunction (HF, DF, MBPT, CI etc) and methods based on the DFT that were originally designed to determine the electron density. As mentioned in the introduction, we

shall largely restrict ourselves to nonrelativistic problems and thus nonrelativistic basis sets in this review.

4.1.1. Wavefunction methods. Already Shore [12] noticed that one of the strong points of *B*-splines is their ability to describe excited and continuum states in addition to GSs. A simple way to make use of the effective completeness for both low and excited states is by using CI calculations. There are two ways to approach this problem. One is to expand a set of single-particle states, for example hydrogenic states, with different l-values in terms of *B*-splines (see section 3.1) and use the hydrogenic states to construct the CI matrix in a similar way to how it is done in any other basis. This was the approach introduced by Chang [47, 318]. Another approach is to use the *B*-spline basis directly as has been done for example by Chen and co-workers [34, 58] for two-electron systems using products of *B*-splines as basis functions. The advantage of the first approach is that computer programs constructed for use with conventional basis sets can be used more or less directly when it is kept in mind that the pseudo-continuum functions in the *B*-spline approach keep oscillating to the boundary at $r_{\max}$. Another advantage of the approach is that it is possible to determine directly, in the above example by comparing with hydrogenic eigenvalues, the largest n values that can be described adequately in a given basis. If the Davidson method is used to diagonalize the CI matrix the use of a physical basis can be an advantage. Since the method is based on perturbation theory the accuracy is increased when the eigenvector of the state sought is dominated by a single or a few terms. This is not likely to be the case if the expansion is directly in terms of *B*-splines. Another advantage is that if it is necessary to reduce the size of the matrix it is easy to remove the highest basis states for each value or some values of l since it is clear that they will have a small effect on the lowest states. If on the other hand the expansion is directly in terms of *B*-splines it is less obvious how to reduce the size except by changing the basis completely, i.e. choosing a smaller number of splines, which might necessitate a reduction in $r_{\max}$. However, there are advantages in the alternative approach. One advantage is that higher n-values can be reached with fewer splines than by using a single-particle basis. On the other hand it is necessary to check whether the wavefunction is accurate by changing the box size, for example, separately for all (or at least a few) eigenvalues. To reduce the size of the matrices Chen [58] has used a method in which the splines for one coordinate is restricted to a range above (outside in r) the index of the spline for the other coordinate minus a constant value. The rationale is that for states of the type 1snl the overlap between the two electrons is small, with the valence electron being located outside the 1s electron. Again the value of the constant must be found by trial and error. Chen has reported that compared with wavefunctions of Hylleraas or similar type, the error in properties such as transition probabilities, which test other regions of the wavefunctions than the energies, can be smaller with an *B*-spline basis even if the energies are better in the Hylleraas approach. Although both wavefunctions are obtained variationally, the overall accuracy of the wavefunction is apparently better in the *B*-spline approach, presumably because of the flexibility of the basis at each r. Another notable property of the resulting wavefunctions is that the higher ones deteriorate more slowly than is the case for Hylleraas wavefunctions.

The direct expansion method is close to the usual practice in quantum chemistry but one difference from the use of Gaussian basis states in quantum chemistry is that two-electron integrals involving Gaussians can be calculated 'on the fly', while two-electron integrals involving individual splines are more time consuming to calculate and therefore for reasons of speed should be stored in memory. In practice this means that the difference in speed compared with the use of a hydrogenic basis, say, expanded in splines, for which the same applies, is less significant than could be expected. However, symmetry properties can be exploited to reduce the number of elementary integrals that need to be stored and here *B*-splines have advantages.

Brage and Froese Fischer [20,24] have introduced a direct method called spline–Galerkin in which the wavefunctions of the outer electrons are expanded directly in B-splines . Since this method is less familiar than the CI approach we shall discuss it in some detail. The advantages of the method are the same as above, namely that the resulting matrix size is relatively small and that high Rydberg states can be readily described. The method has been applied to bound and resonance states in Ca but the interaction with the continuum was not included for the latter. In the method, the wavefunction is written in a multichannel form with pseudo-states. In the nonrelativistic case it is

$$\Psi(\gamma LS) = \sum_{i=1}^{M_\mathrm{p}} c(i)\phi(\alpha_i LS) + \sum_{i=1}^{M_\mathrm{c}} \overline{\Phi}(\gamma_i LS). \tag{22}$$

All functions that are overlined are unnormalized. The $\overline{\Phi}(\gamma_i LS)$ 'channel' configuration state functions (CSF) can be written

$$\overline{\Phi}(\gamma_i LS) = |(\mathrm{target}_i \bullet |\overline{n_i l_i}\rangle)\,{}^{(2S+1)}L\rangle \qquad \text{for} \quad i = 1, M_\mathrm{c}. \tag{23}$$

The $|\overline{n_i l_i}\rangle$ denotes an unnormalized channel function

$$|\overline{n_i l_i}\rangle = \overline{P}_{n_i l_i}|l_i s\rangle \tag{24}$$

where $|l_i s\rangle$ denotes the spin-angular part of the one-electron function. In the calcium case, the target is a CSF with one electron outside the closed core. The notation implies that the channel function is coupled to the target according to the usual angular momentum rules, to form a total L and S. The pseudo-states $\phi(\alpha_i LS)$ arise due to orthogonality restrictions on the channels. The only unknowns in equation (22) are the radial channel functions, $\overline{P}_{n_i l_i}$, and the coefficients, $c(i)$, for the pseudo-states.

The B-spline basis is introduced through the expansion of the radial part, $\overline{P}_{n_i l_i}$, of the channel functions

$$\overline{P}_{n_i l_i}(r) = \sum_{j=1}^{N} a_j^i B_j(r). \tag{25}$$

By inserting the approximate form of the wavefunction equation (22) in the Schrödinger equation, we obtain

$$\Psi_\mathrm{res} = \big(\mathcal{H} - E\big)\Psi(\gamma LS) \approx 0. \tag{26}$$

According to the Galerkin condition [339], the residual of the right-hand side should be orthogonal to the solution space. To impose this, a set of basis functions, so-called test functions, τ, that span the solution space is chosen. This set can be chosen to consist of all pseudo-states

$$\tau^p(i) = \phi(\gamma_i LS) \qquad \text{for} \quad i = 1, M_\mathrm{p} \tag{27}$$

and 'spline channels' obtained by replacing the radial part of the channel functions with a B-spline

$$\tau_j^c(i) = |(\mathrm{target}_i \bullet B_j|l_i s\rangle)\,{}^{(2S+1)}L\rangle \qquad \text{for} \quad i = 1, M_\mathrm{c} \qquad \text{and} \qquad j = 1, N. \tag{28}$$

The Galerkin conditions then lead to the equations

$$\langle \tau_i | \Psi_\mathrm{res}\rangle = 0 \qquad \text{for} \quad i = 1, M_\mathrm{p}$$
$$\langle \tau_i^j | \Psi_\mathrm{res}\rangle = 0 \qquad \text{for} \quad i = 1, M_\mathrm{c}; \quad j = 1, N$$

which leads to a generalized eigenvalue problem, of the form

$$(\boldsymbol{H} - E\boldsymbol{S})\boldsymbol{c} = 0 \tag{29}$$

where $\boldsymbol{H}$ is the multichannel interaction matrix and $\boldsymbol{S}$ is the overlap matrix, consisting of diagonal blocks of the overlap matrix, $\boldsymbol{B}$

$$B_{kl} = \langle B_k | B_l \rangle \tag{30}$$

one for each channel. The solution vector, $\boldsymbol{c}$, can be written

$$\boldsymbol{c} = \begin{pmatrix} \boldsymbol{c}^p \\ \boldsymbol{a}(1) \\ \boldsymbol{a}(2) \\ \vdots \\ \boldsymbol{a}(M_c) \end{pmatrix} \tag{31}$$

where $\boldsymbol{c}^p$ is the column vector of pseudo-coefficients

$$\boldsymbol{c}^p = (c(1), c(2), \ldots, c(M_p))^t \tag{32}$$

and each $\boldsymbol{a}(i)$ is a column vector of B-spline coefficients

$$\boldsymbol{a}(i) = \big(a(i)_1, a(i)_2, \ldots, a(i)_N\big)^t$$

for the unnormalized, radial channel functions. Choosing unnormalized channels, thereby including the weights of the channel CSFs in the spline coefficients, leads to an eigenvalue problem in which the pseudo-coefficients and the B-spline coefficients are introduced on an equal footing.

A major complication, for the bound case, occurs when orthogonality conditions are applied to the radial channel functions, since this in principle destroys the eigenvalue character of the problem. This problem can be solved by a transformation described by Landtman *et al* [100]. The one-electron Schrödinger equation with an orthogonality constraint can be written

$$(\mathcal{H} - E)y(r) + \varepsilon x(r) = 0 \tag{33}$$

where ε is the off-diagonal Lagrange multiplier, introduced through the constraint

$$\langle y(r) | x(r) \rangle = 0. \tag{34}$$

For simplicity we shall refer to the $x(r)$s as core orbitals.

By introducing the B-spline representation

$$x(r) = \sum_i b_i B_i(r) \tag{35}$$

and

$$y(r) = \sum_j a_j B_j(r) \tag{36}$$

and using the Galerkin condition, with the B-splines as test functions, equation (33) transforms into the matrix equation

$$(\boldsymbol{H} - E\boldsymbol{B})\boldsymbol{a} + \varepsilon \boldsymbol{B}\boldsymbol{b} = 0 \tag{37}$$

where $\boldsymbol{b}$ and $\boldsymbol{a}$ are the spline expansion vectors for x and y, respectively.

By multiplying equation (37) by the row vector $\boldsymbol{b}^t$ we obtain, assuming $\boldsymbol{b}^t\boldsymbol{B}\boldsymbol{b} = 1$,

$$\varepsilon = -\boldsymbol{b}^t \boldsymbol{H} \boldsymbol{a}; \tag{38}$$

substituting this into equation (37), we obtain

$$(\boldsymbol{H}' - E\boldsymbol{B})\boldsymbol{a} = 0 \tag{39}$$

where

$$\boldsymbol{H}' = \boldsymbol{H} - \boldsymbol{B}\boldsymbol{b} \otimes \boldsymbol{b}^t \boldsymbol{H}. \tag{40}$$

The $\otimes$ notation denotes the outer product between two vectors, in this case $\boldsymbol{b}(k)$ and $\boldsymbol{b}(k)^t$. This equation corresponds to projecting out the $\boldsymbol{b}$ vector, corresponding to the core orbital $x(r)$, from the solution space of $\boldsymbol{H}$, and will introduce a zero eigenvalue [100].

A related approach has been introduced by Lambropoulos and co-workers [245,327] for the calculation of two-photon single and double photoionization and Brage and Froese Fischer [24] have introduced relativistic corrections via the Breit–Pauli CI approach after having calculated nonrelativistic wavefunctions using the B-spline expansions described above for the relevant states. A major shortcoming of the method lies in the choice of HF target states, which means that core–valence and core–core interactions are neglected. Brage and Froese Fischer used a model potential designed to take core polarization into account to compensate for this, as mentioned later.

To return to the CI methods we need to mention several methods which have been used to construct the basis sets for CI calculations. For two-electron systems, the use of a hydrogenic basis is logical in particular if autoionization or other properties associated with the single-particle limits are studied. However, for systems with more than two electrons a hydrogenic basis is fairly far away from the best possible single-particle description, which is the HF approach. Before discussing HF basis sets, it should be mentioned that there is a simpler way to expand the use of B-splines to more than two electrons, namely the use of model potentials to mimic the presence of a core. This is particularly useful if the core has a closed-shell structure. For example, there are fairly simple model potentials available to represent a $1s^2$ core [366] and van der Hart and Hansen [204, 207] as well as Chen [55, 73] used such an approach to calculate the four-electron Be-like $1s^2nln'l'$ systems with B-splines. Van der Hart and Hansen showed that the accuracy was better than obtained by use of an HF potential for the core although it of course is worse than using a real *ab initio* method. However, it is possible to construct more elaborate model potentials. In fact Chang [47] in the first CI calculations using B-splines for bound states in Be combined a frozen-core HF basis with a model potential introduced by Laughlin and Victor [385]. More recently, Laughlin has worked on constructing model potentials for Ca^{2+} which reproduce all the observed energies in the Rydberg series in Ca^+ [61, 364] and such potentials have been used to calculate both pseudo-two- and three-electron systems in Ca [61, 62] and Ca^- [57, 59]. States in negative ions of other alkaline earths [63, 64] have been determined also and Chen has used a similar potential to describe the 1s and the $1s^2$ cores in Li [68] and Be [73] as well as Be-like systems [55]. For maximum accuracy Laughlin and Hansen [61] showed that it can be advantageous to adjust the model potential in such a way as to give a good fit to certain terms in the neutral spectrum. In Ca, for example, where the 3d electron is very sensitive to its environment, it was found that constraining the d potential to fit the energy of the $3d^2\ ^3F$ term in Ca I in addition to fitting the d series in Ca^+ gave a considerably better description of the Ca spectrum than just fitting the potential to the d series and the necessary change in potential was so small that the fit to the d series in Ca^+ was not appreciably degraded [61].

The use of HF/DF basis sets has mainly been concerned with solving the HF/DF equations for one electron outside a frozen core. In this way the differential equation is linear in the orbital sought and no iteration to self-consistency is required. Johnson, Sapirstein and co-workers [99] were the first to obtain DF basis sets for heavy alkali atoms in this way while Chang [47, 318, 319] and Landtman and Hansen and co-workers [56, 118, 328] introduced frozen-core HF basis sets and the latter studied the problem of the orthogonality between the members of the basis set and the frozen-core orbitals with the same l-value [100]. It turns out

that every orthogonality constraint gives rise to a spurious eigenvector with zero eigenvalue in the eigenvalue spectrum, which should be removed from the basis. This was discussed already in connection with the spline–Galerkin method. Bentley [102] has discussed a method to avoid the nonsymmetric matrix obtained in the previous approach. The latter method has been implemented in Amsterdam [103].

Basis sets of *B*-splines for second Born and *R*-matrix calculations of scattering in two-electron systems have also been reported [40, 249, 250]. Since the *R*-matrix approach is based on solving the Schrödinger equation in a box, it could be expected that *B*-splines would be very useful as a basis for the *R*-matrix approach. The use of *B*-splines in *R*-matrix calculations was first explored by van der Hart [250], who showed that excellent results can be obtained for scattering on one-electron systems by use of the *B*-spline approach for calculations of bound and quasi-bound states of the two-electron system [204] described in section 4.3.2. The completeness of the two-electron description meant that no 'Buttle' correction is required which saves a large amount of computational effort. The formulation due to van der Hart was basically restricted to two- or pseudo-two-electron systems. One difficulty is that, conventionally, the *R*-matrix approach has been formulated on the basis of an orthogonal basis set, which the *B*-spline basis is not. Recently, Zatsarinny and Froese Fischer [40] have extended the *R*-matrix approach to allow the use of a nonorthogonal basis and applied such an approach to the calculation of photoionization of Li. The use of nonorthogonal orbitals solves a number of problems with the conventional *R*-matrix procedure, for example the occurrence of pseudo-resonances. The *B*-splines replace numerical basis functions constructed to be orthogonal to the bound orbitals used in the construction of the target states. To compensate for this restriction correlation functions must be introduced, which for a complex atom can lead to a considerable number of extra functions. To use *B*-splines instead has the advantage that they form a complete set on the interval representing the inner region. The disadvantage is that, as mentioned, the basis is nonorthogonal, but efficient programs were developed to calculate the matrix elements of the Hamiltonian so Zatsarinny and Froese Fischer have reduced this problem to manageable proportions. The initial state and the target states were obtained with the numerical MCHF technique, also using nonorthogonal orbitals, if necessary, which reduces the size of the expansions. The bound orbitals were expanded in a *B*-spline basis of order eight with 105 splines in a box determined by the requirement that the value of the orbitals is less than 10^{-5} at the boundary, which also is chosen as the boundary for the inner region. Zatsarinny and Froese Fischer found that the extra time needed for the calculation of the nonorthogonal matrix elements was largely compensated by the smaller expansions required for a good description of the initial and final wavefunctions in the case of Li and there are no conceptual difficulties in extending the approach to more electrons.

The use of *B*-splines for HF/DF calculations as such, i.e. optimizing all orbitals, have also started. Froese Fischer and co-workers have worked on an HF solution for He [21, 22, 44] and on solving the radial Dirac equation [23] in an *B*-spline basis.

Johnson and Sapirstein have pionered the use of *B*-splines for relativistic calculations. They have pointed out that one important advantage is that the spectrum of eigenvalues separates cleanly into positive and negative energies. That this separation exists means that use of projection operators to suppress the negative eigenstates ($\epsilon \leqslant -2mc^2$) as required for example in the no-pair Hamiltonian advocated by Sucher [386] can be accomplished simply by leaving the negative eigenstates out of the basis. On the other hand in QED calculations the negative eigenstates might need to be kept [82].

The necessity to introduce projection operators in MBPT has the consequence that a description in terms of one-electron orbitals has been the preferred method in the past. Also the relativistic CI calculations by Chen, Cheng, Johnson, Sapirstein and co-workers relies

on the 'no-pair' Hamiltonian. In one of the first papers by Johnson, Sapirstein and co-workers [99], these authors gave a number of examples to show the equivalence between the conventional MBPT procedure, which often require an estimate of the 'remainder' not included in integration, and the summation over a finite B-spline basis set. In the calculations, the particular basis set obtained as an expansion in B-splines is converted to numerical orbitals. This linear transformation does not change the completeness but means that the orbital can be used directly with computer programs developed for any other type of L^2 basis defined in a box with the proviso, mentioned earlier, that the algorithms must be able to cope with the rapid oscillation of the high energy B-splines at the outer boundary of the box. Of course in other relativistic applications, it might be possible to work directly in a B-spline basis.

The construction of a relativistic B-spline basis is somewhat different from the nonrelativistic case and is beyond the scope of the present review. However, we mention that the boundary conditions on the outside of the box have to be treated differently because of the so called 'Klein paradox' [99]. Spurious states have also been observed [383] when the Dirac equation is expanded in finite basis sets. To our knowledge similar phenomena have not been encountered in nonrelativistic calculations. The boundary conditions used by Johnson and Sapirstein are chosen to avoid these problems. In relativistic calculations, it is advantageous to use a real nuclear potential instead of the Coulomb potential with its singularity at the origin, which is difficult to handle also with B-splines. However, in general B-splines can be advantageous because it is possible to distribute knot points over and close to the nucleus, which can be necessary for accuracy of the $s_{1/2}$ and $p_{1/2}$ orbitals, in particular [99].

4.1.2. Density functional theory. An approach which has become popular because of the favourable compromise between accuracy and computational cost, allowing the treatment of large many-electron systems, is the DFT. Since extensive presentations of DFT are available [393–395], we shall give here only working equations and comments on the implementation in the B-spline basis.

Employing the usual Kohn–Sham (KS) scheme, DFT amounts to solving the coupled one-particle equations

$$h_{\mathrm{KS}}\varphi_i = \varepsilon_i\varphi_i \tag{41}$$

$$h_{\mathrm{KS}} = -\tfrac{1}{2}\nabla^2 + V_{\mathrm{ext}} + V_{\mathrm{C}} + V_{\mathrm{XC}} \tag{42}$$

$$\rho = \sum_i \varphi_i^*\varphi_i \tag{43}$$

$$V_{\mathrm{C}}(r) = \int \frac{\rho(r_2)}{r_{12}}\,\mathrm{d}r_2 \tag{44}$$

$$V_{\mathrm{XC}}(r) = V_{\mathrm{XC}}[\rho](r). \tag{45}$$

The potential depends on the density ρ through the Coulomb V_{C} and exchange–correlation term V_{XC}, so the equations are solved through a classic SCF iteration scheme.

The total energy is then computed according to

$$E = \sum_i \langle\varphi_i, (-1/2\nabla^2 + V_{\mathrm{ext}})\varphi_i\rangle + 1/2\int \frac{\rho(r_1)\rho(r_2)}{r_{12}}\,\mathrm{d}r_1\,\mathrm{d}r_2 + E_{\mathrm{XC}}[\rho] \tag{46}$$

where E_{XC} is a universal functional of the density and V_{XC} is the functional derivative of E_{XC} with respect to ρ

$$V_{\mathrm{XC}} = \frac{\delta E_{\mathrm{XC}}[\rho]}{\delta\rho}. \tag{47}$$

Usually the local density approximation (LDA) is employed, so that

$$E_{\mathrm{XC}}[\rho] = \int \varepsilon_{\mathrm{XC}}(\rho(r))\rho(r)\,\mathrm{d}r \tag{48}$$

$$V_{\mathrm{XC}}(r) = V_{\mathrm{XC}}(\rho(r)). \tag{49}$$

$\varepsilon_{\mathrm{XC}}(\rho)$ has been accurately computed at selected densities for the free-electron gas, and has been analytically interpolated. The most popular parametrization is the VWN [396], which is widely used. Generalizations include dependence of E_{XC} on the gradient $\nabla\rho$, or more recently on other functionals such as the kinetic energy or the HF exchange, and a wealth of parametrizations are available. Most of these potentials decay exponentially to zero, like the density itself, due to an inadequate treatment of the self-interaction term in the exchange. For GS energies this has little importance, but it has a larger influence on excited states [167] or electromagnetic response properties [405]. A few potentials including the correct asymptotic Coulomb tail have been devised [399–402], notably the LB94 potential [399], that currently is employed in photoionization studies. We note however that development of potentials for excited-state properties has just begun [397, 398, 403, 404, 414]. DFT is rigorously a GS theory. Calculations on excited states may be performed at the single-particle (LDA) level, by promoting one or more electrons from an occupied orbital φ_i to a vacant one φ_a, employing a fixed h_{KS}, defined by the density.

The extension of DFT to the treatment of time-dependent phenomena, time-dependent DFT (TDDFT), offers a more accurate route to the description of excitation processes, as well as of dynamical response properties of the system [406–410]. Although important applications to nonlinear response properties and multiphoton processes have been carried out, we shall limit the discussion here to the linear response equations, employing the so-called adiabatic LDA (ALDA), which gives rise to the TDLDA approach. In the time-dependent KS equations

$$\left\{-\frac{1}{2}\nabla^2 + V_{\mathrm{s}}(t)\right\}\varphi_i(t) = \mathrm{i}\frac{\partial\varphi_i}{\partial t} \tag{50}$$

with

$$V_{\mathrm{s}}(t) = V_{\mathrm{ext}}(t) + V_{\mathrm{C}}(\rho(t)) + V_{\mathrm{XC}}[\rho](t) \tag{51}$$

one first approximates (ALDA)

$$V_{\mathrm{XC}}[\rho](r,t) = V_{\mathrm{XC}}(\rho(r,t)) \tag{52}$$

that is, the complicated functional dependence on the density of the time-dependent V_{s} potential is replaced by the same LDA local dependence in space and time. Then with a periodic external potential V_{ext} one looks for solutions [408]

$$\varphi_i(t) = \varphi_i + \varphi_i^{(1,+)}\mathrm{e}^{\mathrm{i}\omega t} + \varphi_i^{(1,-)}\mathrm{e}^{-\mathrm{i}\omega t} \tag{53}$$

to equations obtained from first-order time-dependent perturbation theory

$$(h_{\mathrm{KS}} - \varepsilon_i \pm \omega - \mathrm{i}\eta)\varphi_i^{(1,\pm)} = P V_{\mathrm{SCF}}^{\pm}\varphi_i \tag{54}$$

$$\delta\rho^{\pm} = \sum_i \varphi_i^{\pm}\varphi_i^{*} + \varphi_i\varphi_i^{(1,\mp)*} \tag{55}$$

$$V_{\mathrm{SCF}}^{\pm} = V_{\mathrm{ext}} + \delta V_{\mathrm{C}}^{\pm} + \delta V_{\mathrm{XC}}^{\pm} \tag{56}$$

$$\delta V_{\mathrm{C}}^{\pm} = \int \frac{\delta\rho^{\pm}(r_2)}{r_{12}}\,\mathrm{d}r_2 \tag{57}$$

$$\delta V_{\mathrm{XC}}^{\pm} = \left.\frac{\partial V_{\mathrm{XC}}(\rho)}{\partial\rho}\right|_{\mathrm{GS}} \delta\rho^{\pm}. \tag{58}$$

The projection operator P orthogonalizes with respect to the occupied states φ_i, which are the solutions of the unperturbed LDA equations for the GS, and ω is the total excitation energy. The infinitesimal imaginary term η originates from the adiabatic switching of the perturbation at $t \to \infty$ [411], and fixes outgoing boundary conditions for $\varphi_i^{(1,-)}$ in the continuum case. Starting with $V_{\rm SCF} = V_{\rm ext}$, equations (54)–(58) are solved self-consistently. Dipole matrix elements can be obtained substituting the converged $V_{\rm SCF}$ for the dipole operator.

The implementation of the atomic problem in the B-spline basis is very simple [167]. Basically only single radial equations, both homogeneous and inhomogeneous, (41), (54), need to be solved. These are easily recast into matrix algebraic problems for the expansion coefficients in the basis. The same basis set expansion is employed to solve the Poisson equation

$$\nabla^2 V_{\rm C} = -4\pi\rho \tag{59}$$

which is an accurate and convenient means for the evaluation of the Coulomb potential given the density ρ, in both the LDA and TDLDA iterations.

A problem arises with the calculation of highly excited Rydberg states, for instance those required to describe autoionization series. Since Rydberg orbitals quickly become very diffuse, they fall outside the range $r_{\rm max}$ of the basis. This is solved by fitting the corresponding radial functions with a pure Coulomb tail at $r_{\rm max}$ [164, 167]. In practice at each energy $E_i < 0$ one evaluates the bound Coulomb function (with a B-spline basis from $r_{\rm max}$ to a suitable infinity, say 10 000 au, and an exponential grid), and imposes the corresponding logarithmic derivative boundary condition at $r_{\rm max}$ to the inner radial solution, which is easily implemented by taking a suitable linear combination of the last two splines. This gives an excellent reproduction of, at least, the lowest 15 members of a Rydberg series.

Problems of stability in the SCF TDLDA iterations are often encountered if the $V_{\rm SCF}$ potential is substituted directly in (54), and much better convergence is achieved employing convergence accelerator schemes, like Aitken δ^2 [412] or the direct inversion of iterative subspace (DIIS) method of Pulay [306, 413].

4.2. Evaluation of continuum states

In contrast to the methods presented in the previous sections, evaluation of continuum states in many-electron atoms is not a standard problem in atomic physics. Therefore, they require a more detailed presentation than is necessary for bound states. Most methods have been developed in the context of discretization techniques and are well adapted for implementation with B-spline basis sets. We shall begin the discussion with the problem of multichannel continua, which is first encountered for many-electron atoms. Then we shall present the basic formalism for the description of Feshbach resonances in the continuum. This is also a feature specific to many-electron atoms.

4.2.1. The multichannel case. In order to simplify the presentation we shall consider the problem of a free electron in the presence of a hydrogen-like ion (extension to many-electron ions is straightforward). If the system has $N_{\rm c}$ open channels, there exist $N_{\rm c}$ independent boundary conditions that correspond to $N_{\rm c}$ independent continuum states with the same energy. For instance, for continuum states above the $N = 2$ ionization threshold of He, there exist three open channels with total angular momentum $L = 0$. In one channel, the bound electron lies in the 1s orbital of He^+ and the unbound electron (with $l = 0$) has an asymptotic kinetic energy ϵ. For the same total energy, we have two additional channels in which the bound electron lies, respectively, in the 2s and 2p orbitals of He^+ and the unbound electron (with $l = 0$ and 1,

respectively) has a kinetic energy $\epsilon - E_{\mathrm{He}^+(1s)} + E_{\mathrm{He}^+(n=2)}$. In the general case, these boundary conditions can be written

$$\Psi^{\Omega-}_{\mu E}(\boldsymbol{r}_1, \boldsymbol{r}_2) \to k_\mu^{-1/2}\Theta\left[\sum_{\mu'=1}^{N_c} \psi^{\Omega}_{\mu'}(\boldsymbol{r}_1, \hat{r}_2) r_2^{-1}\left(\delta_{\mu\mu'}\mathrm{e}^{\mathrm{i}\theta_\mu} - S^{\Omega\dagger}_{\mu\mu'}\mathrm{e}^{-\mathrm{i}\theta_{\mu'}}\right)\right] \tag{60}$$

where

$$\theta_\mu = k_\mu r - \frac{1}{2}l_\mu\pi + \frac{Z-1}{k_\mu}\ln(2k_\mu r) + \arg\Gamma(l_\mu + 1 - \mathrm{i}(Z-1)/k_\mu) \tag{61}$$

and $k_\mu^2 = 2\epsilon_\mu \geqslant 0$, Θ is the symmetrization (antisymmetrization) operator for singlet (triplet) states and the channel functions, $\psi^{\Omega}_{\mu'}$, are combinations of the target states with the angular functions of the scattered electron forming eigenfunctions of $\boldsymbol{L}^2$, $\boldsymbol{S}^2$, L_z, S_z and π (the parity operator). The corresponding eigenvalues are $\Omega = L, S, M_L, M_S, \pi$. The channel index μ is defined as $\mu = n_0 l_0 l$, where $n_0 l_0$ are the principal and the angular momentum quantum numbers of the target, and l is the angular momentum quantum number of the continuum electron. Alternatively, one may use standing wave boundary conditions of the form

$$\Psi^{\Omega}_{\mu E}(\boldsymbol{r}_1, \boldsymbol{r}_2) \to k_\mu^{-1/2}\Theta\left[\sum_{\mu'=1}^{N_c} \psi^{\Omega}_{\mu'}(\boldsymbol{r}_1, \hat{r}_2) r_2^{-1}\left(\delta_{\mu\mu'} f_{\mu E}(r_2) - \pi K^{\Omega}_{\mu\mu'}(E) g_{\mu' E}(r_2)\right)\right], \tag{62}$$

where $f_{\mu E}$ and $g_{\mu E}$ are regular and irregular Coulomb functions of energy E. The K-matrix is related to the usual scattering matrix S through the equation

$$\boldsymbol{I} - \mathrm{i}\pi\boldsymbol{K} = (\boldsymbol{I} + \mathrm{i}\pi\boldsymbol{K})\boldsymbol{S}. \tag{63}$$

Diagonalization of $\mathcal{H}$ in a basis $\{\Theta(\psi^{\Omega}_{\mu}(\boldsymbol{r}_1, \hat{r}_2)\varphi_{\mu k}(r_2)),\ \mu = 1, 2, \ldots\}$, including at least all open channels μ, does not reproduce the asymptotic behaviour of equations (60) or (62) because, as a result of the arbitrary finite normalization of the basis elements, interchannel coupling is not described properly. The continuum states resulting from such a diagonalization satisfy arbitrary boundary conditions of the form [345]

$$\Psi^{\Omega}_{\mu E_n}(\boldsymbol{r}_1, \boldsymbol{r}_2) \to k_\mu^{-1/2}\Theta\left[\sum_{\mu'=1}^{N_c} \psi^{\Omega}_{\mu'}(\boldsymbol{r}_1, \hat{r}_2) r_2^{-1}\left(a^{\Omega}_{\mu\mu' n} f_{\mu' E}(r_2) + b^{\Omega}_{\mu\mu' n} g_{\mu' E}(r_2)\right)\right]. \tag{64}$$

Moreover, only a single independent solution is obtained at each energy. A possible solution is to employ the least-squares approach [164] (see section 3.3), which can be applied to the multichannel case giving the full set of N_c independent solutions at each prefixed energy E. The coefficients $a^{\Omega}_{\mu\mu' n}$ and $b^{\Omega}_{\mu\mu' n}$ define two matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ that can be determined by fitting the obtained solutions to linear combinations of the asymptotic regular and irregular functions with $E = E_n$. Thus, the wavefunction with the correct boundary conditions of equation (62) can be related to linear combinations of those with the arbitrary ones through the linear transformation $\boldsymbol{\Psi}' = \boldsymbol{A}^{-1}\boldsymbol{\Psi}$ (see [164]). The resulting wavefunctions have the proper K-matrix normalization, with $\pi\boldsymbol{K} = -\boldsymbol{A}^{-1}\boldsymbol{B}$. The S-matrix is then obtained from equation (63). A necessary condition for the method to work is that the basis set must reproduce the asymptotic region. This condition can be easily satisfied with B-splines by choosing a box size large enough to contain a significant part of the asymptotic region; in contrast, it is practically impossible to satisfy with conventional L^2 basis sets (see e.g. [345]). This method has been applied successfully to study photoionization of H^- and He [164–166].

A different approach to the multichannel problem has been discussed in [346, 347]. In a first step, one neglects the coupling between the different channels. Thus the multichannel problem is transformed into a sum of single-channel problems that can be trivially discretized following a procedure similar to that described in section 3.3. For instance, one can start by

defining orthogonal uncoupled continuum states $\zeta_{\mu n_\mu}$ (UCSs) which result from diagonalizing $\mathcal{H}$ for each μ in a basis of configurations $\{\Theta(\psi_\mu^\Omega(\boldsymbol{r}_1, \hat{r}_2)\varphi_{\mu k}(r_2))\}$. The UCSs are eigenfunctions of the zero-order uncoupled Hamiltonian:

$$\left(\sum_{\mu'} P_{\mu'}\mathcal{H}P_{\mu'} - E\right)\zeta_{\mu E} = 0 \tag{65}$$

where P_μ is a projection operator defined by $P_\mu\zeta_{\mu E} = \zeta_{\mu E}$ and $P_\mu P_{\mu'} = \delta_{\mu\mu'}P_{\mu'}$. In a second step, the coupling between channels is introduced using a Lippman–Schwinger formalism, which explicitly includes the correct boundary conditions:

$$\Psi^{\Omega-}_{\mu E^\Omega_{n_\mu}} = \rho_\mu^{1/2}(E^\Omega_{n_\mu})\left(\zeta_{\mu n_\mu} + \sum_{\mu' n'_{\mu'}}\sum_{\mu'' n''_{\mu''}} \langle\zeta_{\mu' n'_{\mu'}}|G^-(E^\Omega_{n_\mu})|\zeta_{\mu'' n''_{\mu''}}\rangle\langle\zeta_{\mu'' n''_{\mu''}}|V|\zeta_{\mu n_\mu}\rangle\zeta_{\mu' n'_{\mu'}}\right) \tag{66}$$

where

$$G^-(E) = \lim_{\eta\to 0}\frac{1}{E - \mathcal{H} - \mathrm{i}\eta} \tag{67}$$

$$V = \sum_{\substack{\mu\mu' \\ \mu\neq\mu'}} P_\mu\mathcal{H}P_{\mu'} \tag{68}$$

and ρ_μ is the density of states associated with the UCS in channel μ. This density of states can be evaluated using the procedures described in section 3.3 for the single-channel case. The G^--matrix elements can be easily obtained following the procedures described in [346–348], which we briefly summarize here. One starts by defining the Green operator associated with the uncoupled Hamiltonian of equation (65):

$$G_0^-(E) = \lim_{\eta\to 0}\frac{1}{(E - \sum_{\mu'} P_{\mu'}\mathcal{H}P_{\mu'} - \mathrm{i}\eta)}. \tag{69}$$

Projecting G_0^- onto the UCS basis, the latter operator takes the form

$$G_0^-(E = E^\Omega_{n_\mu}) = \sum_{\mu'}\sum_{\substack{n'_{\mu'} \\ n'_{\mu'}\neq n_\mu}} \frac{|\zeta_{\mu' n'_{\mu'}}\rangle\langle\zeta_{\mu' n'_{\mu'}}|}{(E^\Omega_{n_\mu} - E^\Omega_{n'_{\mu'}})} + \mathrm{i}\pi\sum_\mu \rho_\mu(E^\Omega_{n_\mu})|\zeta_{\mu n_\mu}\rangle\langle\zeta_{\mu n_\mu}|. \tag{70}$$

The Green functions $G^-(E)$ and $G_0^-(E)$ can be related using standard scattering theory

$$G^-(E) = G_0^-(E) + G_0^-(E)VG^-(E). \tag{71}$$

Then, using equation (70) and the closure relation written in terms of the UCS, the above formal relation leads to

$$\sum_{\mu'', n''_{\mu''}} \mathcal{C}_{\mu' n'_{\mu'}\mu'' n''_{\mu''}}\langle\zeta_{\mu'' n''_{\mu''}}|G^-(E)|\zeta_{\mu n_\mu}\rangle = \mathcal{D}_{\mu' n'_{\mu'}} \tag{72}$$

where

$$\mathcal{C}_{\mu' n'_{\mu'}\mu'' n''_{\mu''}} = \delta_{\mu''\mu'}\delta_{n''_{\mu''} n'_{\mu'}} - \Xi_{\mu'}(E^\Omega_{n_{\mu'}})\langle\zeta_{\mu' n'_{\mu'}}|V|\zeta_{\mu'' n''_{\mu''}}\rangle \tag{73}$$

$$\mathcal{D}_{\mu' n'_{\mu'}} = \delta_{\mu'\mu}\delta_{n'_{\mu'} n_\mu}\Xi_{\mu'}(E^\Omega_{n_{\mu'}}) \tag{74}$$

and

$$\Xi_{\mu'}(E^\Omega_{n_{\mu'}}) = \begin{cases} \mathrm{i}\pi\rho_{\mu'}(E^\Omega_{n'_{\mu'}})\delta_{n_\mu n'_{\mu'}} & \text{for} \quad E = E^\Omega_{n'_{\mu'}} \\ 1/(E - E^\Omega_{\mu' n'_{\mu'}}) & \text{for} \quad E \neq E^\Omega_{n'_{\mu'}}. \end{cases} \tag{75}$$

Therefore, the $G^-(E)$-matrix elements appearing in equation (66) are the solutions of the system of linear equations given in equation (72), which can be solved using standard numerical procedures. The coefficient matrix $\boldsymbol{C}$ multiplying the unknowns is the same for all μ, so that each system of equations differs only in the right-hand side column vector $\boldsymbol{D}$. Therefore, only one matrix inversion is required to solve equation (72). The need for the closure relation to write equations (70) and (72) in terms of the UCS basis means that a complete basis is required. This is what makes B-splines superior to other L^2 basis sets because convergence can be easily checked within the box (see also section 5.1.2).

4.2.2. Resonances in the continuum. The continuum of many-electron atoms contains resonant states. These states may arise when two or more electrons are excited simultaneously or when an inner-shell vacancy is created (Feshbach resonances). The corresponding energies lie above the ionization threshold and, consequently, belong to the continuum spectrum. However, the electron density of the resonant states is highly localized around the nucleus, which makes them substantially different from the nonresonant continuum states. As mentioned above, a direct approach to describe a resonant continuum spectrum requires the use of large values of $r_{\max}$ (i.e. a large density of states). An alternative approach to avoid this problem is to work in the framework of the Feshbach formalism. The advantage of this theory is that the resonant and nonresonant contributions to the wavefunctions are handled separately and, therefore, one can use specific methods to calculate each part. Furthermore, the Feshbach theory provides the appropriate framework in which resonance parameters (energy position, autoionization width, Fano profiles, etc) can be accurately defined. Evaluation of these resonance parameters in two-electron and three-electron systems was one of the first successful applications of B-splines.

The theory is based on the definition of two complementary orthogonal subspaces P and Q which contain, respectively, the nonresonant and resonant continuum states. Let us illustrate the meaning of these subspaces in the case of continuum states lying above the $N = 2$ threshold of He. As mentioned above, this energy region contains three open channels with total angular momentum $L = 0$. In the $1s\epsilon s$ channel, the bound electron lies in the 1s orbital of He^+ and the unbound electron has an asymptotic kinetic energy ϵ. For the same total energy, we have two additional channels, $2s\epsilon'$s and $2p\epsilon'$p, in which the bound electron lies, respectively, in the 2s and 2p orbitals of He^+ and the unbound electron has a kinetic energy $\epsilon' = \epsilon - E_{He^+(1s)} + E_{He^+(n=2)}$. Now, in this energy region, there exists a special kind of continuum state for which the dominant configurations are not the ones just mentioned, but configurations in which both electrons occupy excited bound orbitals. These are the so-called *doubly excited states* of He, whose dominant configurations are of the form $3lnl$ with $n \geqslant 3$. Notice that all the dominant configurations defining nonresonant continuum states include a 1s, 2s or 2p orbital, while configurations containing the latter orbitals barely contribute to doubly excited states. One can take advantage of this fact to define the P and Q subspaces according to the projection operators [361]:

$$P = P_1 + P_2 - P_1 P_2 \qquad Q = 1 - P \tag{76}$$

where P_i is a one-electron projection operator of the form

$$P_i = \sum_{n=1}^{2} \sum_{l=0}^{n-1} \sum_{m=-l}^{l} |\psi_{nlm}(\mathrm{i})\rangle\langle\psi_{nlm}(\mathrm{i})| \tag{77}$$

which includes all hydrogenic ψ_{nlm} states of He^+ with $n \leqslant 2$.

In all two-electron or effectively two-electron systems the definition of P and Q is straightforward and follows the above procedures. In many-electron systems, this is more

complicated and has been the object of much work over the years (see e.g. [362]). For the present purposes, let us assume that the projection operators are known in the general case. As shown in [349], for each channel μ, the exact eigenfunction $\Psi^-_{\mu E}$ of the Hamiltonian $\mathcal{H}$ can be written (for simplicity we shall drop the superscript Ω)

$$\Psi^-_{\mu E} = \frac{\langle\phi_s|Q\mathcal{H}P|P\psi^{0-}_{\mu E}\rangle}{E-\mathcal{E}_s-\Delta_s(E)-i(\Gamma_s(E)/2)}\phi_s + (1+G^{(s)}_Q(E)Q\mathcal{H}P) \times\left[P\psi^{0-}_{\mu E}+\frac{\langle\phi_s|Q\mathcal{H}P|P\psi^{0-}_{\mu E}\rangle}{E-\mathcal{E}_s-\Delta_s(E)-i(\Gamma_s(E)/2)}G^{(s)-}_P(E)P\mathcal{H}Q\phi_s\right] \tag{78}$$

where P and Q are orthogonal projection operators satisfying

$$\lim_{r_i\to\infty} P\psi^-_{\mu E} = \psi^-_{\mu E} \tag{79}$$

$$\text{and} \qquad P+Q=1, \tag{80}$$

ϕ_s is a resonant wavefunction which is a solution to a projected Schrödinger equation in the Q subspace,

$$(Q\mathcal{H}Q-\mathcal{E}_n)\phi_n = 0, \tag{81}$$

$G^{(s)}_Q$ is the Green operator in the Q subspace from which the s state has been excluded,

$$G^{(s)}_Q = \sum\!\!\!\!\!\!\int_{n\neq s}\frac{|\phi_n\rangle\langle\phi_n|}{E-\mathcal{E}_n}, \tag{82}$$

$P\psi^{0-}_{\mu E}$ is a nonresonant wavefunction which is the solution of the Schrödinger equation in P subspace

$$\left[P\mathcal{H}P+\sum\!\!\!\!\!\!\int_{n\neq s}\frac{P\mathcal{H}Q|\phi_n\rangle\langle\phi_n|Q\mathcal{H}P}{E-\mathcal{E}_n}-E\right]P\psi^{0-}_{\mu E} \equiv (\mathcal{H}'-E)P\psi^{0-}_{\mu E} = 0, \tag{83}$$

$G^{(s)-}_P$ is the corresponding Green operator in P subspace,

$$G^{(s)-}_P(E) = \lim_{\eta\to 0}\frac{1}{E-\mathcal{H}'-i\eta} \equiv G^{(s)}_P(E) + i\pi\delta(E-\mathcal{H}'), \tag{84}$$

Γ^s is an energy-dependent 'width',

$$\Gamma^s(E) = \sum_\mu \Gamma^s_\mu(E) = 2\pi\sum_\mu |\langle P\psi^{0-}_{\mu E}|P\mathcal{H}Q|\phi_s\rangle|^2, \tag{85}$$

and Δ_s an energy-dependent 'shift',

$$\Delta_s(E) = \mathrm{Re}\,\langle\phi_s|Q\mathcal{H}PG^{(s)}_P(E)P\mathcal{H}Q|\phi_s\rangle. \tag{86}$$

The way of writing equation (78) is not unique; it depends on the particular ϕ_s state selected to build the Green operators $G^{(s)}_Q$ and $G^{(s)}_P$. However, all possible choices provide identical representations of $\psi^-_{\mu E}$, so that, in each particular case, one can single out the ϕ_s state that is more convenient to accelerate convergence in the expansions of the Green operators. Also, equation (78) can be used in both resonant and nonresonant regions. The lowest eigenfunctions of $Q\mathcal{H}Q$ are L^2 representations of the resonances and, therefore, they can be evaluated with B-splines using standard procedures for bound states. To evaluate the wavefunctions in P subspace one can apply the discretization techniques described in the preceeding section.

From the formal point of view, there is no difference in using equations (64), (66) or (78) for the evaluation of continuum states. All of them provide the exact continuum wavefunction, including the possibility of Feshbach and shape resonances. Methods based on equations (64) and (66) do not require a special treatment of the resonances: they just appear as a result of

the coupling between the different channels. Therefore, attention must be paid to the selection of the channel wavefunctions and to the energy grid chosen to perform the calculations (e.g., using a too broad energy grid could lead to narrow resonance structures being missed). In contrast, methods based on equation (78) allow one to use specific basis sets for resonant states. Furthermore, from the eigenvalues of $Q\mathcal{H}Q$, one knows exactly where the resonances are expected to appear, so resonance structures cannot be missed by accident and the computational effort can be better optimized. Thus, the choice of any of the above methods depends on the particular problem under consideration and on the kind of information that one is interested in.

An additional advantage of the Feshbach formalism is that it allows one to obtain the autoionization width and the energy shift of the resonance using approximate expressions for equations (85) and (86). The usual approximation is to evaluate these parameters for $E = \mathcal{E}_s$. This is a reasonable approximation because all matrix elements appearing in equations (78), (85) and (86) do not vary significantly with energy, even in the resonance region. This is because the strong energy dependence of the wavefunction in the resonance region is explicitly written into the denominator of equation (78). This is not the case for methods based on equations (64) and (66): the matrices $\boldsymbol{A}$, $\boldsymbol{B}$ and $\boldsymbol{G}^-$ are strongly varying functions of energy near a resonance.

An approach of the Feshbach type is the truncated diagonalization method (TDM). In this method, one constructs hydrogenic states as linear combinations of B-splines. The use of the hydrogenic basis is particularly suitable for the calculation of autoionizing states in two-electron atoms, since the final state of the ion after the decay is hydrogenic. Two-electron configurations built from these orbitals are orthogonal, and can be used to define the two complementary P and Q subspaces. Thus, in practice, the Q subspace is built by excluding all configurations containing the hydrogenic orbitals of the ion that can result after the decay of the resonance state. In three-electron and more complicated systems, the situation is less advantageous, as mentioned later, since the final ionic state is a two-electron state which cannot be determined without approximations. This introduces an error in the definition of the P and Q subspaces (hence in the calculation of the autoionizing states), which is not present for two-electron systems. In any case, multiply excited states can be obtained by standard diagonalization of the Hamiltonian in the Q basis. Since only the lowest eigenvalues in an B-spline approach correspond to real physical states the use of diagonalization schemes where only a few of the lowest eigenvalues are determined (such as the Davidson procedure [371]) is advantageous, since it results in a considerable saving in computer time.

To exploit the accuracy which can be obtained with the B-spline basis set it is necessary to determine the energy shifts. These can be calculated approximately by imposing $E = \mathcal{E}_s$ in equation (86) and using $\mathcal{H}' \simeq P\mathcal{H}P$. In the case of a single open channel, the contribution of bound states to the shift is given by

$$\delta E_{\mathrm{b}} = \sum_{n=1}^{N_{\mathrm{b}}} \frac{|\langle \psi_n | 1/r_{12} | \phi_s \rangle|^2}{\mathcal{E}_s - E_n} \tag{87}$$

where ψ_n is a bound state of $\mathcal{H}'$ and N_{b} is the number of states below the ionization threshold. The shift due to the continuum belonging to the open channel is given by

$$\delta E_{\mathrm{c}} = \wp \int \mathrm{d}E \, \frac{|\langle \psi_E^{0-} | 1/r_{12} | \phi_s \rangle|^2}{\mathcal{E}_s - E}. \tag{88}$$

Here $\wp$ indicates the principal part of the integral. Since the matrix element in the numerator is obtained at discrete energies in the CI approach, van der Hart and Hansen [205] approximated this function by a straight line between consecutive energies, starting with the highest pseudo-bound state and ending with the highest continuum state. On each interval (88) was evaluated

analytically. Alternatively, the integral can be replaced by a quadrature in which only the energies arising from the discretization in the B-spline basis appear:

$$\delta E_c = \sum_{E_i \neq \mathcal{E}_s} \frac{|\langle \psi_i^{0-} | 1/r_{12} | \phi_s \rangle|^2}{\mathcal{E}_s - E_i} \tag{89}$$

where ψ_i^{0-} is normalized to unity. The total shift is then given by

$$\Delta_s(E = \mathcal{E}_s) = \delta E_b + \delta E_c. \tag{90}$$

In most cases the main contribution originates from the lowest bound states and leads to a decrease in the binding of the state. The shift due to continuum states can have either sign.

In the TDM approach and for a single open channel, the autoionization width defined in equation (85) can be approximated by Fermi's golden rule:

$$\Gamma_s(E = \mathcal{E}_s) = 2\pi |\langle \psi_{E=\mathcal{E}_s}^{0-} | 1/r_{12} | \phi_s \rangle|^2. \tag{91}$$

The continuum $\psi_{E=\mathcal{E}_s}^{0-}$, normalized on the energy scale, has the same parity and LS symmetry as the resonance state. In the multichannel case, the continuum state appearing in equation (91) must be evaluated as described in section 4.2.1. However, Martín *et al* [376] has discussed a first-order correction, Γ^1, to equation (91), which can be used within the UCS approximation:

$$\Gamma_R^1 = 4\pi \langle \psi_{E=\mathcal{E}_s}^C | 1/r_{12} | \phi_s \rangle \sum_{D \neq C} \wp \int_{E'} \frac{\langle \phi_s | \frac{1}{r_{12}} | \psi^D(E') \rangle \langle \psi^D(E') | \frac{1}{r_{12}} | \psi_{E=\mathcal{E}_s}^C \rangle}{(\mathcal{E}_s - E')} \tag{92}$$

where ψ^D is a continuum series different from $\psi_{E=\mathcal{E}_s}^C$. The first summation is over all such series, while the second summation (integration) is over all levels in a specific series. From equation (92) a series of first-order corrections for each open channel, ψ^D, is obtained that is added to Γ_s to give the decay rate to a particular final state. For the total decay rate, the effect of the first-order correction, in fact of the coupling between continua, is not expected to be particularly large except for neutrals and negative ions [345], while the corrections to decay rates to individual final states can be large even for ions.

The use of Fermi's golden rule implies that the continuum wavefunction is calculated at the resonance energy. Since the B-spline basis is determined in a box, the energies of the continuum wavefunctions will be determined by the boundary condition at $r = r_{max}$. If the boundary condition is that the wavefunction is set to zero, then it will be a coincidence if the resonance energy is obtained and several approaches have been used to get around this problem. The simplest approach is probably to use a knot sequence which is dense enough in the appropriate energy range. Then it can be expected that the inner part of the wavefunction, which overlaps the wavefunctions of the bound electrons and determines the value of the interaction integral, will change smoothly between adjacent energies. Van der Hart and Hansen [205] calculated the strength of the interaction at five adjacent energies in the region of the resonance and fitted the five values to a straight line in order to determine the value of the interaction integral at the resonance energy. With this procedure the relative numerical accuracy obtained for the widths was estimated to be better than 1% of the total width. The main advantage is that all relevant autoionization rates can be determined in one calculation. The same five-point interpolation procedure can be used to determine the first-order correction Γ^1 in equation (92) [207].

Another approach, in principle more accurate, is to determine the size of box necessary to give the correct energy. The disadvantage is that if many resonances are considered a separate calculation is required for each one. The same applies to the approaches used by Chang and Tang [162] to obtain continuum functions for photoionization calculations and

Chang [217] to calculate resonance widths. Instead of the composite knot distribution for the B-spline basis set described above, they use the sinelike distribution described in the appendix, while the continuum wavefunctions are obtained by matching an asymptotic expression when approaching the boundary. From this matching procedure, a series of phase shifts is obtained as a function of energy and to obtain resonance widths Chang and Tang fit these phase shifts to an expression due to Burke and McVicar [369] for the phase shift in terms of the eigenenergy and autoionization width,

$$\varphi(E) = \sum_{n=0}^{2} \alpha_n E^n + \tan^{-1} \frac{\Gamma_s}{2(\mathcal{E}_s - E)} \tag{93}$$

in which the first term represents the nonresonant background. To have enough phase shifts as a function of energy in the region of the resonance, where the phase shift is varying fast, results are obtained for a number of slightly different cavity radii.

Chang [217] has proposed an alternative, very transparent, method to determine the resonance width in which the density of continuum states is determined as function of energy. In brief, for a two-electron system, the continuum associated with decay to 1s is described by a product of a hydrogenic 1s and a wavefunction of the appropriate symmetry with any energy both negative and positive. This CI in the continuum, CIC, basis for the continuum function is combined with the wavefunction for the resonance state (expanded suitably) and the combined system diagonalized in a box with $r_{\max} = R$ corresponding to an energy E in the neighbourhood of the resonance. The sum of the squares of the expansion coefficients of the continuum wavefunction represents the continuum density, which was found to have a minimum, as expected, at the resonance energy and from the variation in continuum density with $E(R)$ the halfwidth of the resonance can be deduced directly. The difficulty is that a basis size of 120 splines per l-value was estimated to be required for an accurate determination and this leads to a large matrix size, which is unattractive when a number of diagonalizations are required. However, Chang [217] proposed an procedure to reduce the matrix size and obtained very accurate results in a number of cases for two-electron systems in the single-channel case.

The complex rotation (CR) approach has also been used to obtain energies and widths [65, 270]. This approach is usually based on a basis set so the method can fairly easily be adapted to use a B-spline basis, but Brandefelt and Lindroth [65] comment that a smaller matrix suffices when B-splines are used.

In the case of three- or more-electron systems, the projection operator technique explained above (for two-electron systems) does not lead to comparable precision. The difficulty arises from two causes: the exact target eigenfunction is not known and furthermore an explicit expression for the projection operators does not exist. The problem of writing down a projection operator arises because of the requirement of antisymmetry (see [362, ch 2]). Nevertheless a zero-order approximation for the P and Q operators has been shown to be efficient in certain cases. This technique, combined with a B-spline-based CI approach, has been successfully applied to calculate positions and partial autoionization rates for Li I, Be II and B III [101, 223, 225, 333].

The stabilization and complex scaling techniques can be also employed to calculate triply-excited-state positions and total widths. Bachau [222] has used a method for calculating total decay rates of three electron states in which the eigenvalues are determined as a function of box size. This method, introduced by Mandelshtam *et al* [363], does not require any explicit attention to boundary conditions but makes it possible to determine the total decay rate taking the interaction between channels into account by calculating results for different box sizes, which means that a large number of diagonalizations are required. Bachau reduced the calculational work by not actually changing the box size, which would require a recalculation

Table 3. Results for $3s3p^6\ ^2S + 3s^23p^4(^1D)n/\epsilon d\ ^2S$ energies (au) and the squared $3s3p^6$ component (%) in the seven lowest eigenvectors in Ar II from Landtman and Hansen [56]. The 'limited CI' and MCHF frozen-core calculations are from Smid and Hansen [381, 382].

Term	Energies relative to $3s^23p^4\ ^1D$			$3s3p^6\ ^2S$ component		
^2S	*B*-spline	MCHF	CI	*B*-spline	MCHF	CI
$3s3p^6$	−0.5668	−0.5667	−0.5312	0.5981	0.5978	0.5645
$3s^23p^4(^1D)3d$	−0.2181	−0.2181	−0.2086	0.0824	0.0824	0.0857
4d	−0.1255	−0.1255	−0.1188	0.0419	0.0419	0.0431
5d	−0.0811	−0.0811	−0.0764	0.0233	0.0233	0.0239
6d	−0.0565	−0.0565	−0.0530	0.0139	0.0140	0.0141
7d	−0.0416		−0.0389	0.0089		0.0086
8d	−0.0318		−0.0298	0.0060		0.0054

of the *B*-splines, but including a barrier in the single-particle potentials, which has the effect of restricting the wavefunction to a box of variable size without having to change the basis between different calculations. The basis set used by Bachau was nevertheless fairly small.

Complex scaling has been applied for three-electron states by Madsen *et al* [224], who used a correlated basis constructed in term of *B*-splines to calculate triply-excited states in Li and laser-induced transitions between Li hollow states.

4.3. Applications

4.3.1. Bound states. One of the simplest procedures for evaluating bound states consists in solving the single-particle Schrödinger equation (hydrogenic, HF, DF or DFT) for an electron with a particular l and then use this set of one-electron basis functions with selected l-values to carry out a CI calculation. Chang [47] was the first to see the advantages of using a *B*-spline basis for CI calculations and he emphasized in particular the effective completeness of the basis as well as the technical advantage that several states can be determined in one diagonalization.

To demonstrate the completeness of the *B*-spline basis, Landtman and Hansen [56] considered a case for which it is well known that there is very strong CI, namely the interaction between the $3s3p^6\ ^2S$ term and the $3s^23p^4(^1D)n/\epsilon d\ ^2S$ series in the Cl I isoelectronic sequence. Here n and ϵ are used to indicate the bound and the continuum members of the d series, respectively. In Ar II and heavier members of the isoelectronic sequence, the interaction is predominantly with the 3d member of the Rydberg series, but Smid and Hansen [382] have shown that more than 15% of the strength of the perturber is mixed into the ϵd continuum in Ar II. This means that to obtain the correct admixture of the $3s3p^6$ perturber in the individual bound states, it is necessary to be able to include the interaction with the whole $3s^23p^4n/\epsilon d$ series, in particular with the continuum part of the series. In fact, Smid and Hansen [381] have shown that failure to include the continuum results in a considerable error in both energy and eigenvector composition in addition to resulting in a spurious eigenvalue. Table 3 shows the results of a calculation by Landtman and Hansen [56] for the lowest ^2S states in Ar II using *B*-splines compared to a series of frozen-core MCHF calculations reported by Smid and Hansen [382] as well as a CI calculation by Smid and Hansen [381] in which the 12 lowest members of the d series are included.

The comparison is based on the fact that an elegant and compact way to include the continuum in a bound-state type calculation is by use of the MCHF approach. In the present case, it is possible to use a two-state calculation including the $3s3p^6$ and one $3s^23p^4d$ state, where the d orbital is a bound orbital in so far as it goes to zero immediately outside the atom,

but its shape and the number of nodes have no physical significance and the orbital must be optimized for each state separately [372]. For higher states this makes such calculations fairly capricious. This is the reason that only the five lowest MCHF solutions shown in table 3 were reported. The B-spline calculation on the other hand involves simply a diagonalization of a 36×36 matrix. The matrix is constructed by solving the HF equation for the d electron in the $3s^2 3p^4(^1D)d$ state with 35 B-splines of order seven using a logarithmic knot sequence in a box with $r_{max} = 150$ au with a frozen HF core for $3s^2 3p^4\ ^1D$ to simulate the MCHF calculation and using the HF 3s and 3p orbitals to construct the wavefunction of the perturber and, together with the d orbitals, to calculate the CI matrix elements. The choice of core means that the only nonzero matrix elements are those between the perturber and the members of the d series e.g. the matrix in the B-spline representation is a bordered one. The same is the case for the CI calculation which is based on numerical frozen-core HF wavefunctions for the d states obtained using the Froese Fischer computer program.

The agreement between the B-spline and the MCHF calculations is so good that the example provides convincing evidence for the completeness of the B-spline basis. The results of the CI calculation illustrate the effect of leaving out the high bound states and especially the continuum in this case (Smid and Hansen [381] report an approximate calculation including all bound states with very similar results to the 12-d-state calculation reported here) and underscores the need to include the continuum part of the d series as is done in both the B-spline and MCHF calculations although in very different ways. For completeness, it should be mentioned that the CI calculation is nonrelativistic, while the B-spline and MCHF calculations include a correction for relativity. However, this difference is unimportant in the present context.

Also for H^- the influence of the continuum is important in a hydrogenic basis. The usefulness of a B-spline basis for this problem was first pointed out by Chang and Wang [53].

The choice of B-spline basis depends on the problem under study. In the Ar case the effect of the continuum on the low-lying states is taken into account i.e. only the lowest eigenvalues represent real states. The nine lowest eigenvalues of the B-spline representation of the d series equal the numerical frozen-core HF eigenvalues, calculated individually to better than five significant digits. The higher orbitals in the basis set do not represent physical states but averages over the high bound states and slices of the continuum (see section 3). The completeness of the basis for a particular application can be checked by varying the size of the box and the number of splines and/or the knot point distribution for them.

For bound states several CI calculations of excited states in (pseudo-) two-electron systems have been published, combined for example with calculations of transition probabilities [34, 35, 47, 48, 50–55, 58, 60, 67, 68, 70, 73, 174] or with calculations of dipole polarizabilities [71], while Chen, Johnson and co-workers [74–80, 86] have used a similar approach in relativistic calculations for bound low-lying two-electron states in heavy atoms. A model potential approach has been used by Laughlin, Hansen and co-workers [61, 62] as well as Brage and co-workers [24, 45, 46] to calculate energy levels and oscillator strengths for neutral Ca as a pseudo-two-electron problem.

Chang's [47] first calculation concerned transition probabilities for Rydberg series in Be. He used 27 splines of order seven with an exponential knot sequence in a box with $r_{max} = 150$ au, which was found to be sufficient to describe the lowest four or five states in the ^{1,3}S, P and D Rydberg series with an accuracy of seven figures. The importance of the continuum was noted and it was found that the approach was as accurate as any other that had been applied to the system. The calculations were extended to Mg I [51] and good agreement with experiment was obtained for f-values and lifetimes of the $3sn\text{d}\ ^1D$ states, which are strongly perturbed by the $3p^2\ ^1D$ term. f-values for many other series were also

published [50] and the calculations extended to members of the isoelectronic sequences [52,54]. In calculations of f-values for He and the Al^+ ion, Chang and Fang [174] demonstrated the importance of including the continuum in the basis. Subsequently the method was applied to Ne [60] using the frozen-core HF approach with an additional (fitted) model potential, one rationale being that a model potential method would be easier to implement for MPI than to start from a multiconfiguration approach. In a very early calculation, Chang [48] combined MBPT with CI in order to determine the fine-structure level splitting of the 1s4f configuration in He. The electrostatic exchange interaction and the spin-dependent interactions are of the same size for the 4f states and Chang was able to improve earlier MBPT calculations significantly, attributing the improvements to the properties of the B-spline basis.

Chen and Hsue [34] reported a calculation of $1sns\ ^3S$ states in He from $n = 8, \ldots, 13$ in order to show that high n-values could be reached, while using conventional methods, even of Hylleraas type, the higher states are more difficult to calculate than the lowest ones. Chen and Hsue used box sizes from 385 to 485 au with a logarithmic knot sequence and expanded their wavefunctions directly in products of B-splines. The wavefunctions were optimized by varying the number and the order of the splines and it was found that the highest energy values agreed with the values published by Accad *et al* [388], in which the dependence on r_{12} is taken into account explicitly. B-splines of order ten were used with up to 44 splines per l-value, leading to a maximum matrix size of 1264×1264 with up to four l-values included. The number of terms is comparable to those used by Accad *et al*. Chen and Hsue calculated radial expectation values and noted that these were consistently more accurate than the energies, indicating that overall the B-spline wavefunctions were the more accurate. Chen and Hsue also noted that the same basis could be used for a number of different n-values, again in contrast to the Hylleraas technique. This is an advantage for example for subsequent MBPT calculations. Chen [70] extended the calculations to higher n-values and to the ^1S series and found that the calculations were more accurate than those of Accad *et al* above $n = 15$ for ^3S and above $n = 10$ for ^1S when the same number of l-values was used while the number of terms in the expansions was less than used by Accad *et al*.

In subsequent papers, Chen [35] has calculated transition probabilities in He between S and P terms with $n \leqslant 5$ and obtained good agreement with calculations including the interelectronic distance explicitly although the energies are still inferior, in particular for the lowest terms. For the latter the number of l-values was increased to seven, which is known to be insufficient for the lowest singlet terms, in particular. The calculations were extended [58] to higher n-values and to D terms by reducing the number of B-spline products to reduce the size of the expansions, assuming that the outer electron does not significantly penetrate the 1s core as explained in section 4.1.1. Larger calculations in which this assumption was relaxed were used to calculate dipole polarizabilities for the two lowest ^1S and the lowest ^3S term [71]. Later, the approach was extended [315] to calculate the long-range dipole–dipole retardation potential between two He atoms.

More recently Chen has used a model-potential method including dipole and quadrupole polarization to calculate bound states in Be [73] and Be-like ions [55]. Radiative transition probabilities for a large number of transitions have been reported.

As an example of the accuracy of the model potential methods, we show in table 4 the results of the calculation reported by Laughlin and Hansen [61] for the $^{1,3}G$ terms in neutral Ca. The $4sng\ ^1G$ series is weakly perturbed by the $3d^2\ ^1G$ term that is located somewhere between $n = 5$ and 6. (The term has not been found experimentally.) In an unperturbed series the ^1G term is expected to lie above the accompanying ^3G term and the NBS compilation [389] conforms to this expectation for the pair of terms with $n = 5$ although the energy difference is less than 0.1 cm^{-1}. However, Geller [390] has interpreted infrared lines in the spectrum

Table 4. Comparison of term energies (in cm^{-1}) for the 4s5g–7g ^{1,3}G and 3d^2 ^{1}G^e terms of neutral calcium. From [61].

	Energy		
	Experiment		Calculation
Term	[389]	[390]	[61]
4s5g ^{1}G^e	44 875.95	44 874.42	44 875.1
3d^2 ^{1}G^e			45 509.5
4s6g ^{1}G^e		46 231.70	46 232.2
4s7g ^{1}G^e		47 048.89	47 049.5
4s5g ^{3}G	44 874.86	44 874.83	44 875.7
4s6g ^{3}G		46 231.03	46 231.7
4s7g ^{3}G			47 049.2

of the sun to mean that the order is inverted and that 5g ^{1}G is 0.41 cm^{-1} below 5g ^{3}G. This is confirmed by the B-spline calculation that places 5g ^{1}G 0.6 cm^{-1} below ^{3}G. On the other hand the distance between the 6g terms is slightly increased although the two terms lie in the normal order. This is also confirmed by the B-spline calculation. Such an accuracy for a highly excited term is very unusual and testifies to the accuracy which can be obtained with the B-spline approach.

Brage and Froese Fischer [24,45,46] have used the spline–Galerkin approach to calculate bound and doubly-excited states in Ca using a fairly simple model potential in which a cut-off parameter is used to adjust a particular ionization limit to agree with experiment. This necessitates several calculations when there are several low-lying limits as is the case in Ca^+, 4p and 3d, but aside from that the accuracy is very good and there is fairly good agreement with the calculations by Hansen and co-workers [61, 62], who has used a more complicated model potential but only calculated bound states. The differences can be associated with the simpler model potential and the fact that Brage and Froese Fischer did not take into account the dielectronic polarizability that the Amsterdam group has found essential for close agreement with experiment [57, 59, 61, 62]. For the doubly-excited resonance states, Brage and Froese Fischer did not include the interaction with the continuum. The advantage of the spline–Galerkin approach is that the matrices required are fairly small, fewer than 1500 basis states were used and it is possible to obtain wavefunctions for rather high n-values if a sufficiently large box is used. Brage and Froese Fischer used a box size of 1000 au and B-splines of order eight with about 100 B-splines for each l-value. An exponential knot sequence was found to be sufficient for the bound states up to about 25.

Relativistic CI calculations have been reported by Chen, Cheng, Johnson and Sapirstein [74–80, 86]. Since this application is so close to the nonrelativistic case we include a more detailed discussion than of other relativistic applications. The first application [74] was to the 1s2l ^{3}L states in the He isoelectronic sequence from $Z = 5$ to 100. The purpose was to check earlier MBPT calculations [127], also obtained using a B-spline basis, which differed slightly from an earlier calculation by Drake [384] using a so-called 'unified' approach, i.e. including both correlation and relativity, where the former is especially important at low Z while the latter becomes dominant at higher Z. The CI calculations basically confirmed the MBPT calculations but were a little more accurate by including higher-order effects. The calculations were carried out in the 'no-pair' approximation where the negative solutions to the Dirac equation are neglected. The retarded Breit interaction was calculated but QED and mass polarization corrections were taken from Drake. Test calculations were carried out for

Ne with 40 B-splines for each l-value up to $l = 5$. Not all splines in each series were used, leading to matrices of order 5000. The Hamiltonian matrices are dense and Chen *et al* [74] comments that the evaluation of the Hamiltonian matrix is the most computer intensive part of the calculation, requiring typically several hours of supercomputer time (in 1993) even though use was made of symmetry properties in the calculation of the two-electron integrals. The Davidson method was used for the diagonalization since basically only the lowest eigenvalue is required. In a subsequent calculation [75] the corresponding singlet levels were considered. A larger basis was required in this case ($l \leqslant 8$) since the Breit interaction converges slowly with l, in particular for the GS. Despite several simplifications up to 20 hours of supercomputer time was needed for each ion but the results were as accurate as for the triplets and the CI results were judged to be the most accurate available. Also relativistic polarizabilities of He and He-like ions, $2 \leqslant Z \leqslant 30$, have been reported by this group [78]. The wavefunctions were also used to calculate hyperfine structure constants and hyperfine quenching of radiative lifetimes for the $1s2p\ ^{1,3}P_J$ levels in the isoelectronic sequence [81].

The CI techniques used in the two-electron calculations have been extended to more than two electrons [76, 77, 79, 80]. Chen *et al* [76] have discussed the approach in connection with calculations of the $n = 2$ states of Li-like ions with $10 \leqslant Z \leqslant 92$. The physical relevance of these calculations was high-precision measurements of the $2s–2p_{3/2}$ transitions in Li-like to Ne-like uranium [387], for some of which Johnson *et al* [133] had carried out RMBPT calculations, again making use of a relativistic B-spline basis. The main difficulty in the CI approach is the explosion in the size of the Hamiltonian matrices. For the $1s^2 2s$ GS, including only single and double excitations, 13 952 CSFs from $1snln'l'$ and $2snln'l'$ configurations with $l = l' = 0–6$ were used, neglecting contributions from states with $|l - l'| = 2$, which were found to be insignificant. For the $1s^2 2p_{1/2}$ state the matrix size was 24 155 CSFs while for the $1s^2 2p_{3/2}$ state further approximations were introduced to reduce the matrix size. The B-spline expansions were made with 30 positive energy orbitals, of which the lowest 19–20 were used in the diagonalizations. A box size of 8 au was used for $l \leqslant 3$ for Ne^{8+} and the radius was gradually increased to 12 au for $l = 6$ orbitals. For other ions $1/Z$ scaling was used to determine the box size, which was chosen large enough that the first few s and p eigenvalues agree precisely with the actual eigenvalues of the unconstrained potential i.e. not restricted to a box. A Hartree (model) potential for $1s^2$ was used to obtain the basis and Chen *et al* [76] comment that the choice of potential is immaterial due to the completeness of the basis. In fact, in a publication about Be-like ions [79], a Dirac–Slater potential with KS exchange and a Dirac–Hartree potential are compared and it is found that, while individual CI contributions, for example, can be different, the final energy is the same in the two cases. For a correct treatment of QED the situation is a little more complicated [82]. The main difficulty was the evaluation of the (large) Hamiltonian matrix. To reduce the computation time of the angular part of the matrix elements, the concept of a angular momentum channel was introduced, where a channel is defined as an electronic configuration including coupling schemes but ignoring principal quantum numbers except to distinguish between identical and nonidentical electrons with the same angular symmetry. Only angular matrix elements between different channels need to be calculated explicitly. The same tactic has been used by Verbockhaven and Hansen [223, 225] for three-electron systems. This reduced the number of matrix elements that needed evaluation in the calculation by Chen *et al* from hundreds of millions to tens of thousands. To speed up the evaluation of radial integrals it was possible by careful ordering of the CSFs to reuse the Z-integrals (see appendix A.3) involved in the calculation of the two-body integrals. Finally the work involved in evaluating the matrix elements of the Breit interaction was reduced by using the retarded interaction for the diagonal matrix elements while the nondiagonal elements were obtained with the unretarded Breit operator. To obtain

an estimate of the frequency dependence of the nondiagonal elements, smaller calculations were made comparing the use of the unretarded and the retarded version of the operator for the nondiagonal matrix elements. It was found that inclusion of l-values up to three was sufficient in such comparisons. The use of the Davidson procedure for obtaining the few lowest eigenvalues of these large matrices is helpful also because the gigabyte fast memory needed for conventional storage of such large matrices is not required with the Davidson approach.

The calculations were extended to the $n = 2$ states with $J = 1$ in Be-like ions with $10 \leqslant Z \leqslant 92$ in [79]. Single and double excitations were included but contributions from triple excitations were estimated by comparing smaller calculations with and without triple excitations included and also some types of double excitation were estimated in this way. Matrix sizes were between 10 000 and 36 000. Calculations for $2s–2p_{3/2}$ transitions in Li- to F-like uranium were presented by Cheng and Chen [77]. In the calculations for B- to F-like systems, a frozen-$1s^2$-core approximation and $l \leqslant 3$ were imposed to restrict the matrix size to less than 30 000. The assumption is that excitations from the $1s^2$ core would cancel in the transition energies. An accuracy of about 0.25 eV was obtained in the transition energies, which are larger than 4000 eV for all the ions involved. The main remaining uncertainty was estimated to be due to lack of knowledge of higher-order QED and nuclear polarization effects. While the accuracy obtained for some of the simpler ions was comparable to that achieved in RMBPT [133], RMBPT results did not exist for B-, N- and O-like ions. The open-2p-shell structure for these elements can more readily be described using the CI technique.

Finally, Chen and Cheng [80] have used the same technique on the $3s^2\ {}^1S_0–3s3p\ {}^{1,3}P_1$ transition energies in some magnesium-like ions up to Mo^{30+}. Valence–valence as well as some core–valence correlation was included while core–core correlation was neglected. To be able to include core–valence correlation, separate calculations were made for excitations involving 2p, 2s and 1s together with a valence electron. These calculations included all the valence–valence correlation and the core–valence contribution from each inner shell was obtained separately by differencing the appropriate calculations. Finally the three contributions were added, together with some other effects obtained in the same way such as three-electron excitations. An accuracy of 100 cm^{-1} was obtained, which led the authors to conclude that the feasibility of carrying out accurate CI calculations on many-electron systems had been demonstrated.

A large number of the B-spline studies have been concerned with bound or quasi-bound states of negative ions. A number of studies of He^- have been published; however, since most of them concern photodetachment they are discussed in section 4.3.4. For heavier systems, Laughlin, Hansen and co-workers have made use of the model potentials constructed by Laughlin [61, 62, 364] for the Ca^{2+} ion. Van der Hart *et al* used this potential to calculate the binding energies of states in Ca^- [57, 59] in particular the GS. This is a very weakly bound three-electron system. The binding is due to correlation and a HF calculation does not predict that a 4p electron is bound. The binding has been notoriously difficult to calculate, and the calculation by van der Hart *et al* was the first to obtain an electron affinity which was in line with a remeasurement [365] that had reduced the earlier experimental value by a factor of more than two to below 20 meV. The success of the calculation compared to earlier calculations was ascribed to the importance of a complete set of f states in the calculation.

In a model potential method it is essential to include polarization effects, not only the dipole and quadrupole polarizability but also the dielectronic polarizability, which is the reduction in polarizability due to the presence of another electron outside the core. In the Ca^- calculation [57], the B-splines basis consisted of 22 B-splines of order nine in a box with $r_{max} = 60$ au and l-values lower than four. To reduce the size of the matrix only some of the B-splines were used. The angular momenta were ssp, sdp, ddp, ppp, sdf, ppf,

ff(^{1,3}S)p and dd(^{1,3}D)f. All intermediate couplings are included, except for ffp and ddf, where only the indicated couplings are included. If the configurations are ordered $nln'l'n''l''$, the maximum allowed values were five ($l \geqslant 2$) and six ($l \leqslant 1$) for n, eight for n' and 14 for n''. Configurations of the type ddp, sdp and ffp which have a 6p, 6d or 6f orbital and an orbital with a higher n-value were also excluded. The resulting matrix size was 6343×6343 and the excluded configurations were taken into account by second-order perturbation theory. Today much larger matrices could be used. The approach is different from the approach used by Verbockhaven and Hansen [221, 223, 225], who for triply excited states, see later, selected configurations for the final CI on the basis of size, while in Ca^- the selection was made on n-value (and experience).

In a later publication Froese Fischer *et al* [59] used the same potential to calculate the position of the $4s4p^2\ ^4P$ state in Ca^-. The prediction turned out to be very close to a later experimental measurement [367]. Van der Hart has published similar calculations [63, 64] for other alkaline earths.

We have mentioned that the purpose of the introduction of B-splines by Johnson and Sapirstein [104] was the use of the B-spline basis for RMBPT. The bibliography shows at a glance how successful this approach has been [104–159]. The first application was to parity violation in heavy elements, such as Cs, where Johnson and co-workers calculated a number of properties of ground and excited states, such as radiative transition probabilities, in order to pin down the accuracy of the wavefunctions obtained and the procedure applied. For Cs it was possible in the end to calculate atomic properties with an accuracy better than 1%, which is very satisfactory for such a heavy system.

Another application has been to the determination of energy level values and transition probabilities in medium and highly ionized systems using a Z-expansion approach. A third application, pioneered by Blundell and Sapirstein, has been to the calculation of QED effects in few- and many-electron systems [88–98]. Also properties of clusters have been determined. In these calculations, the completeness of the B-spline basis has been the property of interest and the B-spline wavefunctions have in general been converted to numerical form so that B-splines are mainly used for the initial determination of the basis.

Liaw and co-workers have used a variant of MBPT in which the Dyson equation is truncated after second-order contributions. The approximation is called the Bruckner equations, which can be solved self-consistently to obtain higher accuracy. Liaw has used the approach with a B-spline basis to calculate properties of alkali atoms [72, 160, 161] and the approach has also been used to calculate *ab initio* the attachment energy of Ca [69]. In this calculation no bound state was obtained for this weakly bound system after the first iteration. Johnson *et al* [111] did obtain a bound state of Ca^- in a similar approach but using a semiempirical dipole polarizability to describe the core.

Hansen and co-workers [118, 119, 321] have used B-spline basis sets for nonrelativistic MBTP calculations of orthogonal parameter sets, defined to be able to describe the structure of configurations to infinite order in PT. Orthogonal sets of effective parameters are linearly independent of each other and therefore well suited for systematic calculations. So far only calculations to second order have been reported.

4.3.2. Doubly excited states. Van der Hart and Hansen [204, 205] have shown that the accuracy of the CI approach can compete with the most accurate approaches that have been introduced for doubly excited states of two-electron systems. These workers and their colleagues used a nonrelativistic approach for both high- and low-lying states in ions with two electrons up to Ne^{8+} [192, 193, 204–212]. Some of their results in the UCS approximation (see section 4.2.1) are described in the following together with some later calculations beyond this approximation.

Table 5. Types of configuration included in calculations for $3l3l'$ terms in N^{5+} in [205] for each type of LS term. Also the number of basis states of each type used in the calculations for the resonance states is indicated. The B-spline sets each consist of 25 splines.

LS	Configurations	s	p	d	f	g	h
$^1S^e$	s^2, p^2, d^2, f^2, g^2	25	25	25	25	25	0
$^1P^o$	sp, pd, df, fg	22	22	22	22	22	0
$^1D^e$	sd, p^2, pf, d^2, dg, f^2	21	21	21	21	21	0
$^1D^o$	pd, df, fg	0	21	21	21	21	0
$^1F^o$	sf, pd, pg, df, dh, fg	20	20	20	20	20	5
$^1G^e$	sg, pf, ph, d^2, dg, f^2	20	20	20	20	20	20

For doubly excited states of the type $nln'l'$ the effects of CI can be very large, especially when $n \approx n'$, so large in fact that the conventional single-particle labelling in many cases loses is usefulness and other schemes, in particular the (K, T) classification, which has its origin in the $O(4)$ symmetry of the hydrogen Hamiltonian, have been found to be useful. The strong interaction between configurations with the same distribution of n-values and the same parity, for example between 2s3p and 2p3s, was first discussed in detail by Layzer [375] and the set of configurations is referred to as the 'Layzer complex' or simply 'the complex'.

Technically an important difficulty is related to the fact that the lowest states in an $nlnl'$ manifold have the electrons located far from each other, i.e. on opposite sides of the nucleus, to reduce the interelectron repulsion, while for many of the higher terms the electrons have a large probability for being on the same side of the nucleus and thus close to each other. The two types of state are often referred to as Wannier and anti-Wannier states, respectively. Both types of state often require an expansion in a large number of l-values, one reason being that the electron repulsion leads to a 'Coulomb hole', which gives a cusp in the wavefunction for $r_1 = r_2$. Here r_i is the position of electron i relative to the nucleus. The Coulomb hole is particularly prominent for the anti-Wannier states and to describe the cusp using an expansion in angular momentum is especially difficult. The calculation of doubly excited states provides therefore a good test of the quality of a particular approach.

In an early paper about N^{5+}, van der Hart and Hansen [205] reported the energies obtained in the B-spline approach for some $3l3l'$ and $3l4l'$ terms and compared with results obtained by other methods. Table 5 shows details of the B-spline sets used for the different singlet symmetries.

The table shows the configurations included for each symmetry. In many cases the effect of the highest splines, which corresponds to very high energies, is very small, and these splines can be neglected, if computer resources so dictate, with little effect on energies and wavefunctions. The table shows the number of splines actually used for each symmetry. The box size was 40 au, with an exponential grid comprising 21 intervals. This leads to 25 splines of order seven when the boundary conditions are enforced by removing the first and the last spline. Some configurations involving electrons with large l-values, for example ph in the case of $^1G^e$, have little influence on the energy but a considerable effect on the autoionization rate. The parameters of the basis sets, such as the knot sequence, were defined so that the eight lowest hydrogenic eigenvalues correspond to real physical states, which means that the $nln'l'$ states with (n and n') < 8 are expected to represent real states. The size of the matrices varied between 1300 and 2100. Thus most of the eigenvalues do not represent real physical states and the Davidson approach was used to obtain the appropriate eigenvalues and eigenvectors. Today much larger matrices can be handled without difficulty.

Results were reported for $3lnl'$ singlet terms with $n = 3$ and 4 in N^{5+} with and without

the shift due to the open channels. The shift was calculated using equations (87) and (88). The results were compared with results obtained from CR [374] and CC [378] calculations as well as with results obtained by Martín *et al* [376], who also used a discretization technique to include continuum effects in the Feshbach projection formalism (FPF) based on STO orbitals. The most accurate results are expected from the CR and CC methods and comparison showed that the B-spline method is slightly less accurate than the CR method when a Hylleraas basis is used, roughly comparable to the CC method and more accurate than the FPF of [376], where a much smaller basis was employed. The remaining deficiencies in the B-spline method seem to be due mainly to the cut-off in angular momentum for the basis sets. As mentioned, with more modern computers a considerable improvement could easily be realized.

One advantage of using the TDM is that all the eigenvalues can be determined simultaneously, whereas in the CR and CC methods it is necessary to stabilize each (complex) eigenvalue separately. In addition the interaction between the open channel and the resonances needs to be reasonably large in both of these methods. Otherwise, the energy and the width of an autoionizing level cannot be determined. The advantage of one simultaneous calculation for all states compensates in most cases for the small loss in accuracy. The ability to obtain small autoionization rates is important when branching between radiation and autoionization is studied since the states with a weak autoionization usually are the only ones for which radiation can compete.

For $3l3l'$ states there are several degenerate continua available as final states (multichannel continuum, see section 4.2.1). The interaction between the continua will influence the total autoionization rate as well as the branching ratio between the different final states. This effect is usually most important for small free electron energies, i.e. close to threshold, in particular for neutral systems. However, Martín *et al* [376] have shown that, particularly for the total cross section, the effect of the interaction is fairly small for highly charged ions and that an approach based on the use of UCSs, in general, works reasonably well. In this approximation, the UCSs are used in equation (91) for each continuum separately and the results are added to give the total cross section. Martín *et al* [377] found for O^{6+} that, even in the presence of a strong coupling between the continua, the effect on the total width is small, particularly for the Wannier states. For $2lnl'$ states the only possible continuum is associated with the 1s limit and no interaction between limits needs to be taken into account.

Van der Hart and Hansen [205] used the UCS to calculate the autoionization rates for the $3lnl'$ singlet terms in N^{5+}. Each open channel consists of a bound electron in a fixed hydrogenic state (1s, 2s or 2p) and a free electron with a particular angular momentum. For this free electron a B-spline basis set with another set of knot points than for the resonance states is used. The number of intervals was increased from 21 to 137, of which 21 are distributed exponentially from 0 to 0.6 au and 116 are distributed linearly from 0.6 to 40 au. Note that the box size is the same for both bound and continuum states. Since the bound-state wavefunctions involved in the interaction have gone to zero before the boundary, it is not critical to use the same boundary for the calculation of the two types of wavefunction, if the density of states for the continuum is correctly taken into account.

The choice of knot sequence is based upon several considerations. At large distances a linear grid is used in order to have enough knot points (more correctly enough splines) per oscillation of the wavefunction in the asymptotic region. This criterion, which depends on the free electron energy, follows from the need to describe the continuum wavefunctions accurately in the appropriate energy range (see section 3). Since the interaction integrals needed for the calculation of energy shifts and resonance widths depend only on the continuum wavefunction in the inner region where it overlaps the bound-state wavefunction, it could be hoped that restrictions on the wavefunction in the outer region would have little influence on

these integrals. However, in a global method, the solution at one point is linked to the solution at all other points through the system of equations and a good solution in the outer region is necessary for obtaining a good solution in the inner region. This was checked by showing that the results were unchanged when the interval size for the linear part of the grid was halved, while on the other hand an increase in the interval size led to inaccurate results.

Finally, it is a condition for the validity of equation (91) that the continuum wavefunction is calculated at the resonance energy. We have already discussed several ways to obtain this condition or to avoid doing so. Van der Hart and Hansen [205] reported autoionization rates obtained by interpolating between values for adjacent energies in a particular box. The density of states in equation (91) was determined using equation (21). They compared their results with the same calculations already discussed in connection with the energies. The agreement was good. The largest differences with the FPF, for example, 20%, were found for the states which also had the largest errors in the calculated energies. For the other states the differences were considerably smaller.

Van der Hart and Hansen have also used B-splines to calculate the competition between radiative and nonradiative decay for $2lnl'$ and $3lnl'$ states in C^{4+} [206] and for doubly excited $1s^2 3lnl'$ states in O^{4+} [207]. Also these calculations were undertaken in order to develop the understanding of double capture in charge exchange collisions.

Several calculations of properties of H^- has been reported. In an early calculation, Chang and Wang [53] demonstrated the importance of the continuum for the energies of the lowest ^1S and the lowest $^3P^e$ states. The possibility to include only part of B-spline basis in the CI calculation makes it easy to determine the effect of bound and continuum states separately. For the GS it was shown that 9% of the electron affinity is due to continuum p states only. It was also found that the convergence with l in the continuum is slow. In addition, the f-value for the transition between the GS and the lowest $^1P^o$ Feshbach resonance was calculated, and a substantial reduction in f-value was found when the effect of the continuum was included. This calculation showed also that the use of the TDM for this system is very sensitive to the completeness of the basis. Chen [202] has modified the saddle-point (SP) complex-rotation method [391] to represent the closed channel part of the wavefunctions with B-splines and applied the approach to calculate doubly excited states in H^- below the $n = 2$ threshold. Autoionizing $2snl$ $^{1,3}S^e$ and $^{1,3}P^o$ terms with n up to five as well as the $2p^2$ $^1D^e$ term were reported using angular momenta up to $l = 6$ and expansion lengths up to 2500 for the resonance states. An exponential knot sequence with box sizes ranging from 640 au ($2s^2$ ^1S, for example) to 4200 au (2s4p ^3P) was used and the results were in good agreement with earlier calculations, in particular for the higher states. Similar results were reported for the equivalent states as well as the $^{1,3}D$ terms in He [199], in all results for 78 terms. Here angular momenta up to $l = 7$ for the ^1S terms were used with box sizes of 250 or 550 au for low and high states, respectively. Fewer l-values were used for the P and D terms. Later [200], the number of basis states was kept down by combining a number of terms in the expansion of the resonance terms for the CR calculation. In [200] this technique was used to reduce the number of terms in the calculations for the resonance terms from a maximum of 2500 to 1000, with the result that another 1000 terms with higher angular momenta, for example, could be added, with a resultant improvement in energies and widths for the lowest terms both in He and in H^-. More recently Chen [198] has determined the number of $^{1,3}P$ terms in H^- that exist below the $n = 2$ threshold in H by including relativistic effects as first-order perturbations to study the lifting of the 2s and 2p degeneracy. Since the binding is very weak, box sizes up to 34 000 au were used.

Chen [68] has reported calculations for doubly excited states in Li using a model-potential approach similar to that developed by Laughlin [364] to calculate energies, transition

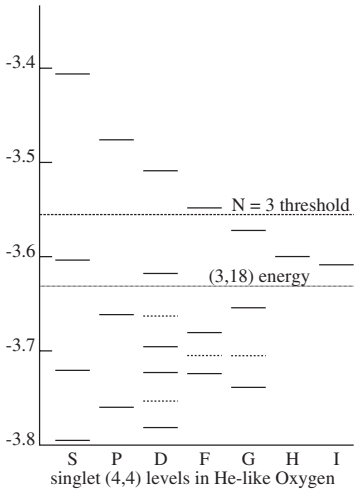


Figure 8. Positions of the singlet terms belonging to the $4l4l'$ complex in O^{6+} relative to the $N = 3$ threshold. The $3lnl'$ series is not included, but the average $3l18l'$ energy is indicated. The difference between dashed and solid lines refers to states with so-called unnatural, respectively, natural parity.

probabilities and lifetimes for quartet levels belonging to the configurations 1s2snl and 1s2pnl with $l \leqslant 4$, where the model potential represents the 1s electron and is fitted to reproduce the energies of the 1snl ^3l observed energies in Li^+. B-splines of order nine were used in a box as large as 600 au with up to 41 splines for the smallest l-value. The wavefunctions were expanded in products of B-splines and the number of terms in the expansions was reduced by allowing only some of the products as explained earlier. Results up to about $n = 15$ were reported. Since there are no quartet continua possible that the levels can autoionize to, only radiative decay is allowed, which makes variational calculations simpler for the quartets than for the doublets.

Another area of research where B-splines have contributed significantly is the study of doubly excited states embedded in a Rydberg series of autoionizing states. In double-capture processes using bare nuclei heavier than He, say, energy conservation has the consequence that the electrons are picked up in higher-n states. This means that autoionization is a very likely decay mechanism and consequently that it is unlikely that both electrons will remain with the projectile. This process is called autoionizing double capture (ADC) in contrast to true double capture (TDC). However, during the 1990s it became clear that sometimes TDC was much more likely than earlier experiments had indicated. Roncin *et al* [379] noticed that TDC seems particularly important when the states which are believed to be involved are close to or straddling a threshold. This is for example the case for the $4l4l'$ states in O^{6+}, which are embedded in the $3lnl'$ series as figure 8 shows.

It can be seen that the $4l4l'$ terms are spread over a considerable range of $3lnl'$ configurations, some terms being above the $N = 3$ threshold. This puts considerable strain on a calculation since it is necessary to be able to represent $3lnl'$ Rydberg states with large n-values. The representation of high Rydberg states is a problem for most types of basis set and to our knowledge the highest n-value that has been used in a calculation using an STO basis is 17, but using quadruple precision, while as we shall see $n = 50$ states can be described without difficulty in double precision using B-splines . However, the B-spline basis sets introduced earlier do not suffice since they do not represent physical states with n larger than about seven or eight.

Figure 9 shows an example in which a hydrogenic 35s function is expressed in B-splines. Keeping in mind the cancellation in the expansion of a 4s hydrogenic orbital in terms of STOs (see figure 3 in section 3), it is noticeable that both the wavefunction and the B-splines can be on the same scale in the B-spline expansion.

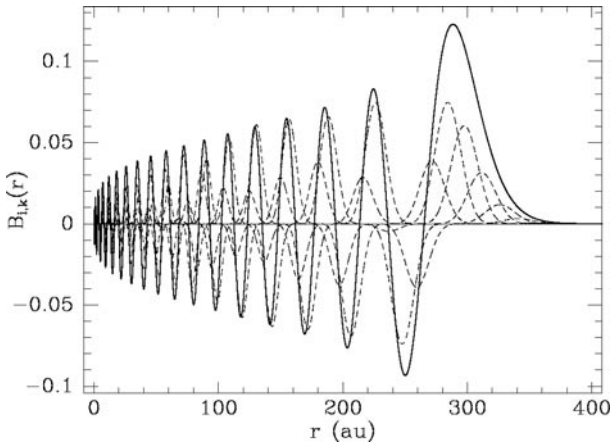


Figure 9. A hydrogenic 35s orbital for O^{7+} expanded in terms of B-splines . The expansion uses 181 B-splines of order nine in a box with $r_{max} = 1000$ au. From [329].

Van der Hart, Vaeck and Hansen [193, 208, 210] have carried out calculations of autoionization and radiative properties of the $4l4l'$ terms in O^{6+} in which they use three different B-spline basis sets. One basis set is used to describe the $4l4l'$ terms. It consists as before of 25 B-splines of order seven in a box with $r_{max} = 40$ au and an exponential knot sequence divided into 21 segments. The second basis set is used to describe the $3lnl'$ Rydberg series. For most of the series this set consisted of 199 splines of order nine in a box with $r_{max} = 500$ au and an exponential knot sequence. This allowed the determination of the series members up to $n = 40$. An exponential knot sequence is used because the Rydberg states, in contrast to continuum states, go to zero without oscillation at large distance. Using a box with $r_{max} = 1000$ au and 181 splines of order nine it was possible to obtain accurate wavefunctions with n up to 50. Finally, to calculate autoionization, a third basis set was used, in fact two different sets. For terms with total angular momentum L between S and G a box with $r_{max} = 40$ au, while for higher L values a value of $r_{max} = 80$ au, was used and a composite knot sequence which is exponential near the nucleus and linear from 0.6 au to the boundary of the box. In both cases a total of 141 B-splines of order seven were used. This makes it possible to describe continuum electrons with energy below 35 au. The reason for the larger box is that the high L values involve high l for the continuum electrons. That the box size for the continuum wavefunctions is much smaller than that for the Rydberg electrons does not matter since the continuum wavefunctions are not needed outside the region where the lowest bound wavefunctions involved in the decay are localized. However, for the determination of the interaction between different $3lnl'$ Rydberg series it is necessary to integrate over the entire box, i.e. 500 or1000 au, respectively.

Since it is necessary to calculate decay properties of the mixed $4l4l'$ and $3lnl'$ series up to high n-values, as the former states can have appreciable amplitude high up in the Rydberg series, it is not possible to use the Davidson diagonalization routines. A complete diagonalization must be carried out and therefore the wavefunction was calculated in two steps. First the eigenvectors of the $4l4l'$ terms were obtained from TDM calculations as described earlier with matrix sizes between 1300 and 2500. Here the Davidson approach was used. In a second step, the interaction between each of the $4l4l'$ eigenstates and the appropriate Rydberg series with the same symmetry was determined, followed by a complete diagonalization of the resulting

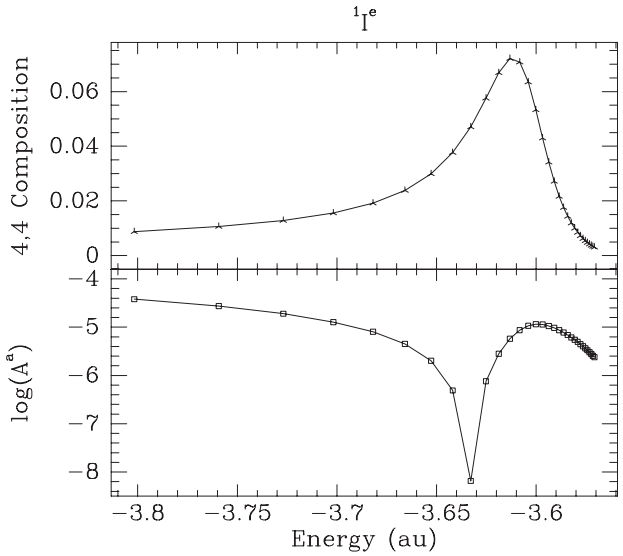


Figure 10. The effect of the interaction between a $4l4l'$ perturber, the ^{1}I^e term, and a Rydberg series on the autoionization decay rates of the combined system. The amount of the perturber in the individual states is shown in the upper panel (crosses) and the corresponding autoionization rates, A in au, in the lower one (squares). Note that the maximum amount of ^{1}I^e character in any state is less than 7%. In the Rydberg series, $n \approx 13$ corresponds to $E \approx -3.76$ au. From [210].

matrix. It is clear that it is an approximation to freeze the $4l4l'$ wavefunctions in this way but it led to matrix sizes between 300 and 1150, which could be diagonalized in a reasonably short time.

As an example we show in figure 10 the effect of the interaction between the $4l4l'$ ^{1}I^e and the $3lnl'$ series. There are six Rydberg series but it turns out that only one of them has an appreciable interaction with the ^{1}I^e term. This makes the results easy to visualize. The figure shows (above) the amount of $4l4l'$ character in the eigenvalues of the combined system and (below) the autoionization decay rates, on a log scale, of the corresponding eigenstates. Notice that no level has more than 7% $4l4l'$ character, so the $4l4l'$ term is completely 'dissolved' in the Rydberg series.

The smooth behaviour of both curves is due to the $4l4l'$ term being located fairly high up in the $3lnl'$ series while the interaction is quite strong. The autoionization rates have a typical 'Fano type' profile with destructive interference on one and constructive interference on the other side of the resonance. The result of the minimum in the autoionization rate is that the probability for TDC is increased from 0.6 to 54.5%, an increase of nearly a factor of 100, assuming that initially the pure $4l4l'$ ^{1}I^e term is created, subsequently distributed over the $3lnl'$ series according to the calculated percentage composition, and the fraction of radiative decay in the total decay rate is determined from the decay properties of the series. A calculation with so many details is probably today only possible using B-splines. This is exemplified by the results in table 6, which compares different calculations of autoionization rates for ^{1}S^e terms belonging to $4l4l'$.

Two sets of B-spline values are shown: one set, $A^a_{\rm NI}$, for which the interaction with the Rydberg series has been neglected, and another, $A^a_{\rm WI}$, for which the interaction was included. Due to the strong mixing, the choice of a particular state as being '$4l4l'\ SL$' can be rather arbitrary. Column 3 shows the amount of $4l4l'$ for the terms identified as $4l4l'$ and we note

Table 6. Comparison of theoretical values for the total autoionization rates (in 10^{12} s^{-1}) of $4l4l'$ ^{1}S^e terms in O^{6+}. The B-spline results correspond to the autoionization rates obtained without, A^a_{NI}, and with, A^a_{WI}, the interaction with the Rydberg series taken into account. The amount of $4l4l'$ character in each term after interaction with the Rydberg series, C_P, is shown also. The results due to Roncin *et al* [380] were obtained for N^{5+}. The results labelled AIS correspond to the autoionization rates for the decay to the $2l$ limits of pure $4l4l'$ terms while the results labelled T are the sum of the autoionization rate and the 'auto-transfer' rate (see text). For terms above the $N = 3$ threshold, this last result corresponds to the sum of the autoionization rates to the $2l$ and $3l$ limits and can therefore be compared with the other results, while this is not the case for the states below threshold (see text). From [210].

	O^{6+}					N^{5+}	
						Roncin *et al* [380]	
$4l4l'$	A^a_{NI}	C_P %	A^a_{WI}	Ho [373]	Chetioui *et al* [370]	AIS	T
^{1}S^e	25.9	57	19.3	120	25.8	25.2	133
	53.7	54	43.7	641	53.9	52.1	537
	3.59	9	0.108	951	4.04	4.22	831[a]
	4.04[b]		4.04	4.96	1.78		0.628

[a] This term is above the $N = 3$ threshold in N^{5+} but not in O^{6+}.
[b] This term is above the $N = 3$ threshold and interacts weakly with the $3lnl'$ series, $A^a_{NI} = A^a_{WI}$.

that for the third term the maximum amount of $4l4l'$ is only 9%. The Feshbach formalism has been used by Chetioui *et al* [370] to calculate positions and autoionization widths for the $4l4l'$ states in O^{6+}. All configurations of the type $4lnl'$ ($4 \leqslant n \leqslant 7$) were included together with the first members of the Rydberg series $3lnl'$ (n up to nine). The results can be compared with the B-spline values that neglect the interaction with the $3lnl'$ states (A^a_{NI}). There is good agreement, which indicates that the effect of the inclusion of the first members of the Rydberg series on the autoionization rates is small.

The autoionization rates vary relatively little along the isoelectronic sequence. Roncin *et al* [380] applied the Feshbach formalism to the calculation of the autoionization rates for the $4l4l'$ singlet terms in N^{5+}. Their results (labelled AIS), neglecting the interaction with the $3lnl'$ series, are in very good agreement with the B-spline values for O^{6+} at the same level of approximation (A^a_{NI}) despite the different atoms involved. In the dynamic approach introduced by Roncin *et al* [380], the $3lnl'$ Rydberg states (for $n \to \infty$) are considered as a 'low continuum' into which the $4l4l'$ states 'auto-transfer' with a characteristic 'width'. The sum (labelled T) of this auto-transfer width and the autoionization width to the $2l$ limits is also given in table 6. For the state above threshold the value can be compared with the A^a_{WI} results, while for the states below the $N = 3$ threshold, they cannot, since here the mixing between the $4l4l'$ states and the Rydberg series is not an autoionization process but a transfer of $4l4l'$ character to the $3lnl'$ states. Bachau *et al* [368] noticed, however, that the T values are in rather good agreement with results of Ho [374] for the singlet $4l4l'$ manifold in N^{5+}. Ho calculated energy and autoionization widths for the $4l4l'$ ^{1}S^e terms in N^{5+} and O^{6+} using the CR method with Hylleraas basis sets [373] and later extended the results to all symmetries for N^{5+} [374]. His results for O^{6+} are given in table 6 and are one or two orders of magnitude larger than the other results for the same ion. However, the agreement between the results of Ho [373] for N^{5+} and the 'total' width of Roncin *et al* [380] for this ion is very good. Bachau *et al* [368] pointed out that this result raised doubts about how the results of the CR method for states overlapping with Rydberg series should be interpreted and it was later confirmed by Bürgers and Bachau [216] that the limit is artificially lowered, i.e. autoionization increased,

Table 7. Sizes of the Hamiltonian matrix for different B-spline bases using a full CI approach (i.e. including all states that can be constructed from the basis functions). The last column, headed 'N', refers to the number of coupling schemes included in the description of the different series. Only s, p, d and f electrons are considered. The number of splines is the same in each series. (From [333]).

	Number of basis functions					
LS	15	20	22	25	30	N
$^4S^o$	6 080	14 140	18 722	27 300	46 810	5
$^2S^o$	11 245	26 660	35 486	52 075	89 990	8
$^4S^e$	12 160	29 280	39 138	57 725	100 370	8
$^4P^e$	22 500	53 000	70 422	103 125	177 750	14
$^4P^o$	22 950	54 800	73 084	107 500	186 300	16
$^2S^e$	25 865	61 320	81 620	119 775	206 980	16
$^4D^o$	31 250	73 900	98 296	144 125	248 800	21
$^4D^e$	35 770	85 260	113 652	167 075	289 340	21
$^2P^e$	43 875	104 000	138 424	203 125	351 000	27
$^2P^o$	43 875	104 000	138 424	203 125	351 000	27
$^2D^o$	61 855	146 640	195 184	286 425	494 960	43
$^2D^e$	73 130	173 340	230 714	338 550	585 010	44

in the CR method if the basis set is truncated. The completeness of a B-spline basis would reduce the risk of this occurring.

Calculations for $4l4l'$ states in N^{5+} [209] and Ne^{8+} [211] have also been reported. In some calculations [207] the UCS approximation was improved by including a second-order correction Γ^1, equation (92), to Γ_s. Autoionization widths have also been evaluated beyond the UCS approximation using the formalism presented in section 4.2.1. In studies of the $3lnl'$ and $4l4l'$ states Bachau and collaborators have included the interchannel coupling in several papers [212–215]. Similarly, autoionization widths obtained including interchannel coupling are reported in a number of papers concerned with photoionization of atoms [164–168, 182] and with MPI of atoms [231, 233, 234, 236, 237], which are discussed in later sections.

4.3.3. Triply excited states. The extension to three and more electrons of the different techniques which have been successful for two-electron systems is in most cases difficult. There are no conceptual difficulties involved in the extension of the direct CI techniques, which is the case with many of the other techniques, but there are practical difficulties associated with the explosion in the size of the matrices. For two-electron systems and considering states which do not involve very large L-values, i.e. a very large range of l-values for the hydrogenic basis, it is possible to include basically all correlation in the calculation using a relatively small workstation as shown in the previous section. However, this is no longer possible when electrons are added. Table 7 shows the size of the full CI matrix for different symmetries using different numbers of B-splines .

The table also gives the number of coupling schemes that contributes for each symmetry using only s, p, d and f electrons, i.e. a fairly small basis set. Nevertheless, the increase in the size of the matrix clearly makes it impossible to include full correlation using a workstation. We noted earlier that Chen and Cheng [80] concluded that CI calculations for many-electron atoms were feasible. However, Chen and Cheng had the lowest excited states of an ion in mind while triply excited states, for example, involve a much larger number of coupling schemes.

Verbockhaven and Hansen [101, 221, 223, 225, 333] have published calculations using the TDM for three-electron states in the Li isoelectronic sequence with either one or two 1s

Table 8. Results for 1s2sns ^{2}S terms in Li using hydrogenic and HF basis functions. The CI matrix in the hydrogenic calculation had a size of about 16 000 × 16 000 while the matrix size using the HF basis was closer to 15 000 × 15 000. The labels in column 1 are based on the single-particle approximation. For the position of the hydrogenic TDM energy of (^{1}S)3s, see the text. From [101].

	Hydrogenic			Hartree–Fock		
^{2}S term	TDM	Shift	Final	TDM	Shift	Final
1s2s^2	56.172	0.288	56.460	56.397	0.010	56.407
1s2s(^{3}S)3s	61.962	0.055	62.017	62.007	0.003	62.010
1s2s(^{3}S)4s	63.175	0.019	63.194	63.172	0.001	63.173
1s2p^2(^{1}S)	63.366	0.038	63.403	63.355	0.000	63.355
1s2s(^{1}S)3s	63.058	0.612	63.670	63.586	0.002	63.588
1s2s(^{3}S)5s	63.727	0.011	63.738	63.738	0.001	63.739
1s2s(^{3}S)6s	63.996	0.008	64.004	63.970	0.000	63.970

electrons excited; the latter leads to a so-called 'hollow atom'. These authors were obliged to restrict the size of the matrices to less than 30 000×30 000, in many cases below 20 000×20 000 in order to obtain reasonable computation times. This was achieved by systematically removing basis states that gave a small contribution to the relevant eigenvectors and although the result now is basis dependent it was possible to obtain very good accuracy.

However, it is clear that in this situation the construction of the basis set can be important. Verbockhaven and Hansen [101] have shown that the use of an HF basis can improve the convergence considerably in the TDM. This is, for example, the case when large single-particle matrix elements are present because the interaction with the continuum is picked up by perturbation theory following the TDM. To be specific, consider the two 1s2sns ^{2}S series in Li for which the TDM approach means that the 'open channel', 1s^2ns/ϵs, is removed from the matrix, where ns/ϵs indicates the bound/continuum states associated with the 1s^2 ^{1}S limit in Li$^+$.

Of the two types of 1s2sns series, the 1s2s(^{1}S)ns ^{2}S series interacts with 1s^2ns ^{2}S while 1s2s(^{3}S)ns ^{2}S does not and this difference turns out to be very important for the quality of the results. In an HF approximation the interaction 1s^2ns ^{2}S $\leftrightarrow$ 1s2s(^{1}S)ns ^{2}S is zero to first order due to Brillouin's theorem but in a hydrogenic approximation the interaction is very large and this interaction must be taken into account in the TDM approach. Table 8 shows the results for both hydrogenic and HF basis functions.

The table gives the result of the TDM diagonalization and the shift due to the 1s^2ns/ϵs ^{2}S bound and continuum states, which is obtained using second-order perturbation theory as well as the corrected energy, 'Final' (the sum of 'TDM' and 'Shift') for the two different basis sets. The most obvious difference is clearly connected with the size of the shift, which is much larger for the hydrogenic than for the HF basis. Since the shift is obtained using perturbation theory, the value of the shift can be expected to be more reliable the smaller it is. In the present case, the size of the shift clearly indicates that second-order perturbation theory is of doubtful value in a hydrogenic basis. For the lowest ^{2}S term, 1s2s^2, the shift, 0.29 eV, reflects the size of the 1s^{2}2s $\leftrightarrow$ 1s2s^2 matrix element in a hydrogenic basis. In the calculation using the HF basis, the shift is much smaller, although not zero, as expected from the Brillouin theorem. One reason is that the HF basis was built on a frozen 1s^2 core while Brillouin's theorem applies in the case of a relaxed core. However, the size of the shift clearly shows the improvement obtained by use of the HF basis. The disadvantage of the HF basis is that single-particle matrix elements of the Hamiltonian are not identically zero as for a hydrogenic basis and the calculation of the elements of the CI matrix is therefore more time consuming than for the hydrogenic basis.

Table 8 shows that the advantage of the HF basis is even more important for the higher terms, which are located closer together and therefore are more sensitive to small shifts in energy. In this case the lowest terms going to the 1s2s ^{1}S limit are embedded in the 1s2s(^{3}S)ns series going to the (lower) 1s2s ^{3}S limit. Also perturbers, such as 1s2p^2(^{1}S)^{2}S, are mixed into the 1s2s(^{3}S)ns series. In fact from the third term, these three components are mixed together very strongly and in particular the naming of the third and fourth terms is not unambiguous. In this situation, it is particularly important that second-order corrections are small. In table 8 the terms are arranged in the order of the corrected energies. It is seen that the ordering of the hydrogenic TDM energies deviates from the final one. The offender is the state with the label 1s2s(^{1}S)3s, i.e. the second state in the series going to the 1s2s ^{1}S limit and interacting directly with 1s^2 ns. In fact this interaction is so large that the term is moved from third to fifth place in the series. In such a situation, if the eigenvalue interacts with its neighbours, the effects of moving it 'across' others can lead to larger errors in the calculation as indeed is found in this case. In contrast, the lowest term 1s2s^2 ^{2}S is also considerably perturbed but this term is located well below the next ^{2}S term and the wavefunction is only slightly modified, so the accuracy of the calculated shift is higher. This example shows that the use of HF basis functions can be very advantageous, although the case is probably somewhat special, the interaction 1s^2 ^{1}S^e↔ 1s2s ^{1}S^e being particularly large and notoriously difficult to control.

Verbockhaven and Hansen have studied several types of doubly and triply excited states in Li and Li-like ions [101, 221, 223, 225, 333] using the TDM. As mentioned above not all configurations were included; a selection was made in smaller exploratory calculations based on the size of the contributions to the appropriate eigenvectors. The remainder in the case of the energies was taken into account by second-order perturbation theory. 22 splines of order seven were used within each series with l-values from zero to three. This corresponds to the observation, mentioned earlier, by Chen and co-workers [76], who found in relativistic CI calculations that l-values larger than three were less important, in particular for high Z [77]. The accuracy obtained by Verbockhaven and Hansen was found to be comparable to that obtained by Chung and co-workers [392] with the SP approach. For three-electron states the orthogonality of initial and final states must be taken into account explicitly for those electrons, 'spectators', that do not take part directly in the interaction. For three electrons this is possible to do by adding overlap factors between spectator electrons explicitly in the calculation of the CI matrix elements, although the nonorthogonality increases the number of matrix elements that must be calculated very substantially compared to an orthogonal set of initial and final states.

The emphasis in these studies was on obtaining autoionization decay rates. It was mentioned earlier that for the three-electron systems the TDM has an additional problem for the calculation of autoionization rates, since the Feshbach approach for its accuracy assumes that the ionization limit is correctly described. Verbockhaven and Hansen used an HF approximation for the limits but obtained good agreement in the UCS with the SP approach, which did include continuum channel coupling and also used a better approximation than HF for the limits. In contrast it turned out that there is a source of error in such calculations, which had not been noticed in earlier calculations, an error related to the lack of orthogonality between initial and final states. In autoionization decays of Rydberg series of the type $2lnl'$, for example, it is well known that the decay rate in an unperturbed series will decay with n roughly as n^{-3}. Verbockhaven and Hansen [223] showed that for Rydberg series of the type $2l2l'nl''$, for example, the autoionization decay rate is determined by the decay of the inner $2l2l'$ electrons and thus constant independent of the nl'' electron, which basically is a spectator to the decay of the inner electrons. This independence turned out to hold for certain types of Rydberg series even if there is appreciable CI with other series and also to be independent of the

lack of orthogonality between initial and final states. However, to calculate the autoionization decay rate correctly it is necessary to take nonorthogonality into account. Since the largest contribution has the nl'' electron as a spectator, the orthogonality between initial and final nl states must be taken into account and the latter can be low because the final state in the above example has a 1s electron screening the nl'' electron, while the initial state with a $2l2l'$ core has considerably less screening of the nl'' electron. Thus it is important to include a range of 1snl limits in the calculation instead of just 1snl'' or, worse, only the lowest 1snl limits. The systematic study possible with a B-spline basis made this observation possible. However, also when a B-spline basis is used, it should be remembered that only Rydberg states up to a certain n-value are accurately described. The same applies to the limits, and calculations within a Rydberg series can therefore only be pursued up to a certain n-value.

Bachau [220, 222] has reported results for positions and widths of a few Li $2l2l'2l''$ autoionizing states, the lowest $3l3l'3l''$ $^2S^e$ states in N^{4+} and its isoelectronic sequence and some Ne^{7+} $4lnl'n'l''$ states using the Mandelshtam *et al* [363] approach of varying the box size discussed in section 4.2.2 to help in the interpretation of multiple electron pick-up in atom–ion collisions. The results were compared with the TDM and other calculations and good agreement was obtained despite the rather small basis sets employed.

4.3.4. Photoionization. The first theoretical attempt to study photoionization with B-splines was made in 1989 by Froese Fischer and Idrees [18, 19]. These authors used an extended multiconfiguration Hartree–Fock (MCHF) method in which the SCF procedure allows bound orbitals to vary with energy. They applied it successfully to study photoionization of the He atom below the $N = 2$ ionization threshold from both the $1s^2$ ^1S and 1s2p ^1P states, and analysed the $2lnl'$ Feshbach resonances appearing in the spectrum. Later on Brage *et al* [20] studied He photoionization using a modification of the original approach that leads to an energy-independent Hamiltonian matrix, which, therefore, must be computed only once. In the same work, the method was also applied to study photodetachment of H^-; the cross sections were in excellent agreement with results obtained by other authors.

In a series of papers [162, 163, 170], Chang and co-workers have also studied photoionization of He and H^- using an extension of the CI method (configuration interaction in the continuum, CIC), which is similar to that described in section 4.2. In a first paper [162] Chang and Tang studied photoionization of these two systems below the $N = 2$ threshold, paying special attention to the analysis of the $2lnl'$ resonance series (see also [170]). They found excellent agreement with previous theoretical works. In a subsequent paper, Chang and Zhen [163] considered photoionization of He from excited states 1sns $^{1,3}S$. This was the first prediction of photoionization cross sections from 1sns states with $n \geqslant 3$. The doubly excited states below the $N = 2$ threshold were studied in more detail later [195].

Chang and co-workers have extended the CIC method to study photoionization of alkaline-earth atoms. They used an effective two-electron Hamiltonian in which the core potential is described in terms of a frozen-core HF approximation. In practice, the CI expansion is built from single-particle orbitals representing an electron in the presence of the same core potential. The method has been used to study MPI of the Mg atom [226] as well as photoionization below the 2p threshold of the Be isoelectronic series from the ground and a few selected ^1S excited states [171, 172]. This group has shown that, except for photoionization from the GS, most of the strong transitions can be linked directly to one-electron bound–bound excitations with a nearly symmetric structure profile. Also photoionization in the presence of doubly excited autoionization structures has been studied in Al^+ and Si^{2+} [194].

We should also mention a paper by Chang and Fang [173], who studied near-threshold photoionization cross sections of He from bound excited states. They showed that the cross

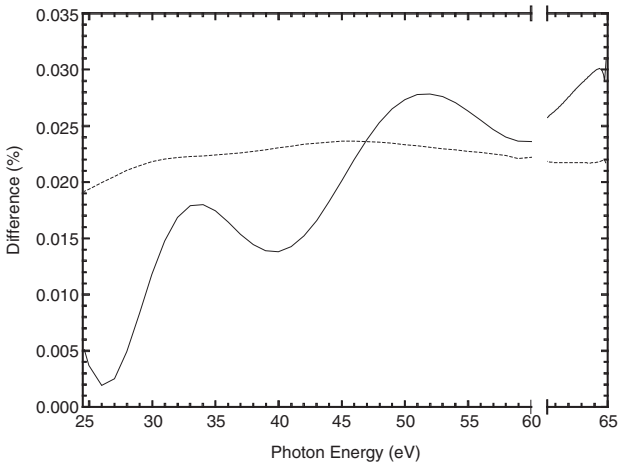


Figure 11. Percentage difference between length (l), velocity (v) and acceleration (a) cross sections for helium photoionization. Full curve, $(l - v)/v\%$; broken curve, $(v - a)/v\%$.

sections in the immediate vicinity of the ionization threshold depend linearly on the wavelength of the incident light, which supports earlier experimental observations [358].

The first study of multichannel photoionization using B-spline functions was carried out by Decleva *et al* in 1994 [164]. They studied photoionization of He and H^- below the $N = 4$ and 3 ionization thresholds, respectively, using a multichannel CC expansion method (see section 4.2.1) which includes orthogonal open and closed channels. In addition to total photoionization cross sections, they computed the partial cross sections and the angular distribution of photoelectrons. The results were in very good agreement with previous theoretical and experimental works. Later on, Venuti *et al* [165, 166] improved the accuracy of the results by using a combined CC and CI expansion. This consists in using a direct CI expansion in a region close to the nucleus and a CC expansion in the outer region. For He photoionization, the accuracy was better than 0.001 Mb for both total and partial cross sections. A stringent test of the completeness of the B-spline basis is provided by the gauge invariance of the results. Figure 11 illustrates the agreement between the length, velocity and acceleration results for the nonresonant total cross section as the percentage difference between length–velocity and velocity–acceleration forms. It is remarkable that even the generally inaccurate acceleration results are in excellent agreement with the velocity data, which are preferred as the length form displays a slightly oscillating behaviour. In any case, the differences are well below 0.03%, which is much better than reported in previous works. As shown in [165], the agreement with other accurate calculations [350, 351] is excellent, the difference being below 0.0013 and 0.0005 Mb, respectively. In contrast, the experimental results [352–354] are notably underestimated. A similar accuracy has been obtained for photoionization of H^- below the $N = 3$ threshold. Figure 12 shows a comparison of the theoretical results of [166] with the experimental data of Halka *et al* [355]. The theoretical results were convoluted with the experimental resolution of 7 meV. The agreement is very good, which is remarkable in the case of the two resonance structures observed in the vicinity of the $N = 2$ ionization threshold (a Feshbach resonance below the threshold and a shape resonance above the threshold). A similar good agreement was found with other theoretical calculations [347, 356, 357]. Very recently, Fang and Chang [182] have also used a multichannel K-matrix method as described in section 4.2.1 to study photoionization of H^- and He.

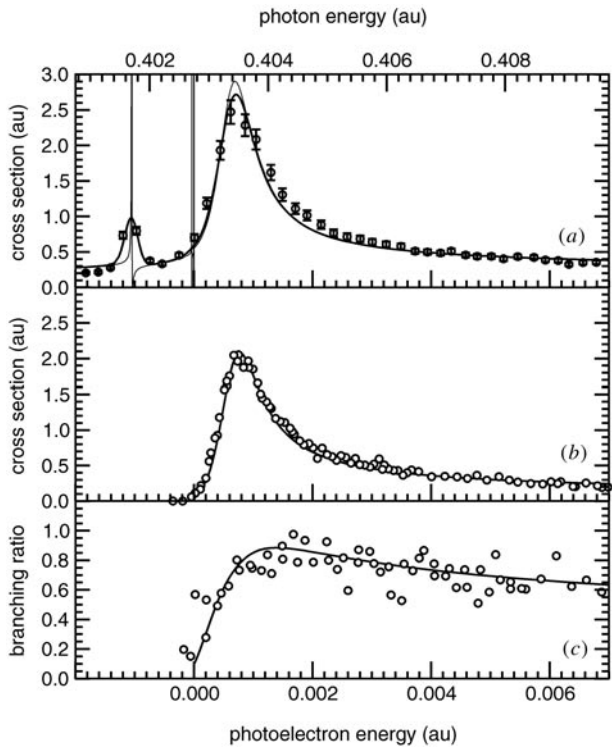


Figure 12. Comparison of *B*-spline calculations of Venuti and Decleva [166] with experimental data of Halka *et al* [355] for H^- photodetachment. The theoretical curves have been convoluted with the experimental resolution of 7 meV. Results before convolution are represented with the light line. The photoelectron energy for the $n = 2$ channels is indicated on the bottom axis. (*a*) Total cross section. (*b*) Partial $N = 2$ cross section. (*c*) Branching ratio $\sigma_{N=2}/\sigma_{\text{total}}$.

Bachau and Martín [178] have extended the *B*-spline techniques to study photoionization in the presence of a constant electric field. This is a multichannel problem in which the different angular momenta are coupled due to the lack of spherical symmetry. The field induces a Stark mixing of resonances with different values of L, thus leading to dramatic changes in the photoionization spectrum. These authors studied photodetachment of H^- in the presence of an electric field and analysed the behaviour of the Feshbach resonances lying below the $N = 2$ and 3 thresholds. In particular, they found that, for small fields, the main resonant structure below $N = 3$ corresponds to excitation of the lowest $3lnl'\ ^1P$ autoionizing state and that its amplitude decreases with increasing fields (in agreement with the experimental observations [359]). For stronger fields, the calculations exhibit new resonant structures resulting from a strong dipole mixing of autoionizing states.

Xi and Froese Fischer have extended the above techniques to study photoionization of atoms with more than two electrons [179–181]. In their approach, the target states, i.e. the states representing the remaining ion, are represented by multi-configuration state functions, with orbital functions obtained via an MCHF procedure. This is especially important for negative ions because, in many cases, a single-configuration target approach may lead to an inaccurate representation of the threshold. They have applied the method to study photodetachment of $He^-(1s2s2p\ ^4P^o)$ below the He ($n = 4$) threshold [179]. In contrast to earlier theoretical calculations, in addition to the sharp $1s2p^2\ ^4P$ resonance, they found

new ^{4}P Feshbach resonances and a ^{4}S one in the same energy region. Two Cooper minima were also found in the $1s2p(^3P^o)kp\ ^4P$ channel and the calculated angular distribution of the photoelectrons substantially improved on earlier predictions. In a subsequent paper [180], they have studied photodetachment of $He^-(1s2s2p\ ^4P^o)$ in the region of the 1s detachment threshold. Detachment of both the inner 1s electron and the outer 2s, 2p electrons was investigated. Although the calculation of the $2s2p^2\ ^4P$ Feshbach resonance was in agreement with other theoretical results, they predicted the existence of new resonance structures, in particular a very narrow 2s3s4s ^{4}S Feshbach resonance. Also recently, Xi and Froese Fischer have studied photodetachment of $Be^-(1s^22s2p^2\ ^4P)$ [181] for which no previous theoretical studies exist and experimental information is scarce. The calculations predict the existence of a resonance below the $Be(1s^22s3s\ ^3S)$ threshold in the $1s^22s2p(^3P^o)(ks, kd)\ ^4P$ channels and the result for the angular distribution agrees well with the few available experimental data.

Also the CR approach using B-splines has been applied to He^-. Brandefeldt and Lindroth [65, 331] have reported calculations for the four lowest ^{4}S resonances using a hydrogenic basis with $Z = 2$ expanded in up to 24 B-splines of order five for each l-value in a box with $r_{max} = 200$ au. The size of the matrices was around 9000×9000 and the authors comment that only the use of B-splines allowed them to keep the matrix down to this size. The largest l-values included were f and g. The latter were included only when the third electron was restricted to being a hydrogenic 1s. This is adequate in this case since these outer electrons are so far away from the 1s that screening of the inner electron is very weak. In addition to calculating the energies of the resonances their widths were also calculated and the results obtained were in good agreement with experiment and for the lowest resonance with the calculation of Xi and Froese Fischer [179].

Photoionization of many-electron atoms has been also studied with B-splines in the framework of the DFT (see section 4.1.2). Stener *et al* [167] have pointed out the importance of using an exchange–correlation potential with the correct coulombic behaviour at large distances in order to reproduce the series of Rydberg states and, therefore, the Feshbach resonances associated with one-electron excitations. These authors have used a time-dependent local density approximation (TDLDA) like that described in section 4.1.2 with a VLB exchange–correlation potential to study photoionization of noble gases [167], Zn [168] and alkaline-earth metal atoms [169]. Table 9 shows that the method provides a good description of singly excited Feshbach resonances, although it is not able to describe double excitations at this level of approximation. As an illustration, figure 13 shows the calculated photoionization cross section of Ar in the TLDA and LDA approximations from [167] compared with the experimental results [360]. The LDA results are very far from the measured values, while the TDLDA results are in good agreement. The small discrepancy at low photoelectron energy may be ascribed to shortcomings in the exchange–correlation potential [414]. Notice that at high energies the theory reproduces the Cooper minimum. In the case of Ca, the theory describes the $3p \rightarrow 3d$ giant resonance, which involves dramatic screening effects [169].

B-splines have also been used [175, 176] to describe double and triple photoionization of Li in the high-energy limit (but nonrelativistic). In this limit, the ejected electron can be described by a plane wave, but the inclusion of the two-electron interaction in both the initial and final state is required in order to estimate the triple-ionization cross section. Very recently, Li photoionization has been investigated using B-splines and nonorthogonal orbitals in the framework of the R-matrix theory [40]. The results are in good agreement with the experimental data. The method has the advantage of requiring much smaller bases for the accurate representation of target wavefunctions and to achieve convergence in the CC expansion.

Table 9. Theoretical and experimental values of Fano parameters relative to the three lowest 3s $\rightarrow n$p autoionization resonances in Ar. (From [167].)

	3s → 4p		3s → 5p		3s → 6p	
	TDLDA	Exp.	TDLDA	Exp.	TDLDA	Exp.
Γ (meV)	183.4	76(±5)	42.8	25(±7)	18.2	16(±7)
q	−0.164	−0.249(±0.003)	−0.107	−0.168(±0.005)	−0.086	−0.200(±0.005)
σ_0 (Mb)	30.95 (100%)	100%	27.66 (89%)	62(±1)%	26.47 (85%)	59(±1)%
E_R (eV)	26.846	26.606	28.115	27.993	28.532	28.506
ρ^2	0.9006	0.890(±0.005)	0.8989	0.873(±0.007)	0.8979	0.526(±0.007)
$\sum_i s_i^2$	6×10^{-4}	—	4.5×10^{-4}	—	9.5×10^{-3}	—

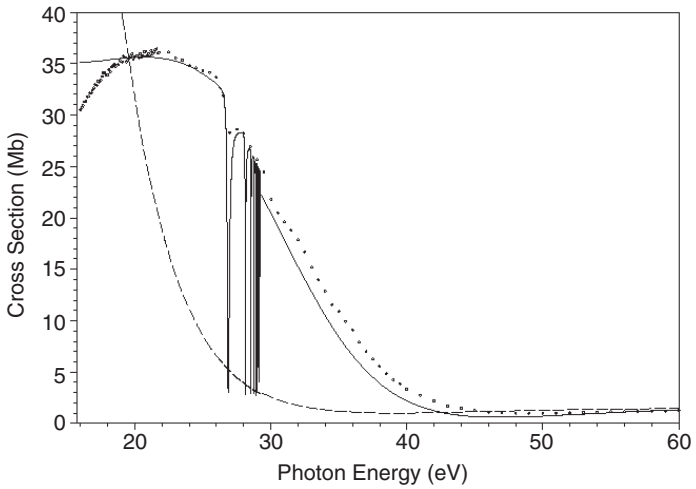


Figure 13. Total photoionization cross section of Ar. Full curve, B-spline TDLDA calculations [167]; broken curve, B-spline LDA calculations [167]; circles, experimental data [360].

5. Atoms in strong fields

The use of finite basis set expansions for the study of atoms in the presence of laser fields has received constant attention for more than 20 years. For example, MPI of atomic hydrogen using perturbation theory has been extensively studied with Sturmian functions [416]. In the last ten years, B-splines have been applied to treat the problem of one- and two-active-electron atoms and ions in the presence of strong laser fields. They were first introduced to treat laser–atom interactions in perturbation theory, but they have been shown also to be very efficient in the nonperturbative context (where one has to solve the time-dependent Schrödinger equation). We present here the general procedures used to study these nonlinear processes, with emphasis on B-spline peculiarities and discussions of some recent applications. In this section, the bibliography mainly refers to calculations involving B-splines. However, extensive discussions of the knowledge accumulated in both phenomenology and theoretical techniques can be found in recent reviews (see [327, 415, 417]).

5.1. Perturbative approach for multiphoton processes

We recall first the general procedure for the calculation of multiphoton processes in perturbation theory. The expressions for the transition amplitude $M_{\rm FI}^{(N)}$, the dynamic (or ac) Stark shift $\Delta_{\rm I}$ and the susceptibility $\chi^{(q)}$ will be given. From the numerical point of view, the difficulties inherent in the calculation of these three quantities are similar. The main point is to perform intermediate summation(s) over the bound and continuum spectrum and, in certain cases, to treat correctly the expressions at the pole(s). Therefore, we use similar discretization techniques in all cases. We shall put emphasis on the calculation of the transition amplitude. It is illustrated with the case of two-photon detachment of H^- with simultaneous excitation of the residual hydrogen atom. This is a good example of the efficiency of B-spline expansions since we have to deal with several difficulties: (i) the correct treatment of electronic correlations in initial, intermediate and final states, (ii) the appearance of a pole in the expression of the transition amplitude and (iii) the presence of several open continua in final states.

5.1.1. Multiphoton ionization and bound–bound transitions. We recall first the basic equations leading to the perturbative expressions of strong laser field effects on atoms. One has to solve the equation

$$[\mathcal{H} - E_A]|A\rangle = 0 \tag{94}$$

where $\mathcal{H}$ is the nonrelativistic Hamiltonian of the atomic system considered. In the case of the low-intensity regime with a slowly varying laser pulse shape (i.e. $T \gg 2\pi/\omega$, where T is the pulse duration and ω the photon frequency), the ionization probability can be calculated within the lowest order of perturbation theory (LOPT) [418]. The probability that the system ionizes through N-photon absorption is directly related to $|M_{\rm FI}^{(N)}|^2$, where the N-photon transition amplitude $M_{\rm FI}^{(N)}$ reads

$$M_{\rm FI}^{(N)}(\text{au}) = \lim_{\epsilon_1,\ldots,\epsilon_{N-1}\to 0^+} \sum\!\!\!\!\!\!\int_{A_{N-1}\ldots A_1} \times \frac{D_{FA_{N-1}}\cdots D_{A_1 I}}{(E_{\rm I} + (N-1)\,\omega - E_{A_{N-1}} + {\rm i}\epsilon_{N-1})\ldots(E_{\rm I} + \omega - E_{A_1} + {\rm i}\epsilon_1)}. \tag{95}$$

$|I\rangle$, $|{\rm F}\rangle$ and $|A_i\rangle$ are eigenfunctions of $\mathcal{H}$ (see equation (94)) with eigenenergies $E_{\rm I}$, $E_{\rm F}$ and E_{A_i}, respectively. $|{\rm F}\rangle$ refers to the final continuum state (normalized on the energy scale) with energy $E_{\rm F} = E_{\rm I} + N\omega$. The N-photon transition amplitude can be evaluated either in the length gauge (${}^lM_{\rm FI}^{(N)}$) with $D_{A_iA_j} = \langle A_i|\sum_{k=1,n} \boldsymbol{e}\cdot\boldsymbol{r}_k|A_j\rangle$, or in the velocity gauge (${}^vM_{\rm FI}^{(N)}$) with $D_{A_iA_j} = \langle A_i|\sum_{k=1,n} \boldsymbol{e}\cdot\boldsymbol{p}_k|A_j\rangle/\omega$, where n is the number of electrons. Formally, the two values are related by

$${}^lM_{\rm FI}^{(N)} = \frac{1}{\omega^N}\,{}^vM_{\rm FI}^{(N)}. \tag{96}$$

In practice, an indication of the quality of the results is provided by calculations of the transition amplitude in both gauges and by checking that equation (96) is satisfied.

The transition probability per unit time (i.e. the rate) for absorption of N photons from the initial state $|{\rm I}\rangle$ to the final continuum state $|{\rm F}\rangle$ is given by

$$\Gamma^{(N)}(\text{au}) = 2\pi\, I^N |M_{\rm FI}^{(N)}|^2 \tag{97}$$

where the intensity I and $M_{\rm FI}^{(N)}$ are given in au (the au for intensity is in this case $I_0 = 14.0378$ W cm^{-2}). In many applications, it is more useful to determine the generalized ionization cross section, which is an intensity-independent quantity given by

$$\sigma^{(N)}(\text{cm}^{2N}\text{s}^{N-1}) = \frac{2\pi}{\tau_0}\left(\frac{\omega}{F_0}\right)^N |M_{\rm FI}^{(N)}|^2 \tag{98}$$

where τ_0 is the au of time expressed in seconds ($\tau_0 = 2.418\,88 \times 10^{-17}$ s), F_0 is the unit of flux expressed in cm^{-2} s^{-1} ($F_0 = 3.219\,86 \times 10^{34}$ cm^{-2} s^{-1}) and all other quantities (ω and $M_{\mathrm{FI}}^{(N)}$) are expressed in au. Equation (95) is valid provided that the multiphoton process is nonresonant. When the photon frequency is such that a combination of absorbed photons from the initial state $|\mathrm{I}\rangle$ reaches an intermediate bound or autoionizing state $|A_i\rangle$ of the atom, the N-photon process is called resonant and must be treated using a different formalism [419]. In the case where an intermediate bound state $|A_i\rangle$ is resonant with the absorption of N' photons, whatever formalism is used (resolvent operator, density matrix etc), one must calculate the Rabi frequency Ω_{R} of the relevant bound–bound transition. This is achieved through the evaluation of the amplitude $M_{A_i I}^{(N')}$ given by equation (95) where the final continuum state $|\mathrm{F}\rangle$ has to be replaced by the resonant bound state $|A_i\rangle$. In addition, one has to evaluate the bound–continuum amplitude $M_{FA_i}^{(N-N')}$ as well.

We focus now on numerical aspects of the discretization procedure using B-splines. Assuming an initial state with total angular momentum $L = 0$, the eigensolutions of $\mathcal{H}$ must be determined for $L = 0, 1, \ldots, N$, N being the order of the process. $\mathcal{H}$ is diagonalized in the B-spline basis set as explained in the previous sections. Using the discrete set of eigenfunctions $|A_i\rangle$ (see equation (94)) in expression (95), the integrals are replaced by summations over the discrete set of eigenfunctions (equivalent quadrature). When no pole appears (e.g. in the case of nonresonant MPI), equation (95) is written

$$M_{\mathrm{FI}}^{(N)}(\mathrm{au}) \simeq \sum_{A_{N-1}\ldots A_1} \frac{D_{FA_{N-1}} \ldots D_{A_1 I}}{(E_{\mathrm{I}} + (N-1)\,\omega - E_{A_{N-1}}) \ldots (E_{\mathrm{I}} + \omega - E_{A_1})}. \tag{99}$$

Thus, this method requires us to solve an eigensystem for each total angular momentum L (M is fixed for linear polarization) and to perform the summation (99). These operations are performed by using standard numerical routines. All intermediate states are normalized to unity, while the final continuum state is properly normalized on the energy scale by either calculating the density of states $\rho(E_{\mathrm{F}})$ or fitting the numerical wavefunctions on the asymptotic expressions (see sections 3.3 and 4.2). Thus, the problem of intermediate summations is extremely simplified in this context. As in the case of atomic structure studies, the convergence of the calculation must be checked by varying the box size and the average density of B-splines. In general, the order of B-splines is chosen between seven and ten (see sections 3.2 and 3.3 for convergence issues).

Another quantity of interest is the ac Stark shift. It plays an important role in many situations, for example when studying the resonant behaviour of complex atomic systems under strong, short, near-UV pulses [423]. For a specified level $|\mathrm{I}\rangle$, with unperturbed energy E_{I}, the shift (to second-order LOPT) is

$$\Delta_{\mathrm{I}}(\mathrm{au}) = \frac{I(\mathrm{W\ cm^{-2}})}{I_0(\mathrm{W\ cm^{-2}})}\, \mathcal{P} \sum\!\!\!\!\!\!\int_{A_n,\pm} \frac{|D_{IA_n}|^2}{E_{\mathrm{I}} - E_{A_n} \pm \omega} \tag{100}$$

where E_{I}, E_{A_n}, ω and D_{IA_n} are expressed in au. $I_0 = 14.0378$ W cm^{-2} and $\mathcal{P}$ refers to the Cauchy principal part which applies when a pole appears in the above expression, i.e. when one-photon ionization of the level $|\mathrm{I}\rangle$ is energetically allowed. With the discretized B-spline basis set, the above expression is written

$$\Delta_{\mathrm{I}}(\mathrm{au}) \simeq \frac{I(\mathrm{W\ cm^{-2}})}{I_0(\mathrm{W\ cm^{-2}})} \sum_{A_n,\pm} \frac{|D_{IA_n}|^2}{E_{\mathrm{I}} - E_{A_n} \pm \omega} \tag{101}$$

where all states are normalized to unity. If a pole appears at the energy E_{A_i} in the continuum ($E_{\mathrm{I}} - E_{A_i} + \omega = 0$), the above summation must exclude the state $|A_i\rangle$. If the resonant state $|A_i\rangle$ is a bound or autoionizing state, it must be also excluded from the summation and its

effect should be treated nonperturbatively. As in the case of the resonant multiphoton process, discussed above, the latter situation makes it possible to treat explicitly the resonant states and to evaluate the Ω_R Rabi term associated with the resonant transition.

5.1.2. Above-threshold ionization. We consider now the case of ATI. ATI appears when the number of photons that are necessary to ionize the system is smaller than the number of photons really absorbed. In this case one or several photons are absorbed in the continuum and, therefore, one or several 'zeros' (poles) appear in the denominator of the transition amplitude $M_{FI}^{(N)}$ (see equation (95)). A possible method to calculate the N-photon transition amplitude is to isolate the dipole matrix elements associated with the poles before replacing the integral by a quadrature [228, 420]. Thus, in a first step, the integral in equation (95) is split into two parts

$$\lim_{\epsilon\to 0^+}\int \mathrm{d}E\,\frac{1}{E_0-E+\mathrm{i}\epsilon}=\mathcal{P}\int \mathrm{d}E\,\frac{1}{E_0-E}-\mathrm{i}\pi\delta(E_0-E) \tag{102}$$

where $\mathcal{P}$ is the Cauchy principal part. This approximation is valid under the assumption that the dipole matrix elements in the numerator are slowly varying functions of the energy in the region of the pole (it is the condition for the validity of the 'pole approximation' [421]). In the case of one-active-electron systems this condition is fulfilled since the continuum is 'smooth' (no resonant structures are embedded in the continuum). In a second step, the integral in equation (102) is replaced by the quadrature provided by the discrete basis of eigenstates

$$\mathcal{P}\int \mathrm{d}E\,\frac{1}{E_0-E}-\mathrm{i}\pi\delta(E_0-E)\simeq\sum_{n'\neq 0}\frac{1}{E_0-E_{n'}}-\mathrm{i}\pi\rho(E_0)\delta_{0n'} \tag{103}$$

where $\rho(E_0)$ is the density of states at $E=E_0$. The normalization of intermediate states to be used in the right-hand side of the above expression is one for all n'. Notice that the discretization procedure described here was initially performed with Slater functions by Bachau *et al* [420] in the early 1990s. The authors showed that these functions have two important limitations in the present context: (i) the system is not strictly confined in an arbitrary box with a Slater basis set and (ii) the continuum function evaluated at the pole has a limited radial extension. Due to these limitations, the calculations of (i) the summation and (ii) the pole contribution in expression (103) are difficult to converge. It is worth mentioning that such limitations are not inherent to the choice of a Slater basis set; they also exist with Gaussian functions for example. This difficulty led Bachau *et al* to choose B-spline functions, which have been widely used in this context since then. A difficulty of the discretization procedure is that, for a given box length, the discrete energies do not necessarily match the continuum energies at the poles. In practice, one overcomes this difficulty by varying the box size until an eigenenergy matches the required value. This approach works well for two-photon ATI but becomes cumbersome when many absorptions (i.e. many poles) occur in the continuum. The other difficulty is that this treatment involves isolated free–free dipole matrix elements (associated with the imaginary part of equation (103)) whose calculation requires the evaluation of a contribution from outside the box. (We note in passing that these contributions are automatically eliminated in the context of the TDSE procedure developed later in this section—see section 5.2.5 and [261]. Indeed, the continua produced through multiphoton absorption are stationary states within LOPT and the properties based, within TDSE, on the finite dimension of the wavepacket and interaction time are not valid here.) In order to keep the advantages of the equivalent quadrature, an alternative approach has been proposed by Cormier and Lambropoulos [235]. This method (referred to as the extrapolation method) consists in evaluating the expression (95) with a finite value of ϵ. The analysis of the dependence of the transition amplitude on ϵ shows that it converges to

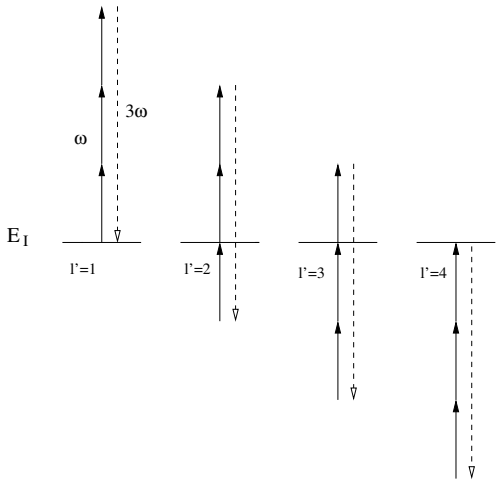


Figure 14. Schematic representation of the four different paths involved in the calculation of the nonlinear susceptibility $\chi^{(3)}$. The horizontal line represents the energy of the initial state $E_{\rm I}$. Note that each l' path results in the emission of an harmonic photon of energy 3ω, represented by a dashed arrow. The short full arrows represent the absorbed photon energy ω.

the correct value, provided that ϵ is larger than the discretized energy spacing in the region of the pole. Somehow, by putting a 'width' ϵ to the discretized continuum states, one avoids the divergence at the pole and the consistency of the equivalent quadrature is preserved. Then the difficulties inherent in the treatment of the pole, mentioned above, disappear. This approach is consistent with the formal equation (95) in the limit of an infinite box, i.e. when the energy spacing between the continuum states tends to zero.

5.1.3. Nonlinear susceptibility. Another important quantity in physics is the nonlinear susceptibility $\chi^{(q)}$. It appears for instance in the evaluation of the harmonic generation within LOPT [424, 425]. We consider here a nonlinear process where the emission/absorption of q photons with energy ω from the initial state $|{\rm I}\rangle$ results in the emission of an harmonic photon with energy $q\omega$. In the latter context, the expression for $\chi^{(q)}$ is, in the length gauge (we assume a linear polarization of the field)

$$\chi^{(q)}({\rm au}) = \sum_{l'=1}^{q+1} \sum\!\!\!\!\!\!\int_{A_{n_1}, A_{n_2}, \ldots A_{n_q}} \times \frac{D_{I A_{n_q}} \ldots D_{A_{n_2} A_{n_1}} D_{A_{n_1} I}}{(E_{A_{n_q}} - E_{\rm I} - l_q\omega) \ldots (E_{A_{n_2}} - E_{\rm I} - l_2\omega)(E_{A_{n_1}} - E_{\rm I} - l_1\omega)}$$

where $l_j = q + 1 - j$ if $l' \leqslant j$ and $l_j = -j$ if $l' > j$; all quantities are given in au. The first summation ($l' = 1 \cdots q + 1$) corresponds to different paths for the absorption or emission of incident photons and the emission of the harmonic photon. Each path is associated with an absorption/emission mechanism. This is illustrated in figure 14 for the case of the calculation of $\chi^{(3)}$, where four paths are involved. For each path, there are q intermediate summations. Note that, when $q\omega$ is greater than the binding energy of the initial state $|{\rm I}\rangle$, one or several poles appear in the above expression, which becomes complex. From the technical point of view, the treatment of this expression with the discrete basis set does not differ significantly from the calculation of the transition amplitude given in equation (99). An illustrative example of LOPT calculation of $\chi^{(3)}$ with B-splines is given in [243] for the helium case.

5.1.4. Applications. We present now, for the case of H^-, an illustrative example of the calculation of the transition amplitude $M_{\rm FI}^{(N)}$ with $N = 2$ [234]. For a two-electron system,

the electronic wavefunction satisfies the equation

$$[\mathcal{H} - E]\Psi_n = 0 \tag{104}$$

where the nonrelativistic Hamiltonian $\mathcal{H}$ reads

$$\mathcal{H} = H_0(1) + H_0(2) + \frac{1}{r_{12}}. \tag{105}$$

The labels 1 and 2 refer to electrons 1 and 2, respectively, and H_0 is the one-electron nonrelativistic Hamiltonian (see section 3.1). For a singlet state with a given total angular momentum L and projection M, the solution of equation (104) is expanded on a basis of two-electron configurations which are products of one-electron orbital functions as follows:

$$\Psi_n^{L,M}(\boldsymbol{r}_1, \boldsymbol{r}_2) = \mathcal{O} \sum_{\alpha=i,j,l_1,l_2} C_{n,\alpha}^{L,M} \Phi_i(r_1)\Phi_j(r_2)\mathcal{Y}_{l_1,l_2}^{L,M}(1,2) \tag{106}$$

where $\mathcal{O}$ is the symmetrization operator, l_i is the angular momentum of electron i and $\mathcal{Y}_{l_1,l_2}^{L,M}(1,2)$ is the usual angular part with a well defined total L and M. The orbitals $\Phi_i(r_1)$ and $\Phi_j(r_2)$ (which are eigenfunctions of the one-electron Hamiltonian H_0) are expanded on a B-spline basis set as explained in section 3.1. We consider two-photon detachment of H^- with simultaneous excitation of the residual H atom within LOPT. In this case, the energy of a single photon is larger than the binding energy of the outer electron. Therefore, this corresponds to the so-called above-threshold detachment (ATD), which is equivalent to ATI in neutral atoms. In addition, one has to deal with other difficulties related to the excitation of autoionizing states and the fact that the final continuum state is above the second ionization threshold (multichannel problem, see section 4.2.1). Thus it is an illustrative example of the efficiency of the B-spline basis set in a case where correlations play an important role. The amplitude to determine is given by

$$M^{(2)}_{F_\mu(E_F)I} = \lim_{\epsilon\to 0^+} \sum\!\!\!\!\!\!\int_{n'} \frac{D_{F_\mu(E_F)n'} D_{n'I}}{E_I + \omega - E_{n'} + i\epsilon}. \tag{107}$$

$F_\mu(E_F)$ is the final continuum state, with energy $E_F = E_I + 2\omega$, in the channel μ and n' refers to the solutions of equation (104) with $L = 1$ and $M = 0$ (linear polarization is assumed). In [234], the calculations were performed in the velocity gauge (where convergence is faster) using a basis of $N = 190$ B-splines with $k = 10$ and $r_{max} = 150$ au. Equation (104) was solved by expanding the wavefunction as in equation (106). A significant difficulty of this approach is that, due to truncation of expansion (106), the position of the GS of H^- may differ significantly from the 'exact' one. Thus, in addition to the usual hydrogenic orbitals, some selected STOs were represented as linear combinations of B-splines. A full CI in this basis leads to an energy of $-0.527\,576$ au to be compared with the 'exact' nonrelativistic Pekeris value $-0.527\,751$ au. The final states $\Psi^{\Omega-}_{\mu E_F} \equiv F^-_{1skl}(E_F)$ and $\Psi^{\Omega-}_{\mu' E_F} \equiv F^-_{2lkl'}(E_F)$ with correct asymptotic conditions were built following the procedure presented in section 4.2.1. In this case, it was necessary to build uncoupled continuum states (UCSs, see section 4.2.1) associated with the 1s, 2s, 2p, 3s, 3p, 3d, 4s, 4p, 4d and 4f detachment thresholds to evaluate the Green function of the coupled problem. For the photon energy range considered, one-photon detachment is energetically allowed and a pole exists in the two-photon detachment amplitude. This pole is located in a 'smooth' continuum region (i.e. far from the autoionizing states lying below the H($2l$) threshold) and the pole approximation is valid. Note that an interesting situation occurs at $\omega = 0.375$ au. At this particular photon energy the hydrogenic transition 1s–2p is resonant and one- and two-photon detachment with $2l$ excitation leads to the same final electron energy. As a consequence, at the pole, the dipole matrix elements involve a divergent term of the type $\langle 1sk\mathrm{p}|r|2lkl'\rangle$ (the final electron has the energy $k^2/2$ in both continua). Thus LOPT cannot be applied around the photon energy 0.375 au.

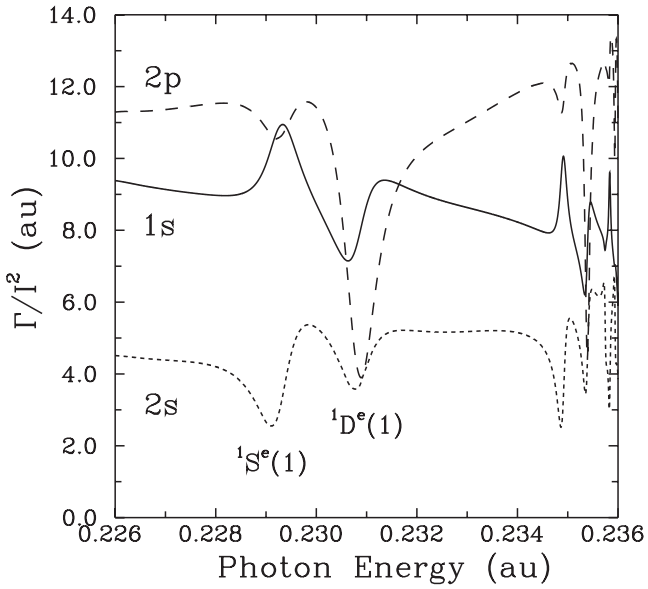


Figure 15. Partial two-photon detachment rates of H^- as a function of photon energy for $\omega = 0.226$–0.236 au. Labels indicate symmetry of the resonances [234].

We show in figure 15 the partial two-photon detachment rates of H^- with simultaneous excitation of the 2s and 2p states of the residual hydrogen atom. Except in the vicinity of the resonance structures, the 2p cross section is larger than the 1s one, and the latter is larger than the 2s one. The rate for leaving the H atom in the GS is in good agreement with the values calculated by Proulx *et al* [426]. In figure 16 we compare the integrated two-photon signal in the region of the lower $^1D^e$ resonance of H^- with recent experimental values [427]. Except for a small shift in the resonance position (which lies in any case within the experimental error bars, see figure 16), the results are in excellent agreement with the experimental signal.

The most recent experimental works on H^- have focused on the angular distribution of photoelectrons produced by two-photon absorption. In [428] two-photon detachment has been performed in the vicinity of the threshold for one-photon detachment, and the angular distribution of the ejected electron has been analysed. In [237] the asymmetry parameters β_2 and β_4 were calculated with the *B*-spline discretization technique and they were compared with the experimental values [428]. The results are shown in figure 17. The very good agreement between experimental and theoretical results is another illustration of the versatility of the method. Besides [234] already cited, other calculations for H^-, based on *B*-splines, can be found in [232, 236, 241]. The method has been also applied to calculate MPI and ATI of Be [229, 231, 233] and He [230, 233, 242, 245].

The case of MPI of magnesium has focused the attention of many scientists for years. As far as we know the first MPI calculations involving *B*-splines were published in the early 1990s [226, 227]. From the experimental point of view, autoionizing states can be reached by absorption of a few photons in the UV range; these photons can now be produced with enough intensity. On the theoretical side, Mg has the advantage of having only two active electrons, which simplifies the atomic structure calculations. Thus Mg is a good candidate to compare theory and experiment in the case where autoionizing resonances are involved. The situation where two autoionizing states are resonantly coupled by the laser field is the

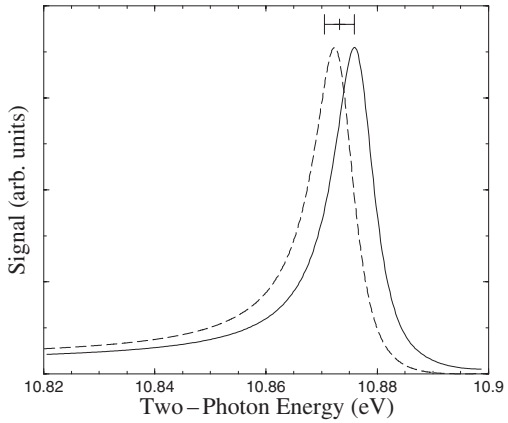


Figure 16. The yield of electrons in the two-photon detachment continuum in the region of the lowest $^1D^e$ resonance of H^-: B-spline results from [234] (solid curve) and the best fit of experimental results (dashed curve) using the Beutler–Fano line shape [427]. We also indicate (above the shape resonance maximum) the experimental energy position of the resonance with the error bar [427].

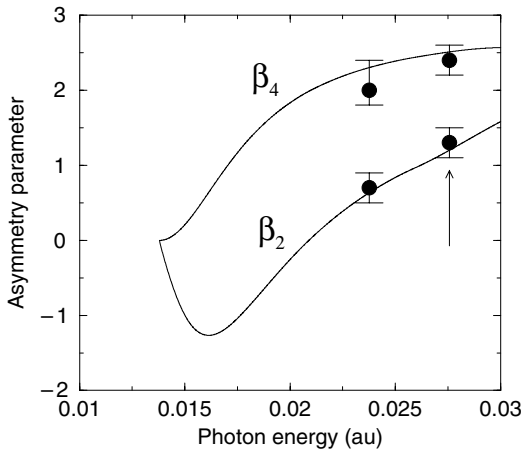


Figure 17. Asymmetry parameters β_2 and β_4 as a function of the photon energy; the theoretical results [237] are represented with the continuous curves and the filled circles with error bars represent the experimental values [428]. The arrow indicates the position of the one-photon detachment threshold.

typical case of a resonant process where perturbation theory is not valid. One example is four-photon excitation of the $^1S^e(4s^2)$ resonance in Mg. The photon energy is about 0.1190 au and a three-photon resonance with the $^1P^o(3p4s)$ autoionizing state occurs [244]. In the latter situation the $^1S^e(3s^2)$, $^1S^e(4s^2)$ and $^1P^o(3p4s)$ states must be treated explicitly while the other (nonresonant) states can be introduced perturbatively. Various formalisms can be used. The resolvent operator approach [421] and the density matrix formalism [422] have been shown to be efficient in this context. An experimental and theoretical investigation of the effects of the strong electromagnetic coupling of two autoionizing states in Mg has been published by Karapanagioti *et al* [238–240]. In this case, the density-matrix approach, developed in the context of the Feshbach formalism [429] with B-spline expansions, has been successfully

applied. Note also that this type of problem can be treated solving the TDSE by expanding the solution on a few complex states. The latter approach, which also uses an L^2 basis, has been applied to treat the electromagnetic coupling of autoionizing states of helium below the $N = 3$ threshold [430].

5.2. Nonperturbative time-dependent methods

The constant evolution of laser technology has reached the point where field strengths in excess of the au E_0 (the Coulomb electric field experienced by an electron in the GS of hydrogen) are commonly achieved with table top devices. As a consequence, during the interaction of an atomic system and a laser beam, the external laser field cannot be considered as a perturbation to the internal field but has to be treated on an equal footing. The domain of laser field intensities roughly ranging, depending on the target and the laser frequency, from 10^{12} W cm^{-2} (upper limit of the perturbative regime) up to 10^{17} W cm^{-2} (lower limit of the relativistic domain) is commonly referred to as the nonperturbative regime. In this intensity domain, various phenomena can be observed by measuring the outgoing electrons or light: to cite a few, ATI [439], HOHG [440], multiple ionization [441] etc. In this section, we report on theoretical investigations based on a direct integration of the time-dependent Schrödinger equation (TDSE) by means of B-spline methods.

5.2.1. Interaction with intense laser fields. In order to account for the explicit time-dependence of the interaction, the pulsed laser electric field is represented using a semiclassical picture in which

$$\boldsymbol{E}(t) = -\frac{\partial}{\partial t}\boldsymbol{A}(t) \tag{108}$$

with a field potential given by

$$\boldsymbol{A}(t) = A_0 f(t)\sin(\omega t + \phi)\boldsymbol{e}_z \tag{109}$$

where A_0 is the amplitude, $f(t)$ is the pulse envelope and $\boldsymbol{e}_z$ is the polarization direction. Notice that in contrast to perturbation theory (see section 5.1.1) the above field leads to the definition of the au of intensity $I_0 = 3.509 \times 10^{16}$ W cm^{-2}.

Realistic pulse shapes can be described using Gaussian envelopes, $f(t) = \exp(-t^2/2\tau^2)$, or cosine square functions, $f(t) = \cos(\pi t/\tau)^2$ with $t \in [-\tau/2, \tau/2]$ (the latter has the practical advantage of getting rid of the long Gaussian tails). Typically, the wavelength of the lasers used in the nonperturbative regime is large enough to justify the use of the dipole approximation in equations (108) and (109), which do not include spatial variables for this reason. For simplicity, we restrict ourselves to the use of linear polarization, which we set along the z-axis.

In such a time-dependent problem, all useful information about the dynamics is extracted from the knowledge of the state of the system during the interaction with the field as well as after the pulse has gone. Therefore, we integrate the time-dependent Schrödinger equation

$$\mathrm{i}\frac{\partial}{\partial t}\Psi(t) = [\mathcal{H}_0 + D(t)]\Psi(t) \tag{110}$$

with the initial condition $\Psi(t = -\infty) = \Psi_\mathrm{g}$, where Ψ_g is the initial field-free atomic stationary state. The field-free Hamiltonian, $\mathcal{H}_0$, does not depend explicitly on time and has the usual kinetic and potential energy terms (see sections 3.1 and 4 for one- and two-electron atoms, respectively). $D(t)$ represents the laser–atom interaction and depends explicitly on time according to the formulae given in table 10. The problem of the choice of gauge is discussed in section 5.2.5.

Table 10. Expressions of the laser–atom interaction in both gauges for one- and two-electron systems.

Gauge	One electron	Two electrons
Length	$D(t) = -\boldsymbol{E}(t) \cdot \boldsymbol{r}$	$D(t) = -\boldsymbol{E}(t) \cdot (\boldsymbol{r}_1 + \boldsymbol{r}_2)$
Velocity	$D(t) = -\boldsymbol{A}(t) \cdot \boldsymbol{p}$	$D(t) = -\boldsymbol{A}(t) \cdot (\boldsymbol{p}_1 + \boldsymbol{p}_2)$

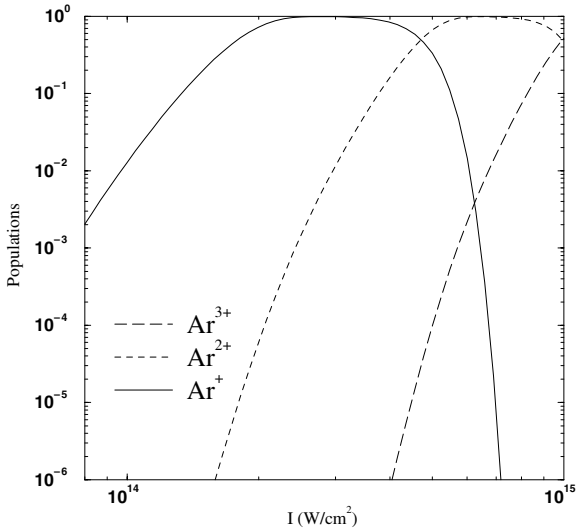


Figure 18. Probabilities for single, double and triple ionization of Ar with a 100 fs pulse as function of the laser intensity. (From [442].)

5.2.2. Solving the TDSE for one-electron systems. This section will be devoted to hydrogen and hydrogen-like ions, but also to many-electron systems that can be treated in the single-active-electron (SAE) approximation [340, 341]. Indeed, it has been shown that many features of the two dominant processes observed in the nonperturbative regime (ATI and HOHG) are well described by assuming that only one electron interacts with the laser field while all other electrons remain frozen. There is, however, a limitation to the SAE model. One has to make sure that the laser intensity is not too high to induce double ionization. To give an idea, we have plotted in figure 18 the probability of singly and doubly ionizing Ar as a function of the field intensity. Clearly, the SAE model does not hold for intensities above 5×10^{14} W cm^{-2}. However, below $I = 3 \times 10^{14}$ W cm^{-2}, most processes should be accurately reproduced within the SAE approximation because double ionization is not significant, even when single ionization has reached saturation.

Since there exists no analytical solution of (110), the integration has to be performed numerically. The procedure is very similar to that described in section 3.1 for solving the stationary Schrödinger equation. Equation (110) is solved by introducing the following expansion:

$$\Psi(\boldsymbol{r}, t) = \sum_{l=0}^{l_{\max}} \sum_{i=1}^{N} c_i^l(t) \frac{B_i(r)}{r} Y_l^0(\theta, \phi). \tag{111}$$

As a consequence of the linear polarization, the value of the quantum number m will be preserved at any time (in the above equation we have chosen $m = 0$). By introducing (111) in

the ordinary differential equation (110) we obtain a system of coupled differential equations:

$$\mathrm{i}\overline{\boldsymbol{S}} \cdot \dot{\boldsymbol{c}}(t) = (\overline{\boldsymbol{H}}_0 + \overline{\boldsymbol{D}})(t) \cdot \boldsymbol{c}(t) \tag{112}$$

with the initial condition $\boldsymbol{c}(t = -\infty) = \boldsymbol{c}_{\mathrm{g}}$ given by the system in its GS and where $\boldsymbol{c}(t)$ is the vector of $Nl_{\max}$ unknown time-dependent coefficients $c_i^l(t)$ to be determined. $\overline{\boldsymbol{S}}$ and $\overline{\boldsymbol{H}}_0$ are diagonal block matrices of individual matrices $\boldsymbol{S}$ and $\boldsymbol{H}_l$ defined in equation (7). $\overline{\boldsymbol{D}}(t)$ is a co-diagonal block matrix of elementary matrices containing the dipolar coupling matrix elements of $D(t)$. The time derivative as well as the value of $\boldsymbol{c}(t)$ are evaluated through the relations

$$\dot{\boldsymbol{c}}(t) = \frac{\boldsymbol{c}(t + \frac{\delta t}{2}) - \boldsymbol{c}(t - \frac{\delta t}{2})}{\delta t} \tag{113}$$

$$\boldsymbol{c}(t) = \frac{\boldsymbol{c}(t + \frac{\delta t}{2}) + \boldsymbol{c}(t - \frac{\delta t}{2})}{2} \tag{114}$$

which leads to the implicit propagation scheme in δt^2 also known as the Crank–Nicholson method (it is simply the Cayley–Hamilton form of the propagation operator):

$$\underbrace{\left(\overline{\boldsymbol{S}} + \mathrm{i}(\overline{\boldsymbol{H}}_0 + \overline{\boldsymbol{D}}(t))\frac{\delta t}{2}\right)}_{\boldsymbol{A}(t)} \cdot \boldsymbol{c}(t + \delta t) = \underbrace{\left(\overline{\boldsymbol{S}} - \mathrm{i}(\overline{\boldsymbol{H}}_0 + \overline{\boldsymbol{D}}(t))\frac{\delta t}{2}\right) \cdot \boldsymbol{c}(t)}_{\boldsymbol{b}}. \tag{115}$$

In practice, from the state $\boldsymbol{c}(t)$ of the system at time t, the value at time $t + \delta t$ is computed by solving the system of linear equations $\boldsymbol{A}(t) \cdot \boldsymbol{c}(t + \delta t) = \boldsymbol{b}$. The sparseness of the matrices due to the B-spline properties is fully exploited by using a biconjugate gradient method with preconditioning. The propagation is then carried out from the initial time (when the pulse is turned on) up to the end of the pulse using expression (115).

5.2.3. Solving the TDSE for two-electron systems. There exist specific processes in which electron correlation cannot be neglected. For instance, autoionization or simultaneous double ionization cannot be explained without an explicit inclusion of the electron–electron interaction. The study of the time evolution of two interacting electrons in the nonperturbative regime is not an easy task and leads to large-scale computational simulations. We summarize here the general methods used to solve the two-electron TDSE using B-splines.

Unfortunately, a method such as that presented in section 5.2.2 in which the wavefunction is directly expanded in a basis of B-splines has not been developed yet. This is mainly due to the fact that the sparse matrices involved in this case have a more complicated structure than the regular bands inherent in the one-electron problem. Rather, the ordinary differential equation is solved by first expanding the time-dependent wavefunction on the bare-atom eigenstate basis and then propagating the wavefunction on that basis. This approach, often referred to as the basis calculation, requires a computation of the basis of eigenstates with B-spline techniques. So far, two alternatives (one originally used by Tang *et al* [260] and reviewed in [327], and the other by Hasbani *et al* [270, 332]) are used to build the basis in which the TDSE is subsequently solved. The CI methods employed in both approaches are described in section 4. However, it is worth recalling the conceptual differences. The first method involves two steps (see e.g. [260, 327].) In the first step, one computes a basis of one-electron orbitals of the ionized system using the techniques described in section 3. In the second step, the two-electron configurations are constructed as antisymmetrized products of one-electron orbitals and the Hamiltonian is diagonalized in a basis of such configurations to produce the atomic eigenstates. Note that there is a total freedom in either including all possible configurations in the basis or only selecting specific ones to build a truncated basis. The second method

directly constructs the two-electron configurations as an antisymmetrized product of B-spline functions instead of one-electron orbital functions. The eigenstates are then provided by the diagonalization procedure. This latter approach is in principle equivalent to the first alternative when all possible configurations are included. Whatever approach is used in the CI procedure, one has to solve equation (110) using the following expansion:

$$\Psi(t) = \Psi(\boldsymbol{r}_1, \boldsymbol{r}_2, t) = \sum_{L=0}^{L_{\max}} \sum_{n=1}^{N} c_n^L(t) \phi_n^L(\boldsymbol{r}_1, \boldsymbol{r}_2) \tag{116}$$

where $\phi_n^L(\boldsymbol{r}_1, \boldsymbol{r}_2) = \langle \boldsymbol{r}_1, \boldsymbol{r}_2 | nL \rangle$ is the nth eigenstate with angular momentum L of the field-free system as computed in section 4. In the atomic basis, equation (110) now reads

$$\mathrm{i} \cdot \dot{\boldsymbol{c}}(t) = (\overline{\boldsymbol{H}}_0 + \overline{\boldsymbol{D}})(t) \cdot \boldsymbol{c}(t) \tag{117}$$

where $\boldsymbol{c}(t)$ is the vector of unknown time-dependent coefficients $c_n^L(t)$ to be determined. The advantage of performing the propagation in the atomic basis is that the probability of finding the system in a particular eigenstate $|nL\rangle$ is directly given by $|c_n^L(t)|^2$. Note here that, unlike the direct expansion in B-splines (112), there is no overlap matrix on the right-hand side of equation (117). This property (due to orthogonality of the basis set), together with the fact that $\overline{\boldsymbol{H}}_0 + \overline{\boldsymbol{D}}(t)$ is a well conditioned matrix, makes it possible to solve the differential equation (117) with explicit propagation schemes (there is no need to solve linear systems as in (115) this time). It is even recommended to use high-order schemes such as Runge–Kutta type methods for example. The disadvantage, however, is that the tridiagonal block matrix $\overline{\boldsymbol{H}}_0 + \overline{\boldsymbol{D}}(t)$ is now a matrix of full $N \times N$ individual blocks, thus raising the problem of memory storage.

There is an accelerating procedure (allowing bigger time steps) to solve (110) which is easily implemented when propagating in the atomic basis. The idea is to remove from the time-dependent wavefunction the oscillations due to the rotation of each eigenstate (see the discussion in section 5.2.5). This is simply achieved by expressing the problem in the interaction picture, that is applying the following unitary transformation: $\tilde{\psi}(t) = \mathrm{e}^{-\mathrm{i}\boldsymbol{H}_0 t} \psi(t)$. Equation (110) now becomes

$$\mathrm{i} \cdot \dot{\tilde{\boldsymbol{c}}}(t) = \mathrm{e}^{\mathrm{i}\overline{\boldsymbol{H}}_0 t} \overline{\boldsymbol{D}}(t) \mathrm{e}^{-\mathrm{i}\overline{\boldsymbol{H}}_0 t} \cdot \tilde{\boldsymbol{c}}(t) \tag{118}$$

where each matrix element on the right-hand side of (118) has the form $\exp[\mathrm{i}(E_{n,L} - E_{n',L'})]\langle nL|D(t)|n'L'\rangle$. Physical quantities are not modified by this transformation since the wavefunction is only multiplied by a phase factor.

5.2.4. Physical observable quantities. Solving TDSE (110) only provides the state of the system at any time and does not give any information on measurable physical quantities. The present section reviews the techniques commonly used to extract the relevant physical quantities.

In the nonperturbative regime, bound states can be extremely distorted by the oscillating field and induce dynamical resonances. It might then be informative to follow the population of particular excited bound states in time. The probability (see e.g. figure 7 in [267]) of finding the system in the state $|nl\rangle$ at time t is given by the projection

$$P_{nl}(t) = |\langle \Phi_{nl} | \psi(t) \rangle|^2 \tag{119}$$

where Φ_{nl} is the bare-atom eigenstate wavefunction as computed in section 3.1 or 4 and $\psi(t)$ is the solution of (110). An important point should be stressed here. In the one-electron problem, the scalar product in (119) is not computed by a numerical integration over space. Instead, it is performed in the B-spline basis and thus involves only matrix–vector operations, which is computationally fast (band structure matrices) and stable. The reader should keep in mind

that, as far as possible, manipulations on wavefunctions should be performed with the B-spline expansion coefficients. The scalar product thus reads $\langle \Phi_{nl}(\boldsymbol{r})|\psi(\boldsymbol{r},t)\rangle = \boldsymbol{c}_{nl}\cdot\boldsymbol{S}\cdot\boldsymbol{c}(t)$. In the two-electron case the probability simply reads $P_{nl}(t) = |c_n^l(t)|^2$.

Because of the dipole transition rules, the dynamics of the interaction results in a permanent, field-driven, transfer of population between angular momenta. Monitoring the population (see e.g. figures 9 and 10 in [253]) in each angular momentum requires summing the coefficients within each l as

$$P_l(t) = \sum_{i=1}^{N} |c_i^l(t)|^2. \tag{120}$$

Everything that is not in bound states at the end of the pulse has ionized. Total ionization probability is therefore computed as

$$P = 1 - \sum_{nl} |\langle \Phi_{nl}|\psi(t=\infty)\rangle|^2 \qquad \text{with} \quad E_{nl} < E_\infty \tag{121}$$

where E_∞ is the energy of the first ionization threshold. Note that, for two-electron systems, this definition of ionization includes population in doubly excited states because time integration usually stops when the pulse has gone and autoionization can occur after that.

Although useful, the total ionization probability does not give much information on the possible mechanisms involved. This is the reason why experimentalists use time-of-flight devices to measure how electrons emitted during ionization are distributed in energy. Computing electron spectra is rather simple for one-electron systems. The following formula developed for one-electron cases is also valid in the case of two-electron systems when only one channel is open for ionization. The case of multiple open channels is however not straightforward and will be discussed after having explained the single-channel case. The electron spectrum is theoretically computed at the end of the pulse with the following expression:

$$\frac{\partial P}{\partial E}(E) = \sum_{l=0}^{l_{\max}} \frac{\partial P_l}{\partial E}(E) = \sum_{l=0}^{l_{\max}} |\langle \Phi_{\mathrm{E}}^l|\psi(t=+\infty)\rangle|^2. \tag{122}$$

In practice, all partial spectra $\partial P_l/\partial E$ are first computed by projecting on each discretized continuum wavefunctions (see section 3.3). Because there are no reasons that discretized states are degenerate between angular momenta, the partial spectra are then interpolated to perform the summation for any value of E. Note again, that the projection method has been preferred to the window operator method [341], because the computation of the eigenstates as well as the projection is very fast. Such a spectrum is referred to as an 'angle-integrated spectrum' and corresponds to what a 4π detector would measure or twice as much as a magnetic bottle would collect [272]. However, most of the detectors have a finite small acceptance angle and therefore only gather electrons emitted in a particular direction. The spectrum at a particular angle (from which the angular distribution can be extracted) is computed from

$$\frac{\partial P}{\partial E \partial \Omega_k} = |\langle f_{\boldsymbol{k}}^-(\boldsymbol{r})|\psi(\boldsymbol{r},t=+\infty)\rangle|^2 \tag{123}$$

where $|f_{\boldsymbol{k}}^-\rangle$ is the Coulomb wavefunction corresponding to an outgoing electron with momentum $\boldsymbol{k}$, which reads in terms of partial waves

$$f_{\boldsymbol{k}}^-(\boldsymbol{r}) = \sum_{l=0}^{l_{\max}} \sum_{m=-l}^{l} (\mathrm{i})^l \mathrm{e}^{-\mathrm{i}\delta_l} \Phi_{E_k}^l(\boldsymbol{r}) Y_l^{m^*}(\theta_k,\phi_k) \tag{124}$$

where δ_l is the coulombic phase and where $\Phi_{E_k}^l(\boldsymbol{r})$ is a discretized continuum wavefunction as computed in section 3.

The problem of multiple open channels (continua) arises when a system is brought into the continuum lying above the second ionization threshold. There exist several final physical states at a particular (degenerate) energy. For instance, one of them corresponds to one electron in the GS of the ion and the other electron having a certain kinetic energy. Another state may correspond to one electron in the first excited state and the other with a smaller kinetic energy. Computing the electron spectrum corresponding to each emitted electron requires a projection onto the eigenstate corresponding to each particular channel. As discussed in section 4.2.1, the discretized continuum wavefunctions obtained from diagonalization of the Hamiltonian do not allow us to separate the contribution from each channel. A special technique described in section 4.2 for computing the discretized continuum state provides the correct final-state wavefunctions corresponding to particular channels. For instance, the partial electron spectrum is obtained in each channel by replacing Φ_{E}^{l} in (122) by the wavefunction given by equation (66).

During interaction with the laser field, the atom also scatters light and possibly converts frequencies. Information on the processes involved can be extracted from the light spectrum generated by the radiating dipole moment. This is achieved by computing the Fourier transform of the time-dependent dipole $d(t) = \langle\psi(t)|\boldsymbol{r}\cdot\boldsymbol{e}_z|\psi(t)\rangle$:

$$S(\omega) = \left|\frac{1}{\sqrt{2\pi}}\int \mathrm{d}(\tau)\mathrm{e}^{-\mathrm{i}\omega\tau}\mathrm{d}\tau\right|^2. \tag{125}$$

Note that as soon as ionization is significant (that is when $d(t)$ differs from zero at the end of the pulse), the spectrum is better computed from the acceleration of the dipole by means of Ehrenfest's theorem $\ddot{d}(t) = \langle\psi(t)| - \partial V(r)/\partial r + E(t)|\psi(t)\rangle$ (where $V(r)$ is the central atomic potential and $E(t)$ the laser field) [443]. The extension to two-electron systems is straightforward, the dipole becoming $(\boldsymbol{r}_1 + \boldsymbol{r}_2)\cdot\boldsymbol{e}_z$.

5.2.5. Convergence. Obviously, any quantity computed by solving the TDSE has to be checked for convergence with respect to the independent discretization parameters. In the B-spline approach for solving the TDSE, these parameters are the box size $r_{\max}$, the average breakpoint spacing (grid spacing) $\Delta r = r_{\max}/N$, the total number of angular momenta $l_{\max}$ and the integration time step δt. In practice, the computation is performed several times by modifying one parameter until the quantity of interest remains unchanged. Applying this procedure to each parameter will determine the validity of the result with respect to the discretization technique. Note that new calculations involving different field parameters (time duration, frequency, intensity etc) have to be checked for convergence as well. As a matter of fact, the processes investigated are highly nonlinear with respect to the field intensity and the parameters needed to achieve convergence are thus closely related to the field strength. Also important is the fact that all physical quantities are different in terms of computational demand. For instance, converged total ionization is easily obtained while the electron spectrum at a particular angle is very demanding. In order to minimize the number of computations related to convergence checks, we briefly give the rules for estimating parameters leading to convergence. This study has been carried out in the one-electron case for the computation of the ATI electron spectrum (the most demanding quantity) of Ar [272] (see also section 5.2.6) and verified on an intensity range beginning beyond the perturbative regime and extending up to saturation of ionization.

In section 3.3 we have explained that a stationary bound state is properly represented if it entirely fits in the box. There is a similar argument for the time-dependent problem. Whatever aspect of the interaction we are focusing on, the physics is the following: the system is initially in a bound state and experiences an external field during a finite time. Ionization mechanisms release electrons with finite kinetic energies, thus expanding the initial wavepacket. At the end

of the pulse, the fastest electrons have travelled the maximum distance from their parent ion. Since no transitions will occur after the pulse has gone (spontaneous emission is not accounted for in this method), all measurements can be made as soon as the pulse is off. At that time, the spatial expansion of the wavepacket is finite and therefore fits in a box. The box size has to be chosen so that no part of the wavepacket has a chance to hit the boundaries and bounce back (the boundary conditions defined in section 3.1 lead to total reflection) towards the nucleus and create artificial interferences. Computing a spectrum in a too small box first affects the highest-energy electrons. By evaluating the time it takes an electron with given energy to travel to the boundaries, we have found that a spectrum converged up to $E_{\rm conv}$ requires a box of

$$r_{\rm max} = 0.55T\sqrt{2E_{\rm conv}} \tag{126}$$

where T is the total duration of the pulse. The physical interpretation is that the ejected electrons start to be emitted slightly before the middle of the pulse. This is in agreement with the physical process (laser-assisted backscattering) involved in the production of these electrons that emits the more energetic electrons when the pulse has reached the peak intensity. Since ATI spectra extend up to $E = 10U_{\rm p}$ (see section 5.2.6) one chooses usually $10U_{\rm p} < E_{\rm conv} < 15U_{\rm p}$. Notice here that, to take advantage of scaling laws, the electron energy is expressed in terms of the ponderomotive energy $U_{\rm p} = I/4\omega^2$ (with the au of intensity defined as $I_0 = 3.509 \times 10^{16}$ W cm^{-2}) corresponding to the quiver energy of a free electron driven by the field. One has to remember that $r_{\rm max}$ also defines the density of discretized continuum states. Therefore, on top of the argumentation about wavepacket expansion, $r_{\rm max}$ has to fulfill the condition that the density of states is compatible with the resolution required in the electron spectrum.

As for the computation of discretized continuum states, the breakpoint average spacing $\Delta r = r_{\rm max}/N$ plays a crucial role and sets the maximum energy at which an electron can be accurately propagated. Because ATI mechanisms are highly interfering processes, the continuum wavefunctions have to have an accurate phase. The spacings required for TDSE calculations are basically the same as for the computation of stationary continuum states (see section 3.3). In [272] for $k = 7$ we have $\Delta r = \lambda_{\rm conv}/3$, that is

$$\Delta r = \frac{1}{3}\frac{2\pi}{\sqrt{2E_{\rm conv}}} \tag{127}$$

still with $10\ U_{\rm p} < E_{\rm conv} < 15\ U_{\rm p}$.

The number of angular momenta to include in LOPT calculations is defined by the initial-state angular momentum and the number of photons exchanged according to the dipole selection rule. In nonperturbative calculations, although transitions are still governed by the dipole selection rule, the actual number of photons exchanged, say for example to bring a system into the continuum, may be more important than the net number of absorbed photons necessary to ionize. This is because beyond the perturbative regime many higher-order virtual processes take place with a significant weight. They would correspond to the higher-order terms in the Born expansion neglected in LOPT. On the one hand, if it takes seven photons to ionize hydrogen, the TDSE calculation may require more than seven angular momenta. On the other hand, the ATI spectrum in figure 19 clearly shows the net absorption of more than 40 photons. Does this mean that the calculation requires more than 40 angular momenta? The answer is yes and no, and opens the door to the gauge issue in TDSE calculations [253]. There are several ways of expressing the interaction with the field by choosing a particular gauge for defining the field. Among them are the so-called 'length' (the interaction reads $D(\boldsymbol{r}, t) = -\boldsymbol{E}(t) \cdot \boldsymbol{r}$) and 'velocity' ($D(\boldsymbol{r}, t) = -\boldsymbol{A}(t) \cdot \boldsymbol{p}$) gauge (see section 5.2.1). Cormier and Lambropoulos have shown [253] that a computation similar to that of figure 19 did not converge even with 100 angular momenta (100 ls) when computed in the length gauge but was fully converged

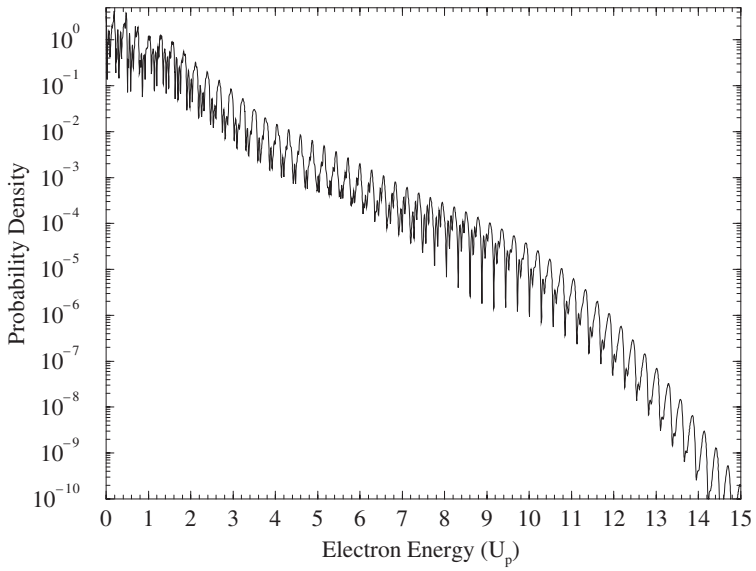


Figure 19. Angle-integrated ATI spectrum of hydrogen from [252]. $\omega = 2$ eV, $I = 2 \times 10^{14}$ W cm^{-2}, FWHM 25 fs ($U_p = 7.18$ eV).

with only ten ls in the velocity gauge. It is then clear that the computation of electron spectra at intensities above the perturbation regime has to be done in the velocity gauge. The physical reason behind the difference in the angular momentum spreading is explained in [253] and originates from the physical meaning of the canonical variable $\boldsymbol{p}$ (different in either gauge) involved in the momentum operator $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$. Note that this argument is independent of the discretization method used in solving the TDSE, being due to the expansion in terms of spherical harmonics. There are no clear rules allowing us to determine the number of angular momenta needed in a calculation. However, no calculations so far have required more than 40 ls even in cases very close to ionization saturation.

The last parameter to be discussed is the time step δt involved in the propagation (115). An unconverged spectrum will show a shift of the highest ATI peaks towards lower energy. Experience has shown that when high-energy electrons are produced, the time step leading to convergence of the spectrum has to be small enough compared with the period of the fast oscillations of the populated discretized continuum states at their own energy. Solving the TDSE for a discretized continuum state (in the absence of the field) with an inappropriate time step propagates the state at a smaller effective energy. From the evaluation of the error on the energy, the time step is chosen as

$$\Delta t = \sqrt{\frac{12\epsilon}{E_{\mathrm{conv}}^3}} \tag{128}$$

where $10U_p < E_{\mathrm{conv}} < 15U_p$ and where ϵ is the maximum error under which a spectrum is considered converged. In other terms, ϵ represents the maximum downward energy shift allowed in the spectrum. Because ATI peaks are separated by the energy of one photon, ϵ is typically set to 5% of the photon energy. The fact that the choice of time step is dictated by the maximum electron energy can be avoided by initially writing the Schrödinger equation (111) in the interaction picture. Then, the time step will be determined by the variations of the interaction term $D(\boldsymbol{r}, t)$. The implementation in the B-spline basis is not straightforward and is currently under development.

5.2.6. Applications. The B-spline method for solving the TDSE has been applied in many different contexts: ATI, HOHG etc. Atomic systems such as H, Ar, Cs, He, K, Na and Mg^+ have been investigated in the SAE approximation for a range of intensities going from the perturbative regime up to ionization saturation. A few examples will be discussed in this section. The time-dependent treatment of real two-electron systems in the nonperturbative regime is much more demanding. Therefore, applications are scarce and we will only report on a selection of successful studies in He, H^- and Mg in situations where electron correlation is expected to play a significant role.

The first case for which the method has been tested was atomic H. When the hydrogen atom is irradiated by strong laser fields, it ionizes, leading to an electron energy distribution which looks like the ATI spectrum [252] shown in figure 19. As already mentioned, the regular spacing of the peaks is due to the simultaneous absorption of many photons in the continuum, which makes the electron increase its energy in units of the photon energy. Another feature almost always present in ATI spectrum is the shape of the envelope of the spectrum. Although not very pronounced in ATI of H, a typical shape is first a decrease of the ionization probability up to say 2–5 U_p, a change of slope giving rise to the so-called plateau [444] and a cut-off and the end of the plateau around 10 U_p [445]. This highly accurate calculation, that reproduces tiny details over ten orders of magnitude and for an energy range exceeding 100 eV, can be considered as an 'exact solution' of the TDSE and used as a benchmark calculation.

Unfortunately, except for a few works, experimental ATI spectra are not recorded in H but rather in rare gases: first of all because atomic H is not easy to produce and separate from H_2 and second because rare gas spectra exhibit more interesting features. Thus, extensive calculations have been performed in Ar within the SAE approximation (using a model potential given in [342]) to simulate recent experiments [272, 446]. Although the experimental results (ATI peak location, substructures, overall shape dependence etc) could not be reproduced quantitatively, the theory provided all features observed in ATI spectra. An accurate quantitative comparison with these experiments is only possible if the theoretical spectra are integrated over the interaction volume where the laser beam intercepts the Ar gas jet. In the experiment, the detector collects electrons originating from atoms located at different positions in the gas jet and, therefore, experiencing different field parameters. If $P_E(I)$ is the probability density for an atom to emit an electron with kinetic energy E_k under a pulsed field with maximum intensity I, the yield measured when collecting electrons from the whole interaction volume reads [450]

$$N(E) = \rho \int P_E(I) \frac{\partial V}{\partial I} \, dI \tag{129}$$

where ρ is the density of atoms in the gas jet (assumed constant). $P_E(I) = \partial P(E)/\partial E$ is obtained (see equation (122)) by solving the TDSE with a peak intensity I. $(\partial V/\partial I)\, dI$ is the partial volume experiencing an intensity between I and $I+dI$. As shown in figures 1–3 in [272], quantitative comparison with experiment is now surprisingly good. Even more impressive is the comparison with spectra recorded at all angles with respect to the laser polarization as shown in figure 20 [446, 447]. Note that these details give information on the parity of intermediate states that have dynamically shifted into resonance as the pulse intensity was rising. Such a simulation is a robust proof that the SAE approximation is perfectly valid in the context of ATI of rare gases up to saturation although it was not intuitive *a priori*. Further investigations [272] of the highly intensity sensitive substructures appearing in the plateau region (see figure 21) have shown that their origin is unrelated to the Freeman resonances which usually show up at the beginning of the spectrum. The underlying mechanisms responsible for the presence of these high-lying 'resonances' are still under debate [342, 448, 449].

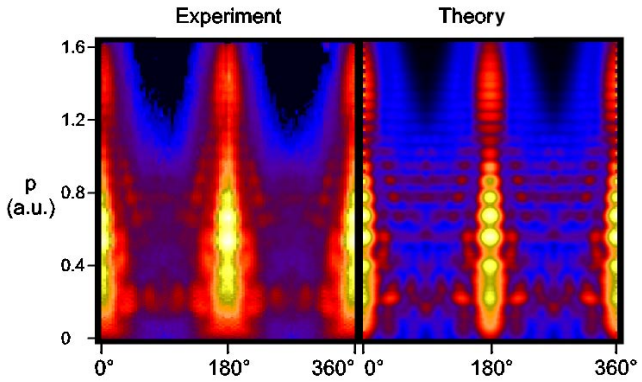


Figure 20. Comparison between theory and experiment of angle-resolved ATI spectra. The plot shows the probability as a function of the collection angle and the electron momentum. The calculation includes averaging over the interaction volume. $\lambda = 800$ nm, $I = 7 \times 10^{13}$ W cm^{-2}, FWHM 40 fs (from [446]).

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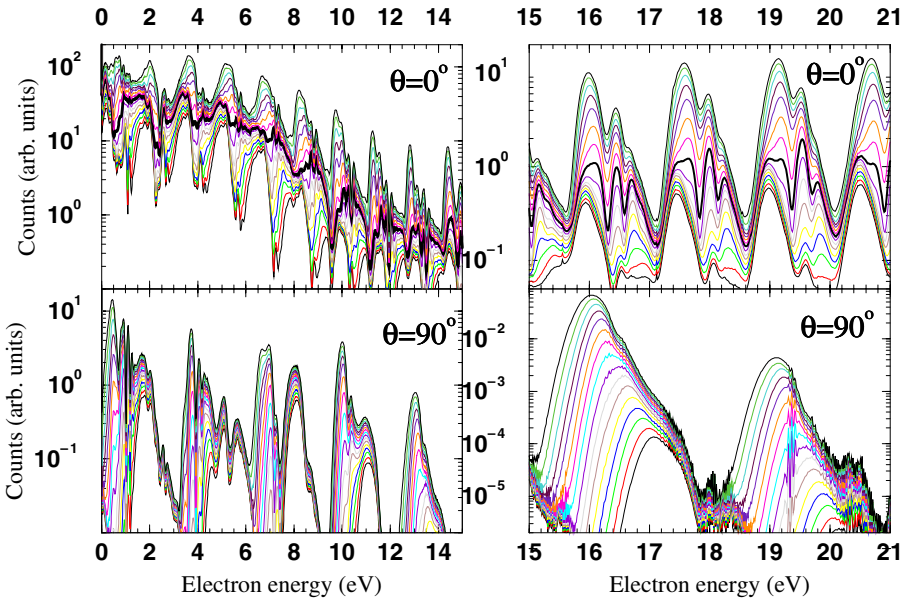


Figure 21. Theoretical low- (left) and high- (right) energy electron spectra of Ar computed at $\Theta = 0^\circ$ and 90° from the polarization axis for 16 intensities varying from 5×10^{13} W cm^{-2} up to 7×10^{13} W cm^{-2}. Black thick solid curve, $I = 6 \times 10^{13}$ W cm^{-2} (from [272]).

(This figure is in colour only in the electronic version)

It is also interesting to study ATI in a completely different context. The above-mentioned work was dealing with a laser wavelength around $\lambda = 800$ nm (Ti:sapphire technology). Because of the dynamic shift of the ionization threshold (the ionization potential is increased by U_p), ionization of Ar requires the absorption of 11–13 photons. In order to investigate the scalability of the mechanisms involved in ATI, theoretical studies (see [267] and experimental

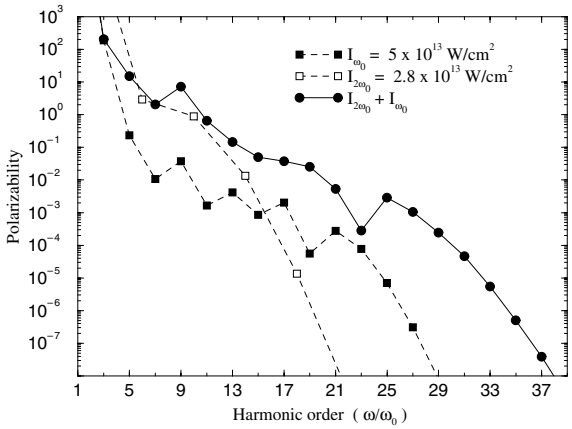


Figure 22. Comparison of the spectrum of odd harmonics produced by the bichromatic field and those obtained using each field component individually ($\hbar\omega_0 = 1.33$ eV, FWHM = 50 fs, $\phi = 0$) (from [265]).

references therein) have been carried out in the 13-photon ionization regime for mid-infrared laser pulses ($\lambda \approx 3400$ nm) in atomic potassium. In this case, the behaviour of the ATI spectra has been found to be very different from that observed in ATI of Ar. The overall shape of the spectrum starts with a quasi-horizontal plateau up to approximately 2.8 U_p and decays continuously afterwards. The absence of the high-energy plateau commonly observed in the optical regime is explained by the very low efficiency of the backscattering process at mid-infrared wavelengths. ATI of Cs atoms by infrared photons ($\lambda = 800$ nm) with moderate field intensity and a few cycles duration is also reported in [255]. This paper discusses the effect of the initial phase of the short pulse on the electron spectrum.

This approach has also been applied in the context of HOHG (see [440] for a review on the subject) to estimate the maximum gain in harmonic generation efficiency [265] when the medium is illuminated by a superposition of a laser beam at the fundamental frequency and another with doubled frequency (see figure 22). The study has involved the computation of harmonic spectra (using equation (125)) for all possible field intensities to determine the best set of parameters. It then has been compared to the most efficient single-field harmonic spectrum. Because the two fields have commensurate frequencies, the relative phase is of importance in the amplitude of the scattered light at harmonic frequencies as demonstrated in an experimental as well as theoretical study [266]. Additional work related to the computation of high-order susceptibility with bichromatic fields is reported in [254]. A study of the evaluation of the phase of high-order harmonics can be found in [256].

Finally, we discuss calculations of electron spectra for very intense ($I \approx 3.5 \times 10^{18}$ W cm^{-2}) and extremely short (FWHM ≈ 0.1 fs) pulses. Under these conditions, electrons are ejected in the opposite direction to the field with very high velocities. The *B*-spline approach for solving the TDSE has been able to reproduce the dynamics of electrons up to energies in excess of 3 keV in excellent agreement with a Coulomb–Volkov type calculation as shown in the electron spectra in [268, 269].

The first attempt to solve the TDSE for a two-electron atom in strong laser [251] fields using *B*-splines has made use of a basis of configurations built from one-electron orbitals (see section 5.2.3) and has been applied to compute MPI of He. The method has been improved over the years until it reached the point where converged ATI spectra and angular distributions

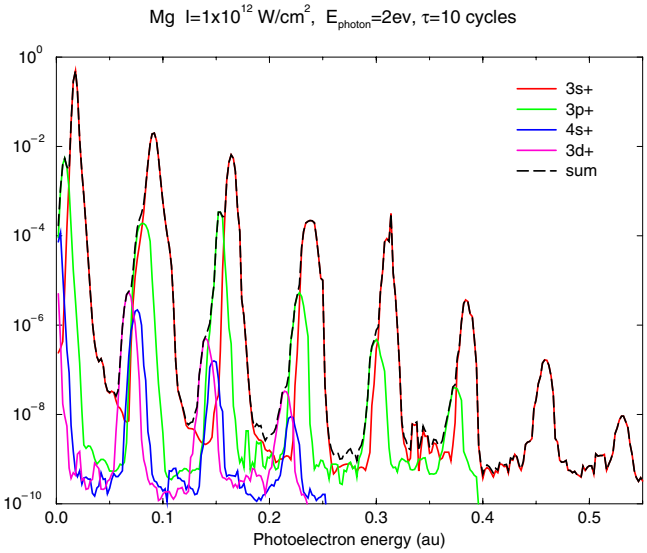


Figure 23. Photoelectron energy spectra correponding to Mg^+ left in the ground as well as a few excited states (from [264]).

(This figure is in colour only in the electronic version)

could be calculated [262] using relations (122) or (123). The case where ATI occurs above the second ionization threshold, that is, when multiple ionization channels are open, has been investigated in Mg and has provided electron spectra corresponding to ions left in excited states as illustrated in figure 23. The efficiency of the alternative approach, constructing two-electron configurations as direct products of B-splines, is emphasized in the case of photodetachment of H^- by strong and ultrashort laser pulses [271]. Figure 24 presents *ab initio* theoretical data for single ionization of H^-. H^- is a difficult but interesting system since its structure is dominated by electron correlation. Single ionization of H^- has been investigated through various numerical approaches over a wide intensity range. Therefore it is an ideal system to test the accuracy of the approach in [271]. The figure clearly shows that the results from [271] agree with other calculations (of very different types: R-matrix Floquet, complex Sturmian functions, correlated basis functions) but differ significantly from the other B-spline calculation when the basis is built with a product of orbitals [259]. As discussed in [271] the difference may arise from the description of the initial GS wavefunction, which is sensitive to the basis truncation. The alternative method (direct product of B-splines) has been also applied to treat the ionization of helium by XUV ($\omega = 5.4$–32.6 au) in short and intense laser pulses [270]. The authors focused on two-photon ionization in the region of autoionizing states (below the $N = 2$ threshold), with a pulse duration smaller than the autoionization lifetime.

In conclusion the two approaches are complementary since one is well adapted to the computation of ATI spectra with a good description of the different channels; nevertheless, due to the basis truncation, the correlations are partially taken into account in the calculation of the electronic structure. The other approach (direct products of B-splines) provides a better description of the electronic structure but it involves a limited number of B-splines (thus a low density of states in the continuum); as a consequence it applies for very short pulses and ATI is described only for a limited number of absorbed photons.

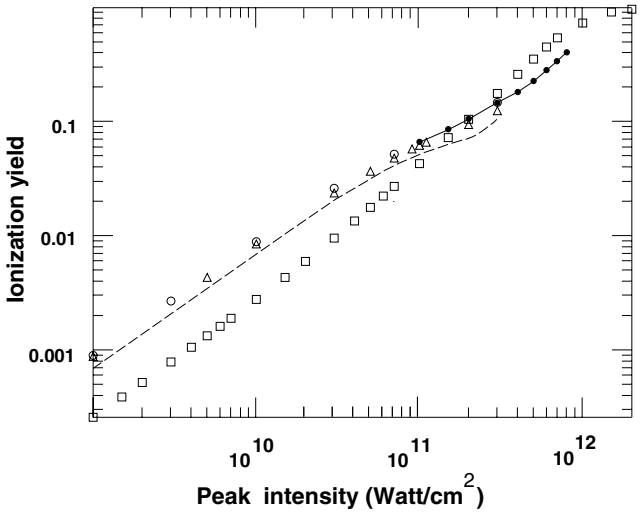


Figure 24. Ionization probability of H^- as a function of the laser peak intensity. $\omega = 0.03$ au, and the pulse has a cosine envelope whose FWHM is equal to four optical cycles. The open triangles are the results obtained with the B-spline functions. The open circles are those obtained with complex Sturmian functions. The open squares are from a basis calculation by Nikolopoulos *et al* [259]; the full circles connected by a full curve are from Scrinzi *et al* (see [271]). The dashed curve is obtained by integrating the R-matrix Floquet rates from Purvis *et al* [271] over the pulse intensity profile. (From [271].)

5.3. Ions and atoms in static fields

B-spline techniques have also been very helpful to study ions and atoms in the presence of static electric and magnetic fields. As an illustration, we consider the hydrogen atom subject to electric and magnetic fields both parallel to the z direction. The nonrelativistic Hamiltonian $\mathcal{H}$ of such a system reads, in au

$$\mathcal{H} = -\frac{1}{2}\nabla^2 - \frac{1}{r} + \beta L_z + \frac{1}{2}\beta^2\rho^2 + \mathcal{F}z \tag{130}$$

where $\beta = B/B_c$ and $\mathcal{F} = F/F_c$, F_c and B_c being the au of electric field and magnetic flux density, respectively. $B_c \approx 4.7 \times 10^5$ T and $F_c \approx 5.14 \times 10^9$ V cm^{-1}. L_z denotes the z-component of the orbital momentum $\boldsymbol{L}$ and ρ the projection of the position vector $\boldsymbol{r}$ onto the plane $z = 0$. The energy spectrum of such a system formally has no bound states. However, when the fields are very weak, ionization can be neglected and the states can be treated as ordinary bound states. Thus, perturbative treatments which make use of the solutions of the field-free problem can be applied to determine the Zeeman and/or Stark effects. As the electric field strength increases, ionization cannot be ignored and the problem becomes much more involved. The bound-state atomic levels shift and broaden, and the issue is now to calculate the positions and widths of resonant states. Since m is the only good quantum number, the problem is two dimensional and, in general, nonseparable.

B-spline basis sets were first applied to calculate the energies of the low-lying states of the hydrogen atom in the presence of magnetic fields of arbitrary strength [191]. Xi *et al* [191] expanded separately the radial and angular wavefunctions R_{nl} and Y_l^m in terms of B-splines. The field-dressed wavefunction may also be expanded as done in [188, 190]

$$\Phi(\boldsymbol{r}) = \sum_{\alpha \equiv i,l} c_\alpha \frac{B_i(r)}{r} Y_l^m(\theta, \phi). \tag{131}$$

In contrast with the case of the isolated one-electron atom, there is now a sum over the angular momentum quantum number l. Apart from this difference, all methods developed to study the structure of isolated atoms can be extended to the present case. For example, the stabilization and the CR methods, both used in (field-free) atomic structure problems, have been shown to be efficient to study the properties of atoms in strong static fields (see the examples given below). Also, the L^2 formalism of Cortés and Martín (see section 4.2.1) has been succesfully applied to study the electric-field effects on the photodetachment of H^- below the $N = 3$ threshold [178]. However, due to the loss of spherical symmetry, the implementation of these methods is computationally demanding.

B-splines have been extensively used in this context. The main contributions are quoted in the text and at the end of this section. We present below three recent illustrations for the case of a static electric field alone ($\beta = 0$ in equation (130)).

The first illustration is the case of the hydrogen atom in an external uniform static electric field (see [183] and references therein). This problem has been extensively studied because the exact solutions are known. Therefore, it provides a useful testing ground for the B-spline approach. Using parabolic coordinates $\zeta = r + z$ and $\nu = r - z$, equation (130) (with $\beta = 0$) leads to two homogeneous differential one-dimensional equations which are coupled via two separation constants [431]. Exact solutions can be determined through a simple diagonalization procedure in a basis of B-spline functions. Indeed, the electron wavefunction can be written $F(\zeta)G(\nu)e^{im\phi}$, where m is the magnetic quantum number. $F(\zeta)$ takes discrete values, while $G(\nu)$ belongs to the continuum. In [183], $F(\zeta)$ and $G(\nu)$ have been expanded on a B-spline basis. Normalization of the resulting discretized wavefunction $G(\nu)$ on the energy scale was possible through the corresponding density of states (see expression (21)). Thus, the problem reduces to the resolution of an eigensystem with a relatively small number of basis functions. The great advantage of B-splines is that the dimension of the box (r_{max}) is explicitly defined, so that it is easy to choose a value such that the classical ionization threshold is well described. In [183], it has been found that $r_{max} \gg \sqrt{1/\mathcal{F}}$ is a necessary condition to ensure convergence of the calculations. This method was used to study photoionization of hydrogen from the ground and excited states in a weak external dc electric field. Oscillator strengths for transitions in the region of the field-free atomic threshold (where the continuum is strongly perturbed by the field) were also calculated. The results compare very well with other accurate calculations [183].

The method presented above suffers from two important limitations: (i) the normalization procedure of $G(\nu)$, based on the determination of the density of states, does not work in the region of resonances and (ii) the Schrödinger equation is separable only in the case of the hydrogen atom or hydrogen-like ions. As mentioned above, the hydrogen atom in a uniform electric field has no bound states and only a continuous spectrum with resonances. Quasi-bound states exist in low fields, but, as the electric field strength increases, the bound states located in the region of the critical energy $E_c = -2\sqrt{\mathcal{F}}$ shift and broaden, thus becoming resonances (usually called Stark resonances) [432, 433]. Although the resonance parameters can be extracted from the behaviour of the continuum at large distances [434], it is much more convenient to calculate these parameters using the complex scaling method [435–437]. One example, involving B-splines, is the work of Rao *et al* [188], who solved the Schrödinger equation associated with the complex-scaled ($r \to re^{i\theta}$) Stark Hamiltonian of hydrogen using a B-spline expansion (131). The non-Hermitian complex Hamiltonian has the form

$$\mathcal{H}(\theta) = -\frac{1}{2}\nabla^2 e^{-2i\theta} - \frac{1}{r}e^{-i\theta} + \mathcal{F}z\,e^{i\theta}. \tag{132}$$

A direct diagonalization procedure leads to complex eigenvalues $E_r = E_{res} - (i/2)\Gamma$, where E_{res} and Γ are the resonance position and width respectively. The advantage of this approach

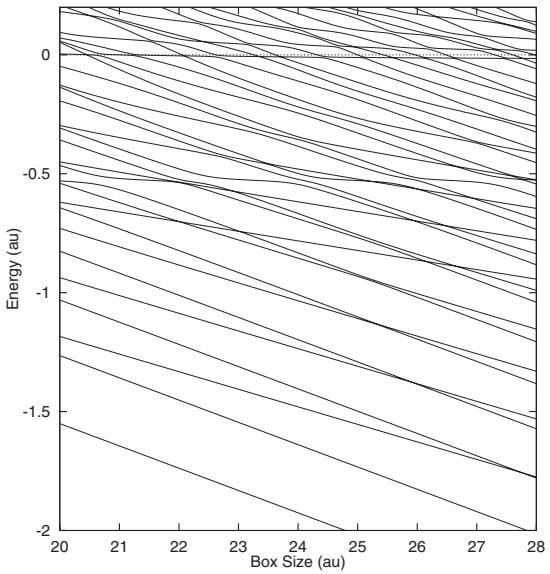


Figure 25. Stabilization diagram of H in a static field $F = 0.1$ MV cm^{-1} [178].

is that explicit consideration of boundary conditions is avoided. It has been successfully employed also to study the hydrogen atom in strong parallel magnetic and electric fields [190]. There are many other recent contributions based on the complex scaling approach. We note in particular the recent work of Ho [438] (see also references therein) where products of Slater orbitals are successfully used to solve the problem of the magnetic-field-induced Stark effect for doubly excited states of H^-.

B-splines are also well adapted to the stabilization method proposed by Mandelshtam *et al* [363] and described earlier. This method consists in repeated diagonalizations of the Hamiltonian for various values of the box size r_{max}. Then, a stabilization diagram is obtained by plotting $E(r_{max})_{i=1,N}$ as functions of r_{max} (N being the number of eigenvalues). According to Mandelshtam *et al* [363], the density of states is evaluated from the corresponding histogram in the energy region of interest. Peaks in the histogram are associated with resonances, whose energy positions and widths can be extracted through a fitting procedure. In [178], Bachau and Martín have applied this method to calculate tunnelling rates induced by electric fields in H and H^-, in the context of a more general study of electric-field effects on the photodetachment of H^- below the $N = 3$ threshold. Figure 25 shows the calculated stabilization diagram of H for a static field of $F = 0.1$ MV cm^{-1} (the B-spline basis contains 1256 terms with l =0–6). Figure 26 shows the calculated histogram in the region of the 1s state and the corresponding fit to a Lorentzian profile. The results for the position and width obtained from the fit are $E_{1s} = -0.5274$ au and $\Gamma_{1s} = 0.014\,54$ au, which are in excellent agreement with those reported by Rao *et al* [188]. As in the case of the CR method, the great advantage of the stabilization method is that it does not require an explicit inclusion of the boundary conditions to obtain the widths. In all cases, the explicit definition of a box size r_{max} ensures a good control of the convergence of the calculations.

Besides the works cited above, there are many other contributions which have made use of B-splines in this context. (In the case of two-electron systems (H^- and He), see [186, 187] for magnetic fields and [184] for electric fields.) The spectrum of the hydrogen atom has also been studied in the presence of magnetic fields [185, 189].

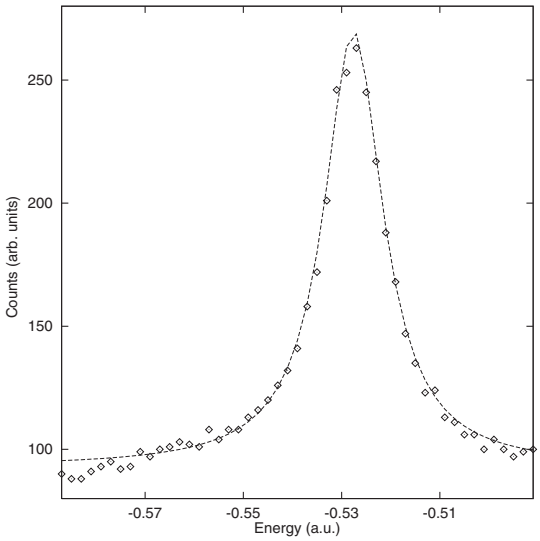


Figure 26. Histogram corresponding to the stabilization diagram shown in figure 25 [178].

6. Diatomic molecules

Applications of B-spline basis sets in molecular systems are very recent. In 1992, B-spline basis sets were used for the first time to describe the electronic states of H_2^+ and HeH^{2+} molecular systems [274–278]. The proof that B-splines could reproduce as accurately as desired the exact result offered a promising perspective and opened the way for subsequent applications in molecules containing several electrons. The subject is now growing very rapidly. In the last three years we have seen a breakthrough in the description of continuum states of two-electron diatomic molecules, which has been possible through the use of B-splines. New theoretical tools have been developed to describe both the electronic and vibrational continuum (including electron correlation), and the interferences between them [284–294]. The theoretical grounds and the most successful applications will be described in this section. Additional details on the use of B-spline basis sets for diatomic molecules can also be found in a recent review article [326]. The case of polyatomic molecules will be considered in section 7.

6.1. General considerations: the Born–Oppenheimer approximation

We illustrate now the general procedure to solve the Schrödinger equation of a diatomic molecule using B-splines. We shall neglect mass polarization terms and relativistic effects. The origin of electronic coordinates will be placed at the centre of mass of the two nuclei. Thus, the Hamiltonian of a diatomic molecule in the body-fixed frame can be decomposed according to

$$\mathcal{H}(\boldsymbol{r}, \boldsymbol{R}) = -\frac{1}{2\mu}\nabla_R^2 + \mathcal{H}_{\rm el}(\boldsymbol{r}, R), \tag{133}$$

where $-\nabla_R^2/2\mu$ is the relative kinetic energy of the nuclei, μ is the reduced mass and $\mathcal{H}_{\rm el}$ is the usual electronic Hamiltonian, which depends parametrically on R and contains all the potential energy terms (including the nucleus–nucleus repulsion). The vector $\boldsymbol{r}$ labels all electronic coordinates, and R is the internuclear distance.

We shall assume the validity of the Born–Oppenheimer (BO) approximation. For a bound state, this implies that the total wavefunction can be written

$$\Psi_{nv}(\boldsymbol{r}, R) = \frac{\chi_v(R)}{R}\psi_n(\boldsymbol{r}, R), \tag{134}$$

where the electronic wavefunction satisfies

$$[\mathcal{H}_{\rm el} - E_n(R)]\psi_n(\boldsymbol{r}, R) = 0 \tag{135}$$

and the corresponding nuclear wavefunction

$$[T(R) + E_n(R) - W_{nv}]\chi_v(R) = 0, \tag{136}$$

where W_{nv} is the total energy of the molecule, $E_n(R)$ is the BO potential energy curve of the nth electronic state of the molecule and

$$T(R) = -\frac{1}{2\mu}\frac{{\rm d}^2}{{\rm d}R^2} + \frac{J(J+1)}{2\mu R^2} \tag{137}$$

with J the total orbital angular momentum (electronic plus nuclear). In equation (134), we have factored out the rotational wavefunction Y_J^M. In the case of one-electron diatomic molecules, the usual BO separation given by equation (134) is approximately valid except for very slow electrons in the continuum. For molecules containing more than one electron, equation (134) has to be modified to account for resonance effects in the continuum. This particular aspect will be discussed in detail in section 6.4.3.

6.2. The electronic Schrödinger equation

In this section we shall focus on the evaluation of bound states in one- and many-electron diatomic molecules. Evaluation of continuum states requires specific methods that will be described in section 6.4.

6.2.1. One-electron systems. For a one-electron diatomic molecule, the electronic Schrödinger equation reads

$$\mathcal{H}_{\rm el}\psi_n \equiv \left\{-\frac{1}{2}\nabla^2 - \frac{Z_A}{|\boldsymbol{r} - \boldsymbol{R}_A|} - \frac{Z_B}{|\boldsymbol{r} - \boldsymbol{R}_B|} + \frac{Z_A Z_B}{R}\right\}\psi_n = E_n(R)\psi_n \tag{138}$$

where $\boldsymbol{R}_A$ and $\boldsymbol{R}_B$ are the position vectors of nucleus A and B respectively, and $R = |\boldsymbol{R}_A - \boldsymbol{R}_B|$. Equation (138) can be solved using a one-centre expansion (OCE), that is expanding the electronic wavefunction ψ_n in a basis of radial B-splines multiplied by spherical harmonics with a well defined value of the quantum number m:

$$\psi_n = \sum_{l=0}^{l_{\rm max}}\sum_{i=0}^{N_l} c_{il}^n \frac{B_i(r)}{r} Y_l^m(\theta, \phi). \tag{139}$$

The B-spline basis is defined in the interval $[0, r_{\rm max}]$. Since different values of m are not coupled, we have dropped the index m from ψ_n. Equation (139) can also be written

$$\psi_n = \sum_{l=0}^{l_{\rm max}} \frac{\mathcal{U}_{nl}(r)}{r} Y_l^m(\hat{r}) \tag{140}$$

with

$$\mathcal{U}_{nl}(r) = \sum_{i=0}^{N_l} c_{il}^n B_i(r). \tag{141}$$

Substitution into equation (138) and projection onto the $B_i Y_l^m$ basis yields the matrix equation

$$\boldsymbol{H}\,\boldsymbol{c} = E\,\boldsymbol{S}\,\boldsymbol{c} \tag{142}$$

where

$$H_{il,i'l'} = \int_0^{r_{\max}} \int_0^{\pi} \int_0^{2\pi} B_i(r) Y_l^m(\theta,\phi) \mathcal{H}_{\rm el} B_{i'}(r) Y_{l'}^m(\theta,\phi)\,\mathrm{d}r \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi \tag{143}$$

$$S_{il,i'l'} = \delta_{ll'} \int_0^{r_{\max}} B_i(r) B_{i'}(r)\,\mathrm{d}r. \tag{144}$$

As in the atomic case, to ensure the continuity of ψ_n in $r = 0$ and $r_{\max}$, one has to exclude the first and the last B-splines from expansion (139) for each l. A standard matrix diagonalization procedure provides the eigenfunctions and eigenvalues from equation (142). The main difference from the atomic case is that terms with different l-values are coupled due to the lack of spherical symmetry.

Coupling terms in equation (143) involve the usual integrals over three spherical harmonics and radial integrals of the form

$$\int_0^{r_{\max}} B_i(r)\, V^{(\mathrm{s})}(r)\, B_j(r)\,\mathrm{d}r \tag{145}$$

where

$$V^{(\mathrm{s})}(r) = \frac{r_<^s}{r_>^{s+1}}, \qquad r_< = \min(r, R_i), \quad r_> = \max(r, R_i) \tag{146}$$

and R_i is the distance from nucleus i to the origin. $V^{(\mathrm{s})}$ has a discontinuous derivative at R_i (nuclear cusp) and becomes highly peaked as s and R_i increase. This is the origin of the slow convergence of the one-centre expansion for off-centre nuclei which are heavy or distant from the origin. Indeed, the exact wavefunction has a discontinuous first derivative at R_i which causes many high angular momenta to give a small contribution in a narrow region centred around R_i. The discontinuity in the first derivative cannot be exactly reproduced by a finite expansion on l-values, but the latter gives at least a discontinuity at R_i in the third derivative of the radial function $\mathcal{U}_{nl}(r)$. This gives rise to a few technical points.

(i) It is beneficial to employ multiple knots at R_i (radial position of each nucleus). Equivalently, closely spaced knots around R_i can be employed. This allows better convergence with increasing order k and allows the use of a larger overall step size.

(ii) Since the contribution of high l-values is very localized around R_i, it is possible to truncate the radial basis for the corresponding $\mathcal{U}_{nl}(r)$, keeping only a limited number of B-splines around R_i. This has a negligible effect on the results, but affords a large computational saving, and allows the use of much higher-l expansions.

(iii) It is important to take care of long-range multipolar potentials when necessary. Both components $V^{(2)} = Ar^{-2}$ and $V^{(3)} = Br^{-3}$ decrease slowly and require larger $r_{\max}$ values to converge. Obviously it is important to choose the origin close to the centre of charge to minimize multipoles.

For the diatomic problem a special coordinate system is available, which takes care of the Coulomb singularities at the two nuclei. These are the elliptical coordinates, defined as (ξ, η, ϕ), where ϕ is the polar angle around the internuclear axis,

$$\xi = \frac{r_A + r_B}{R} \qquad 1 \leqslant \xi < \infty$$

and

$$\eta = \frac{r_A - r_B}{R} \qquad -1 \leqslant \eta \leqslant 1,$$

where $r_{A,B} = |\boldsymbol{r} - \boldsymbol{R}_{A,B}|$ and R is, as usual, the internuclear distance. In these coordinates, the electronic Hamiltonian is separable and the wavefunction may be written

$$\Psi_{n\Lambda}(\xi, \eta, \phi) = F_{n\Lambda}(\xi, \eta)\Phi_{\Lambda}(\phi) \tag{147}$$

with

$$\Phi_{\Lambda}(\phi) = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{\mathrm{i}\Lambda\phi}. \tag{148}$$

$F_{n\Lambda}$ has a square root singularity at the origin of the type

$$(\xi^2 - 1)^{\frac{\Lambda}{2}} (1 - \eta^2)^{\frac{\Lambda}{2}}. \tag{149}$$

There are two obvious possibilities. One is to employ partial waves in η, writing

$$\Psi_{n\Lambda} = \sum_{l=1}^{\infty} f_{nl\Lambda}(\xi) P_l^{\Lambda}(\eta) \mathrm{e}^{\mathrm{i}\Lambda\phi} = \sum_{l=1}^{\infty} f_{nl\Lambda}(\xi) Y_l^{\Lambda}(\theta, \phi), \tag{150}$$

where $\eta = \cos\theta$. This accounts directly for the factor

$$(1 - \eta^2)^{\frac{\Lambda}{2}} = (\sin\theta)^{\Lambda}$$

which is never singular in θ. Then, as proposed in early calculations [479], we can expand $f_{nl\Lambda}$ in a basis of functions which depend on the ξ coordinate. By choosing B-splines, we can write

$$f_{nl\Lambda} = (\xi^2 - 1)^{\frac{\Lambda}{2}} \sum_i B_i(\xi) C_{inl\Lambda}. \tag{151}$$

This kind of expansion has been shown to be very accurate and quickly convergent in the B-spline basis (see below). Alternatively one can employ a tensor product $B_i(\xi)B_j(\eta)$, writing

$$\Psi_{n\Lambda} = (\xi^2 - 1)^{\frac{\Lambda}{2}} (1 - \eta^2)^{\frac{\Lambda}{2}} f_{n\Lambda}(\xi, \eta)\Phi_{\Lambda}(\phi), \tag{152}$$

with

$$f_{n\Lambda}(\xi, \eta) = \sum_{ij} B_i(\xi) B_j(\eta) C_{ijn\Lambda}. \tag{153}$$

Similar tensor expansions have been used in conjunction with finite-difference techniques in a series of very accurate calculations on diatomics [476–478]. The use of B-splines in this context was introduced in [279], where it was shown that results to machine accuracy could be easily obtained. In contrast to finite-difference methods, the latter procedure has all the advantages of basis set expansions. Besides eigenvalues, bound-state properties have been computed, but no application to the continuum has been reported so far, although finite differences have been employed at the HF level [481] and recent results on H_2 and N_2 have been obtained in the framework of the random-phase approximation (RPA) [480].

The classical approach of quantum chemistry is to deal with the Coulomb singularities by employing basis functions centred around the various nuclei. For low-lying bound states the approach with conventional Slater or Gaussian basis sets is very successful and very fast convergence is obtained [482]. Problems may arise when pushing the basis to describe high-lying orbitals, since quasi-linear dependence of the basis sets in. The same approach can be followed with B-splines. In this case one combines the flexibility of the B-splines to describe high-lying or continuum states with the advantage of basis set approaches, which allow building a basis from sets on different centres, a unique possibility not shared by finite-difference approaches. Moreover spatial localization of the B-splines makes it easy to control the radial extent of the expansion around each centre. Essentially two problems have to be solved. The first is the computation of matrix elements between functions on different centres. The second

is to avoid basis set overcompleteness (numerical linear dependence), which irreparably spoils numerical algorithms. The latter puts also high demands on the accuracy requirements in the evaluation of matrix elements to keep numerical stability.

An initial attempt on one-electron diatomics employed three sets of splines, one on each nucleus and one on the centre of charge [276, 278]. Integration was performed in elliptical coordinates for the two-centre integrals, plus a separate spherical integration around each nucleus for the three-centre nuclear attraction term

$$\int \frac{\phi_i(|\boldsymbol{r}-\boldsymbol{R}_A|)\phi_j(|\boldsymbol{r}-\boldsymbol{R}_C|)}{|\boldsymbol{r}-\boldsymbol{R}_B|}\,\mathrm{d}\boldsymbol{r}$$

after separating off the nuclear cusp, giving about ten-digit accuracy. Long-range splines dying smoothly (with simple knots at the outer boundary) were employed, together with a conventional one-centre expansion. Although very fast convergence with angular momenta is obtained, numerical instabilities appear soon, which are very difficult to control. This slowed down development of this approach, and it is not quite clear at the moment how to choose an optimal set. Existing implementations and possible alternatives will be dicussed in the section devoted to polyatomic molecules.

To summarize, two general approaches, OCE and LCAO, and one special method employing elliptical coordinates, have been implemented with B-splines for the electronic problem in diatomics. OCE is simple and robust, the only disadvantage being slow convergence, which is mitigated by the possibility of employing very high-l expansions. The other two are very accurate. Elliptic coordinates require a special purpose program; LCAO still has open numerical problems.

6.2.2. Many-electron systems. For molecules containing more than one electron the Hamiltonian reads

$$\mathcal{H}_{\mathrm{el}} = -\frac{1}{2}\sum_i \nabla_i^2 - \sum_i \frac{Z_A}{|\boldsymbol{r}_i-\boldsymbol{R}_A|} - \sum_i \frac{Z_B}{|\boldsymbol{r}_i-\boldsymbol{R}_B|} + \sum_{i>j}\frac{1}{|\boldsymbol{r}_j-\boldsymbol{r}_i|} + \frac{Z_A Z_B}{R}. \tag{154}$$

As in the atomic case, the usual CI procedure to solve the Schrödinger equation associated with the above Hamiltonian consists in expanding the electronic wavefunction in a basis of configurations built from one-electron orbitals [284]. The latter are molecular orbitals resulting from equation (138) and are described in terms of B-spline basis sets as described in section 6.2.1. Substitution of the wavefunction expansion in the Schrödinger equation leads to a matrix equation similar to equation (142). All matrix elements involve B-splines and can be reduced to products of one-electron integrals like those given in equations (143) and (144), except for contributions arising from the electron–electron repulsion term $1/|\boldsymbol{r}_i-\boldsymbol{r}_j|$. In the OCE approach these are the same as in the atomic case (see appendix A.3). So far this is the route followed in applications to two-electron systems. Integrals in elliptical coordinates, or in the LCAO basis, have not yet been considered, but should be amenable to accurate evaluation with moderate computational effort. Actually all highly developed techniques employed in quantum chemistry for the bound states can be applied in the B-spline basis when the one- and two-electron integrals are available.

An alternative approach is the density functional (DFT) method. For LDA and TDLDA calculations one makes use of the formalisms and general codes (OCE and LCAO) previously developed for polyatomic molecules. For this reason presentation of these methods will be postponed to section 7. Of course, full use of the $C_{\infty\mathrm{v}}$ and $D_{\infty\mathrm{h}}$ symmetries is made and this allows the use of quite large-l expansions, currently up to $l = 100$ (care must be employed in the evaluation of spherical harmonics and angular integrals at such l-values). Moreover,

Table 11. Convergence of the vibrational energies (in au) for a system of reduced mass $\mu = 918.075$ au in a Morse potential $V(R) = D\{1 - \exp[-a(R - R_e)]\}^2$ with $D = 0.1026$ au, $R_e = 1.9972$ au and $a = 0.732$ au. The basis set is formed by N B-splines of order $k = 8$ defined in the interval [0, 12] au. A linear knot sequence was used.

N	E_1	E_2	E_3
30	0.005 477 047	0.016 364 310	0.029 733 428
40	0.005 403 895	0.015 853 941	0.025 622 153
50	0.005 398 917	0.015 759 254	0.025 548 608
60	0.005 398 853	0.015 758 850	0.025 535 281
70	0.005 398 851	0.015 758 825	0.025 535 164
80	0.005 398 851	0.015 758 824	0.025 535 159
Exact	0.005 398 851	0.015 758 824	0.025 535 159

the use of the LCAO expansion also allows an accurate treatment of the core orbitals in heavy systems, so numerically accurate solutions are generally available without limitations.

6.3. The vibrational Schrödinger equation

In the BO approximation, the vibrational wavefunctions are the solutions of the Schrödinger equation (see equations (136) and (137)):

$$\left\{-\frac{1}{2\mu}\frac{\mathrm{d}^2}{\mathrm{d}R^2} + \frac{J(J+1)}{2\mu R^2} + E_n(R)\right\}\chi_v = W_{nv}\chi_v. \tag{155}$$

This equation can be solved by expanding χ_v in a basis of B-splines defined in the interval $[0, R_{\max}]$:

$$\chi_v(R) = \sum_j d_{vj} B_j(R). \tag{156}$$

Substitution of this expansion into (155) leads to a matrix equation similar to (142) that can be solved with standard diagonalization techniques. As in numerical integration, the accuracy of the results is limited by the number of points used to define the potential $E_n(R)$. As an illustration, in table 11 we show the vibrational energies of a Morse potential $E_n(R) = D\{1 - \exp[-a(R - R_e)]\}^2$ with $D = 0.1026$ au, $R_e = 1.9972$ au, $a = 0.732$ au and $\mu = 918.075$ au, which have been chosen to approximately reproduce the potential energy curve of the GS of H_2^+. As can be seen, a relatively moderate number of B-splines leads to results that are practically identical to the exact ones. Similar convergence studies for Morse potentials were performed by Shore [8] using cubic B-splines.

The lowest eigenvalues resulting from diagonalization of equation (155) represent bound vibrational states of the molecule. Those appearing above the dissociation limit correspond to vibrational continuum states that must be renormalized using the appropriate density of states [282]. The discretization method leading to the vibrational continuum is similar to that described in section 3.3 for the single-channel electronic continuum. As for electrons, the best choice for the knot sequence is the linear one. However, notice that, due to the different mass of the nuclei in comparison with that of the electrons, the frequency of the oscillations in the wavefunction is much larger for the nuclei than for electrons with the same kinetic energy. This means that the asymptotic region is reached much earlier for the nuclear continuum, which explains why, in general, one needs a smaller box than that used to describe the electronic continuum.

6.4. Continuum states

Let us turn now to the problem of computation of electronic continuum states. Most theoretical calculations of the molecular electronic continuum have been performed in the *fixed-nucleus approximation* (see, for instance, [284, 451, 452]). The fixed-nucleus approximation makes use of the equilibrium geometry of the molecule to evaluate the electronic wavefunction and neglects completely the nuclear degrees of freedom. This is a reasonable approximation when one is not interested in the vibrational analysis of a molecular property (e.g. to evaluate total photoabsorption rates or total electron yields). However, even in this case, it may fail when the dominant processes take place in a finite range of internuclear distances. This is particularly important when resonant effects in the continuum are important. In this case, inclusion of the nuclear motion is essential to describe the physics of the problem. In this section we review the theoretical methods that have been used in the context of B-spline basis sets in the framework of the fixed-nucleus approximation and beyond.

6.4.1. The fixed-nucleus approximation. In this approximation, the positions of the two nuclei are fixed and, therefore, only electronic degrees of freedom must be considered. For simplicity, we shall start our discussion with one-electron systems. In contrast with one-electron atomic systems, the electronic continuum of one-electron diatomic molecules is a multichannel continuum. Indeed, we can think of the electronic continuum as having an infinite number of channels, one for each value of l of the free electron. For example, the boundary conditions for the continuum states of H_2^+ may be written

$$\psi'_{lE} \sim \frac{1}{r}\left[f_{lE}(r)Y_l^m(\hat{r}) - \sum_{l'} \pi K_{ll'}(E) g_{l'E}(r) Y_{l'}^m(\hat{r}) \right], \tag{157}$$

where f_{lE} and g_{lE} are regular and irregular Coulomb functions of energy E, and $\hat{r}$ denotes the variables θ and ϕ (notice that one must write such an equation for each value of l). The K-matrix is related to the usual scattering matrix S as in equation (63). As discussed in section 4.2.1, the continuum states resulting from diagonalization of the Hamiltonian do not have the above asymptotic behaviour. They satisfy arbitrary boundary conditions of the form

$$\psi_{lE_n} \sim \frac{1}{r} \sum_{l'} \left[a_{ll'n} f_{l'E_n}(r) Y_{l'}^m(\hat{r}) + b_{ll'n} g_{l'E_n}(r) Y_{l'}^m(\hat{r}) \right]. \tag{158}$$

Also, only one state is obtained at each energy eigenvalue, in contrast with the full set of open channels associated with a single value of E in equation (157).

The procedures to evaluate the continuum with B-splines are essentially the same as those described in section 4.2.1 for multichannel continuum states in atomic systems. One can obtain a full set of eigenvectors at each energy E following an approach originally proposed by Froese Fischer [18, 26, 274]. As in section 4.2.1, the coefficients $a_{ll'n}$ and $b_{ll'n}$ define two matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ that can be used to obtain the wavefunction with the correct boundary conditions, $\psi' = \boldsymbol{A}^{-1}\psi$, and the K-matrix, $\pi\boldsymbol{K} = -\boldsymbol{A}^{-1}\boldsymbol{B}$. This method has been used to study photoionization of H_2^+ and HeH^{2+} [274, 275] and, more generally, of polyatomic molecules (see section 7).

Alternatively (see section 4.2.1), the multichannel problem can be transformed into a sum of single-channel problems. The coupling between channels is introduced using a Lippman–Schwinger formalism which explicitly includes the correct boundary conditions. In this case, the uncoupled-continuum states, UCSs, are defined for each channel l:

$$\zeta_{nl} = \frac{\mathcal{U}_{nl}(r)}{r} Y_l^m(\hat{r}). \tag{159}$$

Then the procedure runs parallel to that described in section 4.2.1. The wavefunction with the correct boundary conditions is given by equation (66), where, now, μ must be replaced by l and $\mathcal{H}$ by $\mathcal{H}_{\rm el}$.

For many-electron systems, the procedures are similar. For instance, in the case of the H_2 molecule, for each channel αl_α, one has to define a set of orthogonal UCSs of the form [284]

$$\zeta_{\alpha l_\alpha \epsilon_\alpha}(\boldsymbol{r}_1, \boldsymbol{r}_2) = \Theta(\Phi_{\alpha l_\alpha}(\boldsymbol{r}_1, \hat{r}_2)\varrho_{\alpha l_\alpha \epsilon_\alpha}(r_2)), \tag{160}$$

where Θ is the symmetrization (antisymmetrization) operator for singlet (triplet) spin multiplicities, $\varrho_{\alpha l_\alpha \epsilon_\alpha}$ is the radial wavefunction of the continuum electron and $\Phi_{\alpha l_\alpha}$ is the channel function, which is a state of H_2^+ combined with the angular function of the scattered electron to give the correct channel symmetry (the index α includes all electronic quantum numbers of the molecular ion). This is equivalent to the well known static exchange approximation [455] in electron–atom scattering.

Generalization of equation (160) to more complicated diatomic molecules is trivial. Now, the functions $\Phi_{\alpha l_\alpha}$ can be described in terms of B-spline functions as described in section 6.2.2 and $\varrho_{\alpha l_\alpha \epsilon_\alpha}$ as in the one-electron case. This procedure leads to discrete spectra $\{\epsilon_{\alpha l_\alpha n_\alpha}\}$ and to discretized UCS wavefunctions $\zeta_{\alpha l_\alpha n_\alpha}$ that must be used in equation (66) to write the final continuum wavefunction with the proper asymptotic behaviour (notice that the index μ in equation (66) must be replaced by αl_α in the present case).

6.4.2. Nonresonant continuum states. The simplest description of a continuum state beyond the fixed-nucleus approximation is that of an unbound electron scattered by a positive molecular ion in a given vibrational state and containing all the remaining electrons. In this section we shall neglect the existence of Feshbach resonances, which will be considered in the following section (most applications—either with or without B-splines—to molecular systems containing three or more nuclei have been made using this approximation [295–299, 301, 451, 453, 454]). According to the BO approximation, the wavefunction can be written as a product of electronic and nuclear wavefunctions,

$$\Psi^+_{\alpha v_\alpha l_\alpha E}(\boldsymbol{r}, R) \simeq \psi^{0+}_{\alpha l_\alpha \epsilon_\alpha}(\boldsymbol{r}, R)\chi_{v_\alpha}(R), \tag{161}$$

where $\psi^{0+}_{\alpha l_\alpha \epsilon_\alpha}(\boldsymbol{r}, R)$ is the nonresonant electron-scattering wavefunction for a fixed position of the nuclei. The index α includes all electronic quantum numbers of the molecular ion and v_α stands for either bound or continuum vibrational states of the ion in the α state. The superscript + indicates the usual outgoing boundary conditions in electron–molecule scattering, (for consistency with earlier work [284–294], we use outgoing wave boundary conditions here in contrast to elsewhere in the review. Since calculated quantities involve the square of the appropriate matrix elements, the choice of incoming or outgoing waves is a matter of convenience.) l_α is the angular momentum of the ejected electron and ϵ_α is its kinetic energy. The nuclear wavefunction χ_{v_α} is the solution of the equation

$$[T(R) + E_\alpha(R) - W_{v_\alpha}]\chi_{v_\alpha}(R) = 0, \tag{162}$$

where $E_\alpha(R)$ is the BO potential energy curve of the residual ion, W_{v_α} is its total energy and

$$E = \epsilon_\alpha + W_{v_\alpha}. \tag{163}$$

Equation (161) is often called, in the context of electron–molecule scattering theory, the *adiabatic nucleus approximation* [461]. Evaluation of $\psi^{0+}_{\alpha l_\alpha \epsilon_\alpha}$ can be done with B-spline basis sets following the procedures described in the previous section.

6.4.3. Resonant continuum states. As in atoms, the electronic continuum may include Feshbach resonances associated with multiply excited states of the molecule. Determination of the resonance positions and autoionization widths is extremely important to understand the behaviour of the molecule in many different situations. However, this information does not provide a complete description of the resonant molecular continuum. Indeed, autoionization can lead to the formation of atomic or molecular ions, but it may also lead to dissociation of the molecule in neutral fragments. This is due to the repulsive character of the potential energy curves associated with autoionizing states. These processes may occur simultaneously, leading to complicated interference phenomena such as those recently observed in H_2 and D_2 photoionization experiments [456–458]. It is then clear that, in order to describe all possible ionization and dissociation pathways, the wavefunction cannot be written as a simple product of an electronic function and a nuclear function.

The appropriate theoretical framework has been developed by Sánchez and Martín [286, 288], inspired by the ideas of Bardsley [459] and Hazi *et al* [460]. We summarize here the basic results. In the following, integration over R will be written explicitly, and integration over $\boldsymbol{r}$ will be indicated with the usual bra–ket notation. We assume that there exists a set of orthogonal resonant states $\phi_r(\boldsymbol{r}, R)$ embedded in the electronic continuum of the molecule and define two orthogonal projection operators

$$Q = \sum_{r'} |\phi_{r'}(\boldsymbol{r}, R)\rangle\langle\phi_{r'}(\boldsymbol{r}, R)| \tag{164}$$

and

$$P = 1 - Q. \tag{165}$$

These states can be evaluated with B-splines using standard CI expansions as described in section 6.2.2. The resonance energies are given by

$$E_r(R)\delta_{rr'} = \langle\phi_r|\mathcal{H}_{\rm el}|\phi_{r'}\rangle. \tag{166}$$

Then the complete continuum wavefunction is given by [288]

$$\begin{aligned}\Psi^+_{\alpha v_\alpha l_\alpha E}(\boldsymbol{r}, R) = {} & \sum_{r'} \phi_{r'}(\boldsymbol{r}, R)\xi^{r'}_{\alpha v_\alpha l_\alpha E}(R) + \psi^{0+}_{\alpha l_\alpha \epsilon_\alpha}(\boldsymbol{r}, R)\chi_{v_\alpha}(R) \\ & + \lim_{\eta\to 0} \sum_{r'} \sum_{\alpha' l'_{\alpha'}} \sum_{v'_{\alpha'}} \int\!\!\!\!\!\!\sum \mathrm{d}E' \frac{1}{E - E' + \mathrm{i}\eta} \int \mathrm{d}R' V^{r'*}_{\alpha' v'_{\alpha'} l'_{\alpha'} E'}(R')\xi^{r'}_{\alpha v_\alpha l_\alpha E}(R') \\ & \times \psi^{0+}_{\alpha' l'_{\alpha'} \epsilon'_{\alpha'}}(\boldsymbol{r}, R)\chi_{v'_{\alpha'}}(R).\end{aligned} \tag{167}$$

In this equation, $\psi^{0+}_{\alpha l_\alpha \epsilon_\alpha}(\boldsymbol{r}, R)$ is the nonresonant electron-scattering wavefunction for a fixed position of the nuclei; formally, it is the solution of

$$[P\mathcal{H}_{\rm el}(\boldsymbol{r}, R)P - \mathcal{E}_\alpha(R)]\psi^{0+}_{\alpha l_\alpha \epsilon_\alpha}(\boldsymbol{r}, R) = 0, \tag{168}$$

where $\mathcal{E}_\alpha(R) = E_\alpha(R) + \epsilon_\alpha$, $E_\alpha(R)$ is the BO potential energy curve of the residual ion and ϵ_α is the kinetic energy of the outgoing electron. Also in equation (167), χ_{v_α} is the vibrational function of the residual ion (it is the solution of equation (162)), $E = \epsilon_\alpha + W_{v_\alpha}$ (in the second term), $E' = \epsilon'_{\alpha'} + W_{v'_{\alpha'}}$ (in the third term), $\int\!\!\!\!\!\!\sum \mathrm{d}E'$ represents a sum over bound states and an integral over continuum states of the molecule, $\int\!\!\!\!\!\!\sum_{v'_{\alpha'}}$ represents a sum over all bound vibrational states and an integral over all dissociative states (i.e. the vibrational continuum), $V^r_{\alpha v_\alpha l_\alpha E}(R)$ is a matrix element given by

$$V^r_{\alpha v_\alpha l_\alpha E}(R) = \langle\phi_r|Q\mathcal{H}_{\rm el}P|\psi^{0+}_{\alpha l_\alpha \epsilon_\alpha}\rangle\chi_{v_\alpha}(R) \tag{169}$$

and $\xi^r_{\alpha v_\alpha l_\alpha E}(R)$ is the solution of

$$[E - E_r(R) - T(R)]\xi^r_{\alpha v_\alpha l_\alpha E}(R) = V^r_{\alpha v_\alpha l_\alpha E}(R) + \lim_{\eta\to 0}\sum_{r'}\sum_{\alpha' l'}\sum_{v'_{\alpha'}}\int \mathrm{d}E' \frac{V^r_{\alpha' v'_{\alpha'} l'_{\alpha'} E'}(R)}{E - E' + \mathrm{i}\eta}\int \mathrm{d}R' V^{r'\,*}_{\alpha' v'_{\alpha'} l'_{\alpha'} E'}(R')\xi^{r'}_{\alpha v_\alpha l_\alpha E}(R'). \tag{170}$$

The latter equation represents the nuclear motion when the electrons are in a quasistationary state ϕ_r.

Equation (167) is derived by assuming that the BO approximation is valid within the P and Q subspaces, i.e. for both the resonance states $\phi_r(\boldsymbol{r}, R)$ and the nonresonant continuum states $P\Psi^+_{\alpha v_\alpha l_\alpha E}$. This means that the resonance–background coupling is due to the electronic interaction only. The matrix element in equation (169) represents the coupling between the resonance state and the nonresonant wavefunction of energy ϵ_α and vibrational state v_α. Hence, the two terms in the right-hand side of equation (170) are the result of the autoionizing character of the ϕ_r state. In particular, the last term represents the decay of the resonant state to the adjacent electronic continuum. This term is nonlocal due to the presence of the $\xi_{\alpha v_\alpha l_\alpha E}$-functions and it can be split into the usual delta function term and principal value term. In some cases, equation (170) can be simplified by using a local approximation and can be written in terms of resonance parameters (see [285] for details).

All terms defining the wavefunction given in equation (167) can be evaluated using B-spline basis sets. We have already shown how $\psi^{0+}_{\alpha l_\alpha \epsilon_\alpha}$ and χ_{v_α} can be calculated. The resonant electronic wavefunctions ϕ_r can be obtained using standard CI techniques as explained in section 6.2.2, for instance, using the equivalent of the TDM discussed in section 4.2.2. The most unusual part is the evaluation of the $\xi^r_{\alpha v_\alpha l_\alpha E}$ wavefunctions from equation (170). A numerical solution of this equation is extremely difficult because it is nonlocal and the wavefunctions oscillate strongly. However, equation (170) can be easily solved in the basis set that results from the diagonalization of the Schrödinger equation [288]

$$[T(R) + E_r(R) - \mathcal{W}_k]\tilde{\Upsilon}_k(R) = 0 \tag{171}$$

in a basis of B-spline functions such as that described in section 6.3. The use of the expansion

$$\xi^r_{\alpha v_\alpha l_\alpha E}(R) = \sum_i c^{r,i}_{\alpha v_\alpha l_\alpha E}\tilde{\Upsilon}_i(R) \tag{172}$$

and a short-range absorbing potential near $R = R_{\max}$ leads to a system of linear equations that can be solved using standard methods.

An illustration of the accuracy that can be reached with this procedure is presented in figure 27, which shows the nuclear wavefunction $\xi^r_{\alpha v_\alpha l_\alpha E}(R)$ associated with the lowest $^1\Sigma^+_u$ doubly excited state of H_2, calculated by solving equation (170) both numerically and using an expansion over the $\tilde{\Upsilon}_i(R)$ basis. Figure 27 shows that the wavefunctions calculated with B-splines are indistinguishable from the numerical ones except, of course, in the vicinity of $R = R_{\max}$ and beyond.

It is worth pointing out that although the formalism leading to equations (167) and (170) is general, in the sense that is is completely basis independent, its practical implementation has only been possible by using B-splines. The main reason is that autoionization may occur when the molecule is far from its equilibrium geometry, i.e. at relatively large values of R, and consequently one needs an accurate description of both the electronic and nuclear continua in a wide region of space. As already discussed in other applications of B-splines contained in this review, this cannot be achieved by using conventional L^2 basis functions since they decrease exponentially. We would like to point out that, in spite of the complicated mathematical aspect of the continuum wavefunction given in equation (167), computation of all the nuclear

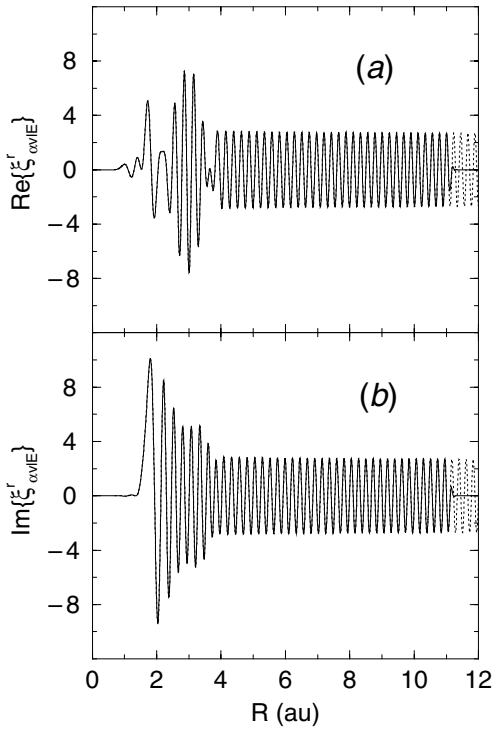


Figure 27. Nuclear wavefunction $\xi^r_{\alpha v_\alpha l_\alpha E}$ of H_2 when the resonance r is the lowest doubly excited state of $^1\Sigma_u^+$ symmetry. The index α represents the $^2\Sigma_g^+(1s\sigma_g)$ state of H_2^+, $E = -0.1723$ au and $W_{v_\alpha} = -0.3824$ au. The wavefunction has been obtained in the local approximation and for the dominant partial wave $l_\alpha = 1$. Dotted curve, numerical solution of equation (170); full curve, algebraic solution in a B-spline basis of order eight with $N = 240$ and $R_{max} = 12$ au, using an absorbing potential at $R = 11.5$ (see text). (*a*) Real part; (*b*) imaginary part.

components (including $\xi^r_{\alpha v_\alpha l_\alpha E}(R)$) takes only a few minutes with a personal computer. This is negligible compared with the time required to evaluate the electronic part of the continuum wavefunction. In fact, for a single value of R, a calculation of the electronic part requires as much computer time as a calculation performed in the fixed-nucleus approximation. The difference is that, in this case one has to repeat calculations for several values of R, while in the fixed-nucleus approximation a single value of R is sufficient.

6.5. Applications

6.5.1. One-electron diatomic molecules. One-electron diatomics represent a useful workbench on which to test different approaches and numerical methods. Initial work on H_2^+ illustrated the possibilities of the OCE approach for the bound and continuum states, the latter obtained via the multichannel least-squares approach [274]. Higher-l expansions with truncation of the radial basis were successfully tested on H_2^+ and HeH^{2+} [275]. Convergence of photoionization parameters was very satisfactory, and bound states are also quite accurate for such light nuclei. Convergence of the lowest $1\sigma_g$ bound state (which is the slowest to converge) is illustrated in table 12 together with dipole moments relative to the lowest transitions to $1\sigma_u$ and $1\pi_u$ states. Very recently, the OCE approach has also been applied to study MPI of H_2^+ with B-splines [280]. This has been done in the perturbative regime using the fixed-nucleus approximation and a reflection approximation which is a simplification of the nonresonant approximation discussed in section 6.4.2. With heavier nuclei, convergence in the lowest eigenvalues is much slower, as shown in table 13 for Ne_2^{19+} ($Z = 10$) at the same internuclear separation.

Extremely accurate solutions can be obtained employing elliptical coordinates. In H_2^+ a test employing quadruple precision on an IBM RISC 6000 machine easily reached convergence to better than 28 digits in the eigenvalues for the whole manifold up to $n = 6$ in the united atom

Table 12. Convergence of the electronic energy (in au) for the lowest $^2\Sigma_g^+(1\sigma_g)$ state of H_2^+ and the dipole transition moments (in au) to the lowest $^2\Sigma_u^+(1\sigma_u)$ and $^2\Pi_u^+(1\pi_u)$ states at the internuclear distance $R = 2.0$ au. Radial grids. OCE, 112 splines of order ten, with initial step $h = 0.125$, which is doubled at 2.0, 10.0 and 20.0 au, $r_{max} = 50$ au, a node of multiplicity seven on the nuclei. LCAO, first value of l_{max} is for expansion on the origin, second on the nuclei.

l_{max}	$-E(1s\sigma_g)$	μ_z	μ_x
OCE			
10	1.101 842	1.050 0826	0.716 429 04
20	1.102 507	1.049 9678	0.715 790 79
40	1.102 616	1.049 9463	0.715 686 27
100	1.102 632 9	1.049 9428	0.715 670 17
LCAO[a]			
0 0	1.102 443		
2 1	1.102 634 13		
4 2	1.102 634 214		
LCAO[b]			
10 1	1.102 634 208	1.049 9426	0.715 668 97
Ellipt.	1.102 634 214	1.049 9426	0.715 668 97

[a] $h = 0.25$ au, $r_{max} = 25.0$ au on the origin, $h = 0.5$, $r_{max} = 6.0$ au on the nuclei (see [276]).
[b] $h = 0.25$ au, $r_{max} = 25.0$ au.

Table 13. Convergence of the electronic energy (in au) for selected one electron states in Ne_2^{19+} at the internuclear distance $R = 2.0$ au. Radial grids as in table 12 for OCE and LCAO[b]. LCAO results converged in the reported digits.

l_{max}	$1\sigma_g$	$7\sigma_g$	$2\pi_g$	$5\pi_u$
OCE				
20	50.461 491	7.534 639	9.735 109	6.734 839
40	53.990 949	7.595 775	9.745 047	6.739 410
100	54.914 850	7.609 729	9.745 520	6.739 580
LCAO				
20 2	55.001 426	7.610 973	9.745 526	6.739 580

$(11\sigma_g, 9\sigma_u, 9\pi_u, 6\pi_g, 6\delta_g, 4\delta_u, 4\phi_u, 2\phi_g, 2\gamma_g, 1\gamma_u)$ [281]. For fun we may report the value

$$E_{1\sigma_g} = -1.102\,634\,214\,494\,946\,461\,508\,968\,945\,31 \text{ au}.$$

Due to the numerical stability of the algorithm it is easy to reach such precision after a little experimentation with grid parameters. Results to machine accuracy have also been obtained employing products of splines in ξ and η [279].

Slow convergence of the OCE expansion prompts the development of an LCAO approach. A first attempt supplemented OCE with additional functions centred on the two nuclei. A rather long range of atomic-centred splines ($r_{max} = 6.0$ au) was employed. This was rewarded by extremely fast convergence with angular momentum expansion [276, 278] but gave rise to numerical instabilities which were difficult to deal with in a systematic way. Currently much shorter LCAO expansions are being adopted, which will be more fully discussed in section 7.1.2. A few results for H_2^+ and Ne_2^{19+} test cases are reported in tables 12 and 13. Further results on the photoionization of diatomic molecules in the DFT framework will be discussed together with polyatomics in section 7.

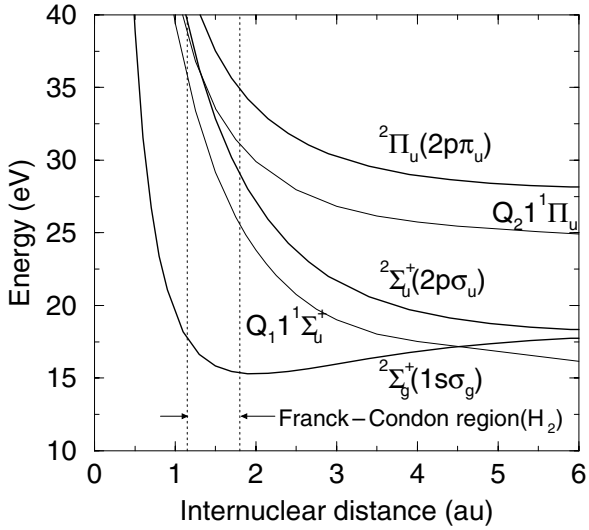


Figure 28. Potential energy curves of H_2 and H_2^+. The curves for autoionizing doubly excited states have been taken from [285, 289]. The energy origin is placed on the lowest rovibrational level of the ground electronic state of H_2.

6.5.2. Doubly excited states of H_2. B-splines have been successfully applied to the study of doubly excited states of H_2. Most of our present knowledge about the exact energy location and autoionization properties of these states comes from theoretical works (see [285] and [289] for an exhaustive list of references). The so-called Q_1 resonance series includes doubly excited states that lie above the first ionization threshold $X^2\Sigma_g^+(1s\sigma_g)$, but below the second one $^2\Sigma_u^+(2p\sigma_u)$. Hence, autoionization of the Q_1 states leads to H_2^+ ions in the GS $X^2\Sigma_g^+$. In contrast, autoionization of resonances converging to higher thresholds may also lead to H_2^+ ions in excited states. This is the case for the Q_2 series, which converges to the third ionization threshold $^2\Pi_u(2p\pi_u)$. The latter resonances autoionize, yielding H_2^+ ions in both the $X^2\Sigma_g^+(1s\sigma_g)$ and the $^2\Sigma_u^+(2p\sigma_u)$ states. The corresponding branching ratios are essential to interpret experimental results. To help the reader in the following discussions, we show in figure 28 the potential energy curves of H_2^+ and the lowest $Q_1\ ^1\Sigma_u^+$ and $Q_2\ ^1\Pi_u$ doubly excited states of H_2. In [285, 289, 293] B-spline basis sets have been used to compute the resonance parameters of many Q_1, Q_2, Q_3 and Q_4 autoionizing states. This is the most extensive and accurate compilation of resonance parameters published to date. B-spline basis sets were also used to evaluate, for the first time, the potential energy curves of the lowest nonautoionizing doubly excited states of H_2. These states have Σ^- symmetry and belong to the Q_2 series. They are equivalent to atomic doubly excited states of unnatural parity, with which they correlate in the united-atom limit. A complete set of results can be found in [290].

In these works, the resonant wavefunctions ϕ_n were calculated by diagonalizing the H_2 Hamiltonian in a basis of configurations built from H_2^+ orbitals. These orbitals were represented in a basis of B-splines using the OCE approach as described in section 6.2.1. For example, in the case of the $Q_1\ ^1\Sigma_u^+$ states, the calculations included the configurations $1\sigma_u n\sigma_g$ (n = 2–75), $1\pi_u n\pi_g$ (n = 1–36), $2\sigma_g n\sigma_u$ (n = 1–36), $2\sigma_u n\sigma_g$ (n = 2–18), $3\sigma_g n\sigma_u$ (n = 1–18), $1\pi_g n\pi_u$ (n = 1–12) and $1\delta_g n\delta_u$ (n = 1–12), which amounts to 200 configurations. (The index n used to label one-electron orbitals in the two-electron basis indicates energy ordering for a given value of R.) Notice that the lowest target orbital $1s\sigma_g$ is excluded from the configuration basis in order to ensure orthogonality with the P subspace. In the case of the Q_2 states, a similar

Table 14. Convergence of the calculated electronic energies (in au) for the lowest five $^1\Sigma_g^-$ states of H_2 (taken from [290]). Basis I, $N_l = 140$, $l_{max} = 8$, $N_{conf} = 95$. Basis II, $N_l = 210$, $l_{max} = 10$, $N_{conf} = 95$. Basis III, $N_l = 210$, $l_{max} = 10$, $N_{conf} = 158$ (see text).

	$R = 1.75$			$R = 3.5$		
	Basis I	Basis II	Basis III	Basis I	Basis II	Basis III
$1\,^1\Sigma_g^-$	−0.515 27	−0.515 27	−0.515 29	−0.436 14	−0.436 16	−0.436 18
$2\,^1\Sigma_g^-$	−0.478 31	−0.478 31	−0.478 33	−0.403 38	−0.403 40	−0.403 42
$3\,^1\Sigma_g^-$	−0.472 56	−0.472 56	−0.472 58	−0.401 23	−0.401 25	−0.401 28
$4\,^1\Sigma_g^-$	−0.463 37	−0.463 37	−0.463 38	−0.389 67	−0.389 69	−0.389 70
$5\,^1\Sigma_g^-$	−0.460 68	−0.460 69	−0.460 70	−0.388 83	−0.388 85	−0.388 86

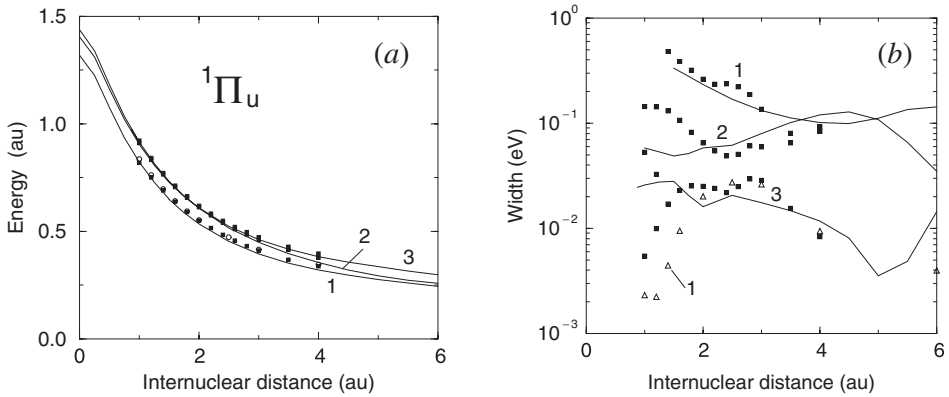


Figure 29. Theoretical energies (*a*) and widths (*b*) of the $Q_2\,^1\Pi_u$ doubly excited states of H_2. The energies are referred to the GS of H_2^+. Solid curves, results obtained with *B*-spline functions [289]; open triangles, [463]; open circles, [464]; full squares, [462]. Numbers indicate energy ordering. Notice the logarithmic scale in (*b*).

basis set was used but the $1s\sigma_g$ and $2p\sigma_u \equiv 1\sigma_u$ orbitals were excluded from the configuration basis. We shall use the results of [290] to show how convergence is achieved in this kind of calculation: first by increasing the number of terms N_l in expansion (139) (basis II), and second by increasing the number of configurations N_{conf} (basis III). Table 14 shows that the results obtained with bases I and II differ by $\sim 10^{-5}$ au at $R = 3.5$ au, and even less at $R = 1.75$ au. Basis III provides energies that differ again in the fifth significant figure. Similar convergence tests have been performed for autoionizing doubly excited states.

For the purpose of illustration, we show in figure 29 resonance parameters for the lowest $Q_2\,^1\Pi_u$ autoionizing states. Since these states play a crucial role in photoionization experiments, they have been also studied using other theoretical methods. There is a reasonable agreement between the *B*-spline results of [291] and those of [462] for both energies and widths. The agreement is also reasonable with [463] and [464] for the energy of the lowest state. However, the width reported in [463] is too small. Concerning the branching ratios (not shown in the figure) the agreement with the results of [462] is reasonable for the lowest two resonances, but there exist significant differences for the third one that might be related to the observed discrepancy for the energy of this state at large R.

6.5.3. Photoionization of H_2 and D_2. Photoionization of H_2 is the simplest process in which molecular continuum states are involved. In molecular systems, the photon energy can be shared between electronic and nuclear degrees of freedom, which contrasts with atomic

photoionization, where the photon energy is entirely absorbed by the electrons. In other words, the outgoing electron may absorb part of the total photon energy and leave the residual ion in an excited vibrational state. For large enough photon energies, the residual ion may also dissociate, thus leading to the process called dissociative photoionization. Due to the large number of available open channels, our knowledge of molecular photoionization is still far from complete. However, the use of B-splines has shed new light on the field, especially in the case of the simplest diatomic molecule H_2.

Among the most successful applications, we shall concentrate here on those that have led to the discovery of new interesting phenomena with important consequences in the interpretation of photoionization experiments. In recent works, the theory presented in section 6.4.3 has been used with B-spline functions [286, 288, 291, 292, 294] to study dissociative photoionization of H_2 and D_2, paying special attention to the role of the doubly excited states embedded in the electronic continuum. According to figure 28, when the photon energy is large enough to populate these doubly excited states ($\simeq$27 eV), the dissociative photoionization spectra exhibit resonant peaks. As a consequence of the coupling between resonant and nonresonant processes, and the interference with the nuclear motion (see section 6.4.3), these peaks are difficult to assign. In principle, one would expect similar resonant effects for the dominant nondissociative photoionization process that leads to the formation of H_2^+. Surprisingly, the experiments [465–469] show no evidence of resonant structures in this case.

As the initial state of H_2 has $^1\Sigma_g^+$ symmetry, application of the dipole selection rules implies that only electronic states of $^1\Sigma_u^+$ and $^1\Pi_u$ symmetries are populated. The total photoionization cross section of [287] is shown in figure 30, where it is compared with the most recent experimental data [468, 469]. As can be seen, the agreement between theory and experiment is very good (it should be pointed out that the comparison is made on an absolute scale). The most striking feature in figure 30 is that no resonant structure is observed, in agreement with the experimental observations. In contrast, the fixed-nucleus approximation used in most previous calculations leads to unphysical resonance structures similar to those observed in atomic photoionization. The B-spline results reported in [287] were the first ones to explain the experimental findings.

Earlier experimental analyses [470, 471] of the final vibrational population had focused on the nonresonant region resulting from excitation with the He 584 Å line ($\simeq$21.2 eV). As an illustration, we compare in table 15 the experimental data of [470, 471] with theoretical results obtained with B-spline functions [287]. The agreement is very good. It is also very good when we compare with the existing theoretical calculations that did not include resonance effects at 21.2 eV [472–475].

Absorption of photons with energy $\hbar\omega > 18.1$ eV leads not only to emission of an electron but also to dissociation of the molecule according to the equation $H_2 + \hbar\omega \rightarrow H + H^+ + e^-$ (dissociative photoionization). The experimental works in this energy region revealed unexpected features that only a theory based on B-splines has been able to explain very recently [286, 291]. For instance, Ito *et al* [456] reported the existence of several peaks in the kinetic energy distribution (KED) of protons that had not been observed in previous works or predicted theoretically. Also, He *et al* [457] observed some structure in the zero-ion-kinetic-energy spectrum of H_2 in the region of low photon energies that could not be interpreted satisfactorily in terms of simple energy conservation arguments. At higher photon energies ($\hbar\omega > 30$ eV) the situation is even more complicated because several ionization channels are open (see figure 28 and [456, 458]). The problem of dissociative photoionization has been systematically investigated by Sánchez and Martín in a series of papers [286–288, 291, 292] in which they made use of the theory described in section 6.4.3 with B-spline basis sets. In these papers they were able to explain most of the experimental observations. Here we reproduce

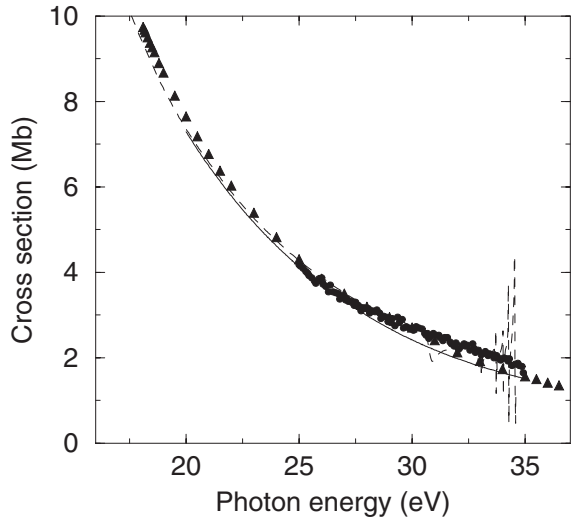


Figure 30. Total photoionization spectra of H_2 obtained with the theory of section 6.4.3 (full curve) [287] and the fixed-nucleus approximation (dashed curve) [284]. Triangles, experimental results of [468]; dots, experimental results of [469].

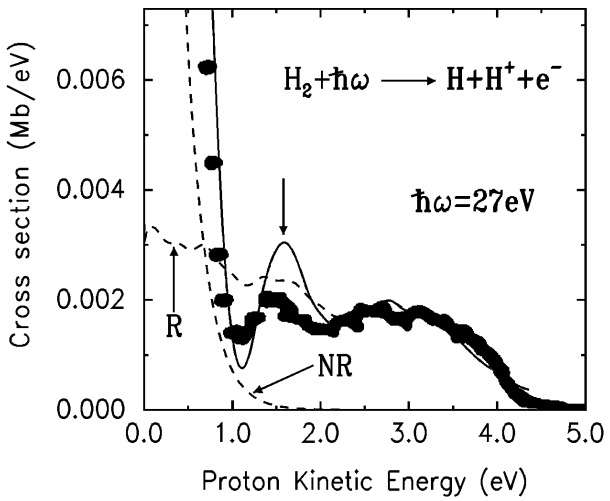


Figure 31. KED spectrum of H_2 for a photon energy of 27 eV protons detected at 0° ($^1\Sigma_u^+$ contribution) [286, 288]. Full curve, theoretical results from [286]; filled circles, experimental results from [456]; dashed curves: (R) resonant and (NR) nonresonant contributions. The arrow indicates a peak not identified before [286].

some of their results in order to illustrate the performance of the theory in this difficult case.

We show in figure 31 the calculated KED for the $^1\Sigma_u^+$ symmetry at a photon energy $\hbar\omega = 27$ eV. In the absence of resonant effects, the cross section would decrease almost exponentially as the proton energy increases [472]. This is in fact the result that is obtained by keeping only the second term in equation (167) (see figure 31). The figure shows the existence of two well defined peaks at $\simeq$1.5 and 3 eV. The high-energy peak arises from the first and third resonant terms in equation (167). It is clear that the origin of the 1.5 eV peak lies in the

Table 15. Comparison of the calculated H_2 ratios, $\sigma(v)/\sigma(v = 2)$, at 21.2 eV [287] with the experimental data of [470] and [471]. Numbers in parenthesis are the absolute values of the cross sections in Mb.

v	B-spline calculations	Pollard *et al* [470]	Ruf *et al* [471]
0	0.458	0.438	0.434
1	0.867	0.850	0.839
2	1. (1.016 23)	1. (1.0249)	1.
3	0.925	0.946	0.937
4	0.766	0.783	0.778
5	0.593	0.598	0.593
6	0.440	0.453	0.440
7	0.319	0.322	0.317
8	0.228	0.231	0.226
9	0.162	0.165	0.157

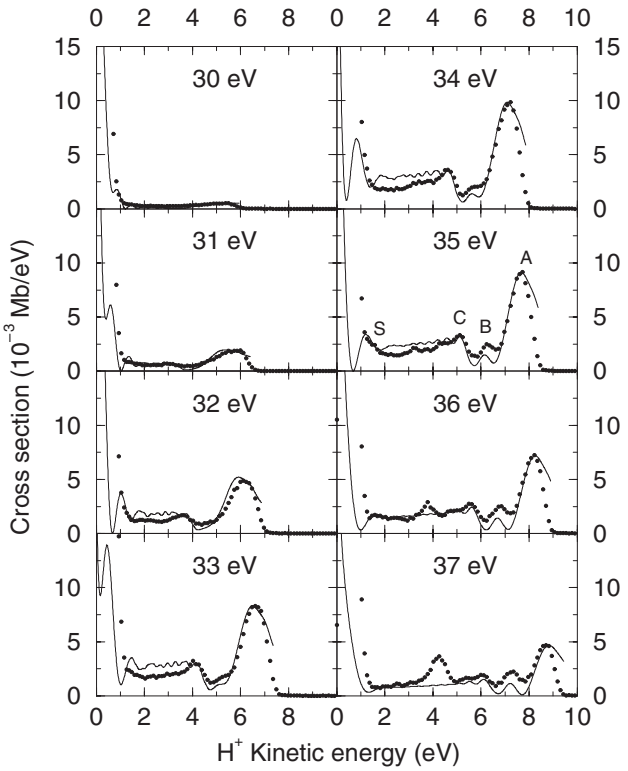


Figure 32. KED spectra of H_2 for protons detected at 90° ($^1\Pi_u$ contribution) [291, 292]. Full curve, theory; circles, experimental results from [456].

strong interference between the amplitudes associated with the resonant and nonresonant parts of the wavefunctions. The calculated proton KED spectra for the $^1\Pi_u$ continuum is shown in figure 32. For $\hbar\omega < 36$ eV, the theory reproduces most of the features observed in the experiment [456]: peaks A, B and C and their positions and relative intensities. The origin of the different structures observed in the spectra was discussed in [292] (see also [294]).

7. Polyatomic molecules and clusters

7.1. Methods

The fixed-nucleus approximation has generally been employed (see section 4.1), so the nonrelativistic Hamiltonian reduces to

$$\mathcal{H} = \sum_i h(i) + \sum_{i>j} \frac{1}{|\boldsymbol{r}_j - \boldsymbol{r}_i|} \tag{173}$$

with

$$h(i) = -\frac{1}{2}\nabla^2 - \sum_{\mathrm{p}} \frac{Z_{\mathrm{p}}}{|\boldsymbol{r}_i - \boldsymbol{R}_{\mathrm{p}}|}. \tag{174}$$

The one-electron operator comprises the kinetic energy term and the electrostatic interaction with the nuclei. A large variety of techniques has been employed for the determination of stationary states, molecular properties and transition amplitudes [411, 483]. In this context the use of B-splines has been restricted to the study of continuum states, with applications to photoionization, for which the properties of the basis appear especially well suited. Most many-body approaches start with the solution of effective one-particle equations

$$h_{\mathrm{eff}}\varphi_i = \varepsilon_i \varphi_i \tag{175}$$

where $h_{\mathrm{eff}} = h + V_{\mathrm{eff}}$ and V_{eff} is a one-electron effective potential operator, which may be local or not. For the lowest bound states, solution of (175) is generally accomplished by the use of an LCAO expansion employing Gaussian type orbitals (GTOs) or STOs [411, 483] centred on the different atomic nuclei. Although there are attempts to employ GTOs for continuum states [521–525], the main limitation stems from the difficulty involved in obtaining a sufficient density of states without running into numerical linear dependence problems. The B-spline basis overcomes this difficulty. Two approaches have been implemented, the one-centre expansion and the LCAO approach, already introduced in section 4, which will be analysed here in more detail in the context of polyatomic molecules. It might be noted that once a practically complete basis is available in three-dimensional space, it provides a convenient and accurate approach not only to the original Schrödinger equation, but also to a variety of general linear equations, like the Poisson equation and the inhomogeneous equations of perturbation theory.

7.1.1. The one-centre expansion. The OCE approach employs an expansion of all functions of interest as products of radial functions and spherical harmonics about a common centre (see 6.2.1), although in the polyatomic case all m values appear in the expansion

$$f(r, \theta, \varphi) = \sum_{lm}^{l_{\max}} R_l(r) Y_l^m(\theta, \varphi). \tag{176}$$

For highly symmetrical molecules a large saving can be obtained by full use of point group symmetry. If $\lambda\mu$ are the labels for the irreducible representations of the group under which h_{eff} is invariant, then equation (175) factorizes in

$$h_{\mathrm{eff}}^{\lambda}\varphi_{k\lambda\mu} = \varepsilon_{k\lambda}\varphi_{k\lambda\mu} \tag{177}$$

and matrix elements (144) are diagonal in λ and independent on μ. Symmetry-adapted angular basis functions are obtained as linear combinations of spherical harmonics

$$X_{j\lambda\mu} = \sum_m Y_{l_j}^m A_{mj\lambda\mu}. \tag{178}$$

Only Ys with the same l are combined, and j counts both l and the number of independent linear combinations for the same l-value. The A-coefficients can be obtained by standard projection operators employing rotation matrices of angular momentum theory. Actually it is computationally simpler to avoid complex algebra and real spherical harmonics and X angular functions are employed. The X-functions are orthonormal

$$\langle X_{j\lambda\mu}, X_{j'\lambda'\mu'}\rangle = \delta_{jj'}\delta_{\lambda\lambda'}\delta_{\mu\mu'} \tag{179}$$

so from

$$f(r,\theta,\varphi) = \sum_{j\lambda\mu} R_{j\lambda\mu}(r) X_{j\lambda\mu}(\theta,\varphi) \tag{180}$$

one obtains

$$R_{j\lambda\mu}(r) = \int_0^{\pi}\int_0^{2\pi} f(r,\theta,\varphi) X_{j\lambda\mu}(\theta,\varphi)\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi. \tag{181}$$

The symmetry-adapted basis is therefore

$$\chi_\mu \equiv \chi_{ij\lambda\mu} = \frac{1}{r} B_i(r) X_{j\lambda\mu} \tag{182}$$

and one-electron matrix elements reduce to one-dimensional radial integrals over B-splines, as in section 6.2.1. Two-electron integrals may be reduced to a linear combination of radial integrals $R^k(ijkl)$, which are the same as in the atomic case, and readily available. These could be used to evaluate matrix elements of V_{eff}, even for nonlocal potentials, like the exchange in the HF equations. Alternatively for the pure Coulomb potential it has been found more convenient to solve the Poisson equation directly

$$\nabla^2 V_{\mathrm{c}} = -4\pi\rho \tag{183}$$

in the same $\chi_{ij\lambda\mu}$ basis

$$V_{\mathrm{c}} = \sum_j V_j(r) X_{j00} = \sum_{ij} v_{ij}\chi_{ij00} \tag{184}$$

$$\rho = \sum_j \rho_j(r) X_{j00}. \tag{185}$$

Both V_{c} and ρ are totally symmetric ($\lambda\mu = 00$). Then

$$\sum_i (\nabla^2)^l_{i'i} v_{ij} = -4\pi a_{i'j} \tag{186}$$

$$a_{ij} = \langle \chi_{ij00}, \rho\rangle = \int_0^{\infty} B_i(r)\rho_j(r) r\,\mathrm{d}r \tag{187}$$

which reduces to the solution of a linear system with the dimension of the radial basis. Accurate solutions are obtained, provided that the correct boundary conditions on the potential are satisfied:

$$\lim_{r\to\infty} V_j(r) = \frac{A_j}{r^{l+1}} \tag{188}$$

$$\lim_{r\to\infty} \frac{(rV_j(r))'}{rV_j(r)} = -\frac{l}{r}. \tag{189}$$

This is easily implemented by forming a linear combination of the last two splines that satisfy the boundary condition at $r_{\max}$.

Radial components of the density $\rho(\boldsymbol{r})$, as well as of other local potentials, like V_{xc} in DFT approaches, are obtained at each radial point $\boldsymbol{r}$ by numerical integration of (181). An angular

grid is obtained as product of two standard GL sets of points and weights. For better accuracy the $[0, \pi] \times [0, 2\pi]$ domain is decomposed into subdomains and GL integration separately employed in each subdomain, in order to have finer grids close to the atomic nuclei. Also only symmetry unique subdomains are considered.

An important feature available with the B-spline approach is the possibility of truncating the radial basis for higher angular momenta. The major drawback of the OCE approach is slow convergence with $l_{\max}$, due to the nuclear cusps in the OCE expansion of the potential (see 6.2). The cusps become sharper and sharper with increasing l, and with the distance R_p of the nucleus from the origin, besides being proportional to the nuclear charge. However they are strongly localized at R_p, so the amplitude of the high-l components of the wavefunction is also localized around R_p. This means that only B-splines in the same range contribute significantly to the wavefunction, and the basis can be truncated according to l. An empirical scheme based on the values of the corresponding component $V_l(r)$, and a selected threshold, has been devised [275]. Since the number of angular components increases quadratically as $(l_{\max}+1)^2$ before symmetry blocking, truncation of the radial basis is very effective in allowing larger expansions in $l_{\max}$ without introducing appreciable errors.

Because of slow convergence with $l_{\max}$ it was believed that OCE was essentially limited to molecules with very light off-centre atoms close to the origin, in practice hydrides XH_n. Computational advances with the B-spline approach, as well as with other numerical methods [529,530], have shown that in fact satisfactory convergence can be obtained in much larger systems, like C_{60} or $Cr(C_6H_6)_2$, with expansions in the range $l_{\max} = 40$–80. So, while it is clear that OCE cannot describe very extended systems, it is able to cover a large range of medium-size molecules. Because of the basic simplicity, the availability of all types of integrals and the ease with which various theoretical models can be implemented, it appears to be an ideal benchmark for the test of new theoretical approaches in the study of polyatomic molecules.

7.1.2. The LCAO expansion. The nuclear cusps in the wavefunction due to Coulomb singularities at the nuclei are optimally described in polar coordinates around each nucleus. Unfortunately no global coordinate system is available which reduces locally to a spherical system around each nucleus, and this essentially hinders the use of finite-difference techniques on discrete meshes, as well as of a basis set obtained as a single tensor product of one-dimensional basis functions in different coordinates. We have seen limitations of the most obvious approach of this type, the OCE expansion. Actually research on more efficient techniques of solving (175) for large systems is being vigorously pursued [514, 517–520]. We may just mention attempts with finite-element techniques [526–528], cartesian grids in real or momentum space, with pseudopotentials to ease the cusp problem [519, 520] and the recent wavelet approaches [515–518]. The standard solution in quantum chemistry is the indirect introduction of local spherical coordinate systems by the use of an LCAO basis, that is a basis which is the union of separate spherical basis sets, each centred about a different nucleus. In standard quantum chemistry LCAO is implemented with global basis sets, usually GTOs [411, 483]. These provide fast convergence for the lowest bound states with a limited number of basis functions. However it is difficult to enlarge the basis towards completeness, because the union of separate complete sets on different centres is overcomplete, and in practice one quickly reaches numerical linear dependence in the set, because of the large overlap between functions on different centres. This problem makes the practical design of a complete three-dimensional basis very difficult.

The local nature of the spherical B-spline functions can be used to control the overlap between functions on different centres, by retaining only a short-range expansion on each

atomic centre. Moreover global continuity up to the pth derivative is retained by selecting only B-splines with the same overall continuity. In practice one deletes from the full set of B-splines on the finite interval around each atom the last $p + 1$ functions. An initial attempt to employ an LCAO type B-spline basis on H_2^+ and HeH^{2+} showed excellent convergence but quickly ran into numerical instabilities [276, 278] (see section 6.2). This was due to the rather long range of the LCAO functions employed, which overlapped considerably. An alternative point of view is based on the observation that a long-range OCE expansion with rather large $l_{\max}$ is needed in any case to reach asymptotic convergence in the continuum. Therefore the off-centre LCAO functions may be limited to a very small range, just to take care of the Coulomb singularities close to the nuclei. Present implementation restricts the range of LCAO functions to mutually nonoverlapping spheres, apart from the long-range OCE set. This restriction is paid for by a somewhat slower convergence, but stability is much improved. Still the basis cannot be arbitrarily enlarged, as in the pure OCE, without running into numerical instabilities. Although little experience is available yet, a region of satisfactory convergence to accurate numerical results is always found, and the approach appears sufficiently robust. Of course accuracy for the core and inner valence states is dramatically improved with respect to OCE, requiring a much smaller $l_{\max}$ in the OCE part, thus allowing the treatment of much larger and low-symmetry systems, well beyond the capabilities of the pure OCE approach. Actually in large systems, $l_{\max}$ of the OCE part appears dominated by asymptotic convergence of the continuum expansion, which at medium energies requires a much larger number of partial waves than was usually believed necessary.

The LCAO approach requires the evaluation of new multicentre integrals for the construction of the matrix elements (143,144). In the present formulation, because of nonoverlapping LCAO functions, only two-centre integrals between OCE and off-centre LCAO functions are required for one-electron matrix elements

$$\langle \chi^0_{ilm}, h\chi^p_{i'l'm'} \rangle \tag{190}$$

where 0 refers to OCE and p to the off-centre functions. Integrals are best evaluated with respect to the spherical harmonic basis and later transformed to symmetry-adapted functions. Integration is performed numerically inside each off-centre atomic sphere p, since the integrand is only nonvanishing inside it and the only singularity is at the centre p. A product of GL grids in r, θ and φ is employed. Actually a local coordinate system with the z-axis from the origin to the centre p is employed, so integration over φ is only required to decompose the potential V into its Fourier components $V_m(r, \theta)$, all remaining φ integration being performed analytically. Satisfactory convergence is easily reached, with results insensitive to further refinement of the integration grids. Since the number of LCAO functions is very limited, as well as the number of overlapping OCE functions, an entire block of integrals $(0, p)$ relative to each centre p is evaluated in parallel. Integrals are then rotated back to the original coordinate system employing rotation matrices for the spherical harmonics [484]. Only integrals with respect to unique symmetry centres are required. Integrals are then symmetry adapted by linear transformation. Linear combinations

$$\chi^q_{ij\lambda\mu} = \sum_{p\in q} \sum_m \chi^p_{il_jm} A_{pmj\lambda\mu} \tag{191}$$

of off-centre functions run over the set of functions on all equivalent centres p. The A-coefficients can be obtained with the same technique as in the OCE case.

The completness of the basis allows an accurate solution of the Poisson equation along similar lines as in the OCE case. Matrix elements of the Laplacian operator in the full LCAO basis are already available, and scalar products of the density with respect to the basis $\langle \chi^p_{ilm}, \rho \rangle$ are evaluated employing the product grid already defined inside each atomic sphere. The

Coulomb potential is then obtained as an expansion in the basis and can be quickly evaluated at grid points, together with any other local potential to obtain matrix elements of the Hamiltonian. This is more efficient and avoids the need of two-electron integrals in DFT approaches. The task of an accurate computation of two-electron integrals does not appear unduly expensive in any case, considering the reduced number of multicentre integrals, and the finite range of integration, with charge densities confined to at most two nonoverlapping atomic spheres, because of the particular choice of the LCAO basis.

In summary the B-spline LCAO basis appears a promising approach to accurate solutions of general linear equations with Schrödinger type operators in the multicentre case, of which application to photoionization processes is but a simple example.

7.1.3. Implementation of DFT techniques. At the LDA level the effective Hamiltonian (175) is completely defined given the initial electron density ρ, which determines $V_{\text{eff}} = V_{\text{c}} + V_{xc}$. The density is obtained by a conventional SCF LCAO type calculation for the initial state with a standard basis. The scalar products $\langle \chi_\mu, \rho \rangle$ are then evaluated by numerical integration and V_{c} is obtained by solution of the Poisson equation. Then matrix elements of V_{eff} are obtained by further numerical integration. Bound-state solutions are obtained by conventional diagonalization of the Hamiltonian matrix, with bound-state boundary conditions. Actually iterative diagonalization based on a block Shavitt algorithm [27] is employed, since it is more efficient when only a limited number of the lowest eigenvectors is required. For the continuum, the least-squares algorithm [18, 26, 274] (see section 3.3) is employed, usually in the original form, employing the matrix $\boldsymbol{A}(E) = \boldsymbol{H} - E\boldsymbol{S}$, in the case of the OCE basis, which is numerically very stable, and the $\boldsymbol{A}^\dagger\boldsymbol{A}$ formulation in the LCAO basis. Since matrix multiplication is expensive, one first computes the energy-independent products $\boldsymbol{H}^\dagger\boldsymbol{H}$, $\boldsymbol{S}^\dagger\boldsymbol{H} + \boldsymbol{H}^\dagger\boldsymbol{S}$ and $\boldsymbol{S}^\dagger\boldsymbol{S}$, which are symmetric matrices, and then at each energy builds

$$\boldsymbol{A}^\dagger\boldsymbol{A} = \boldsymbol{H}^\dagger\boldsymbol{H} - E(\boldsymbol{S}^\dagger\boldsymbol{H} + \boldsymbol{H}^\dagger\boldsymbol{S}) + E^2\boldsymbol{S}^\dagger\boldsymbol{S}. \tag{192}$$

The number of open channels corresponds to the number of angular momentum functions asymptotically open, that is those for which a complete radial set, including the last spline, is kept. A corresponding number of eigenvectors is extracted at each selected energy, following the algorithm already detailed. These are fitted to a linear combination of regular and irregular free solutions at the boundary (see section 6.4.1), giving K-normalized eigenvectors from which dipole matrix elements, cross sections and asymmetry parameters are evaluated using the angular momentum transfer formulas as detailed in [504, 505].

The TDLDA formulation is more complex and has so far been implemented only at the OCE level. The SCF approach originally proposed by Mahan [407, 408] is employed (see section 4.1.2). The inhomogeneous equations (54) are solved in the same B-spline basis, recasting them as a system of linear equations

$$\boldsymbol{A}(E_i)\boldsymbol{c}_i^{(1,\pm)} = \boldsymbol{b}_i^{(\pm)} \tag{193}$$

$$E_i = \varepsilon_i \pm \omega \tag{194}$$

where

$$A(E)_{\mu\nu} = h_{\text{eff}\,\mu\nu} - ES_{\mu\nu} \tag{195}$$

$$b_{\mu i}^{(\pm)} = \langle \chi_\mu, PV^{(\pm)}\varphi_i \rangle. \tag{196}$$

Scalar products of bilinear quantities, like $V^{\pm}\varphi_i$ or the density $\delta\rho$, which are given as bilinear expansions in the basis, give rise to integrals over three basis functions

$$a_\mu^{(\pm)} = \langle \chi_\mu, \delta\rho^{(\pm)} \rangle \tag{197}$$

$$= \left\langle \chi_\mu, \sum_{i\mu'\mu''} (c^*_{\mu' i} c^{(1,\pm)}_{\mu'' i} + c_{\mu' i} c^{(1,\mp)*}_{\mu'' i}) \chi_{\mu'} \chi_{\mu''} \right\rangle \tag{198}$$

$$= \sum_{\mu'\mu'' i} (c^*_{\mu' i} c^{(1,\pm)}_{\mu'' i} + c_{\mu' i} c^{(1,\mp)*}_{\mu'' i}) \langle \chi_\mu, \chi_{\mu'} \chi_{\mu''} \rangle. \tag{199}$$

Specifying all basis set indices one has

$$\langle \chi_{nj\lambda\mu}, \chi_{n'j'\lambda'\mu'} \chi_{n''j''\lambda''\mu''} \rangle = \langle X_{j\lambda\mu}, X_{j'\lambda'\mu'} X_{j''\lambda''\mu''} \rangle \left\langle \frac{B_n B_{n'} B_{n''}}{r^3} \right\rangle. \tag{200}$$

The matrix of integrals over three radial B-splines is very sparse, and can be stored conveniently; the integrals over three angular functions can be evaluated as linear combinations of those over three spherical harmonics, already considered. Also symmetry indices are restricted to the symmetry of the direct products.

Given a set of a_μ matrix elements of $\delta\rho$, expansion of δV_{C} is obtained as usual as the solution of the Poisson equation, while the corresponding expansion of $\delta V_{xc} = \sum_\mu v^{xc}_\mu \chi_\mu$ is obtained from the matrix of $\partial V_{xc}/\partial\rho$

$$M_{\mu\nu} = \left\langle \chi_\mu, \left. \frac{\partial V_{xc}}{\partial \rho} \right|_{\mathrm{GS}} \chi_\nu \right\rangle \tag{201}$$

which is calculated numerically, as

$$\boldsymbol{v}^{xc} = \boldsymbol{S}^{-1} \boldsymbol{M} \boldsymbol{S}^{-1} \boldsymbol{a} = \boldsymbol{Z} \boldsymbol{a}. \tag{202}$$

The matrix $\boldsymbol{Z}$ is computed and stored on disk at the outset.

So, given the expansion of the potential, the $b_{\mu i}$ matrix elements (196) reduce to integrals over three basis functions, entirely analogous to those considered for the density.

The particular solution $\varphi_i^{(1,-)}$ obtained is adapted to outgoing wave boundary conditions by adding a suitable linear combination of the solutions of the homogenous LDA problem (see section 4.1.2). Again TDLDA dipole moments are obtained employing the V_{SCF} potential in place of the pure dipole operator.

7.2. Applications

7.2.1. Convergence of one-centre and LCAO expansion. Let us first discuss convergence of eigenvalues, cross sections and asymmetry parameters in photoionization with respect to basis set expansion. Convergence in the OCE expansion is entirely dependent on angular momentum truncation, since the completeness of the radial basis is easily achieved. As already illustrated in section 6.5.1 convergence of eigenvalues, notably those with s atomic orbital components, is very slow. This depends on the distance and charge of off-centre nuclei, and barely on the symmetry, or actual geometry, of the system. However, symmetry has a dramatic effect on the number of angular momentum components in each symmetry block. In linear $D_{\infty\mathrm{h}}$ molecules, at most $l_{\max}/2 + 1$ components participate, linear in $l_{\max}$, while with no symmetry, all $(l_{\max} + 1)^2$ components are present. Therefore the value of $l_{\max}$ that can be used in practice depends strongly on symmetry, current values being 100 for linear systems, 80 in I_{h}, 70 in O_{h} or $D_{6\mathrm{h}}$, but down to 25 in $C_{2\mathrm{v}}$. This still allows the treatment of most small molecules with second-row atoms as well as rather large highly symmetrical systems. Heavier off-centre atoms are harder. This is mitigated by the fact that convergence with $l_{\max}$ is very smooth, so the qualitative features are generally correct; results at increasing $l_{\max}$ give a good approximation to the convergent results, and could also be used for extrapolation [304]. Moreover, convergence of photoionization profiles is quite different from that of the initial-state eigenvalues; notably,

Table 16. Convergence of the KS eigenvalues (eV) of $(CO)_2$ obtained with B-splines using different OCE expansions and an LCAO expansion. For comparison, STO LCAO (ADF) results are also presented. $\Delta = \varepsilon_{\mathrm{LCAO}} - \varepsilon_{\mathrm{ADF}}$.

	$l_{\max}$OCE				$l_{\max}$LCAO	
MO	50	70	100	ADF	30	Δ
$1\sigma_g$	−347.55	−416.80	−472.26	−539.90	−539.89	+0.01
$2\sigma_g$	−262.64	−278.43	−287.07	−293.97	−293.96	+0.01
$3\sigma_g$	−26.92	−29.48	−31.64	−34.30	−34.32	−0.02
$4\sigma_g$	−16.00	−17.70	−18.84	−19.74	−19.89	−0.15
$5\sigma_g$	−13.68	−14.22	−14.46	−14.54	−14.63	−0.09
$1\sigma_u$	−343.06	−414.12	−471.30	−539.90	−539.89	+0.01
$2\sigma_u$	−261.40	−277.93	−286.91	−293.97	−293.96	+0.01
$3\sigma_u$	−26.76	−29.38	−31.59	−34.30	−34.32	−0.02
$4\sigma_u$	−15.88	−17.64	−18.81	−19.73	−19.88	−0.15
$5\sigma_u$	−13.23	−13.86	−14.15	−14.25	−14.34	−0.09
$1\pi_g$	−16.81	−17.43	−17.66	−17.46	−17.72	−0.26
$1\pi_u$	−16.76	−17.41	−17.66	−17.46	−17.71	−0.25

profiles relative to core states are usually much better than could be expected from the very poor corresponding eigenvalues.

Convergence of the LCAO expansion is much faster. Deep core states are essentially exact, and convergence is dominated by the outer parts of the valence orbitals, which fall outside the atomic spheres and have to be described by the OCE expansion. The net result is that satisfactory convergence in extended systems can be achieved with feasible $l_{\max}$ expansions, although little experience is yet available.

We shall illustrate these aspects with the model system $(CO)_2$, which was suggested by a preliminary OCE investigation on $Cr(CO)_6$. The geometry is the same as that of the two *trans* ligands in the latter, with C and O atoms at 3.6282 and 5.7919 au from the origin. Due to independence of the convergence pattern on the symmetry, the model should represent accurately the convergence of the full metal complex. All the results reported are at the LDA level, employing the LB94 GS potential. The GS density is obtained from a conventional LCAO SCF calculation with the ADF program [493, 494], employing an all-electron DZP basis of Slater orbitals taken from the ADF database.

Eigenvalues are reported in table 16. The slow convergence of O 1s states in OCE is clearly visible, due to the long distance from the origin. The B-spline LCAO with $l_{\max} = 30$ is completely convergent, as checked at $l_{\max} = 40$ (the 0.01 eV lower ADF values are probably numerically inaccurate). Convergence at $l_{\max} = 15$ and 20 is also reported in figure 33 for photoionization parameters. Results agree well at low energies, but discrepancies become large at higher energies, although $l_{\max} = 20$ is still quite good up to $E_k = 100$ eV. Comparison of OCE and LCAO is also reported in figure 33. Even at $l_{\max} = 100$ there are significant errors, especially at low energy. It is nice to observe EXAFS type oscillations, clearly visible in the β-parameter, in both the core and valence regions, which can be easily evaluated given the stability of the B-spline approach up to high energies. It is interesting to analyse the contribution of different angular momentum asymptotic channels to the cross section, as a function of E_k. At each energy the contribution of different l-values reaches a peak and then quickly decays. Defining arbitrarily $l_{\max}$ so that 99.99% of the cross section is obtained, we can plot $l_{\max}$ as a function of E_k. In several cases a square root law is obtained, consistent with the classical impact parameter, that is

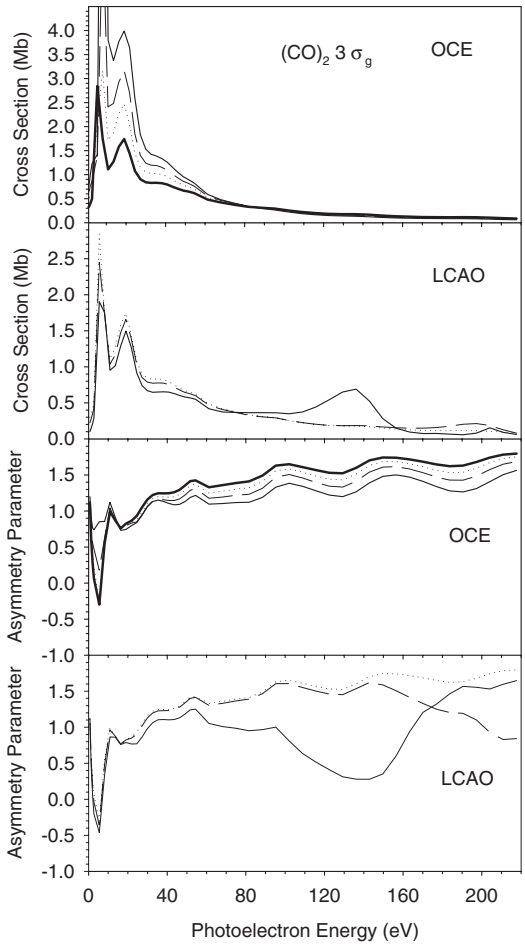


Figure 33. Convergence of OCE and LCAO photoionization parameters of $(CO)_2$ for the $3\sigma_g$ orbital. OCE: (——) $l_{max} = 50$, (– – –) $l_{max} = 70$, (· · · · · ·) $l_{max} = 100$, convergent results are from LCAO $l_{max} = 30$. LCAO: (——) l_{max}=15, (– – –) $l_{max} = 20$, (· · · · · ·) $l_{max} = 30$.

$$l_{max} = ka = \sqrt{2E}a.$$

So, apart from the short-range convergence, the minimum asymptotic l-value grows linearly with the dimension a of the system and the momentum k of the electron, so quite large OCE expansions are necessary to treat extended systems and energy ranges. For instance, in $(CO)_2$, $l_{max} = 18, 23, 30$ is necessary to obtain convergence up to $E_k = 2.0, 4.0, 8.0$ au. This gives a sound basis for the present choice of short-range LCAO functions, since a longer range would then be of little use given the large OCE expansion required, and would probably do harm to the stability of the approach.

7.2.2. Photoionization of molecules and clusters. Several applications to molecular photoionization have been performed at the LDA level: XH_n, X = F, O, N, Cl, S, P [297], X = C, B, Si, Al [298], N_2, $N(CH_3)_3$ [298], Cl_2 [299], ClF, ClF_3 [300], C_6H_6 [295, 309], C_{60} [301,310], M@C_{60} [302,303] and $Cr(C_6H_6)_2$ [308]. Only a few molecules, N_2, PH_3 [304] and CO [311], have been treated at the TDLDA level. We shall review a few selected examples

to illustrate the capabilities of the approach. All results reported are convergent with respect to the basis, so deviations from the experimental results reflect deficiencies of the theoretical scheme employed to describe the many-electron problem (besides neglect of nuclear motion and relativistic effects, which are believed to be of minor importance in the cases considered).

N_2. A comparison of LDA and TDLDA results relative to valence ionizations in N_2 is reported in [306]. The agreement with the experimental data is quite good for the outer valence ionizations $3\sigma_g$ and $1\pi_u$, of mainly N 2p atomic character, and less satisfactory for the inner valence $2\sigma_u$ and $2\sigma_g$, which is a fairly typical situation. Also in the neon atom 2p ionization is very accurate while 2s is overestimated by a factor of two, a clear shortcoming of the theoretical model, possibly associated with strong many-body effects present in inner valence states. The effect of TDLDA is not large, redistributing oscillator strength towards higher energy, widening and shifting shape resonances. This is seen in the $3\sigma_g$ ionization, giving excellent agreement with experiment, and also in $2\sigma_g$, despite the overall overestimate. In any case, the LDA approach is already quite satisfactory and can be used to assist interpretation of the spectra in more complex systems. Satisfactory results are obtained also for the core ionization [304, 307], where σ and π contributions to the cross section have been separately observed [495]. The spectrum can be followed smoothly at higher energy [304] and the weak EXAFS oscillations extracted, showing the excellent numerical stability of the approach over a wide energy range. It is interesting that these results are obtained with the use of the LB94 GS potential, which appears to be able to give a quite good description even of core processes. A further advantage in the case of equivalent core holes is to maintain the full molecular symmetry, which is broken in usual localization procedures [507–510]. For instance, in N_2 one can separate contributions from the $1\sigma_g$ and $1\sigma_u$ initial states, only the former giving the σ resonance, associated with excitation into an antibonding σ_u^* orbital, an effect which has been experimentally detected [506].

PH_3. Here the situation is quite different, resembling the large response effects found in the argon atom [167, 406]. The LDA results for the outer valence ionizations, $5a_1$ and $2e$ are reported in figures 34 and 35. Cross sections are underestimated by a factor of three (although the shape is reasonable) and TDLDA brings a decided improvement, giving very satisfactory agreement with experiment, with the exception of the threshold region in $5a_1$, so in the case of third-row atoms TDLDA is probably necessary to obtain quantitative agreement with experiment. Also the asymmetry parameter profiles show a significant shift towards higher energies, improving the agreement with experiment, although the general shape is unchanged, with a deep minimum corresponding to the Cooper minimum [511–513] in the cross section.

C_6H_6. Only two examples will be reported, taken from the extensive study [309]. The asymmetry parameter for all valence ionizations is reported in figure 36, over an extended energy range from a recent investigation [496]. Although the agreement is sometimes qualitative, the most important features of the spectra are well reproduced, allowing us to discriminate between different orbitals, which may be an important diagnostic tool for the interpretation of complex spectra. Results relative to core ionization are reported in figure 37. Even in this case the agreement is quite good, reproducing well the sharp shape resonances close to threshold, and even the first weak EXAFS oscillations at higher energy.

$Cr(C_6H_6)_2$. Also for a system of this size convergent results can be obtained with an OCE expansion with $l_{max} = 50$, as is verified by selected results at $l_{max} = 70$. We report in figure 38 cross sections and branching ratios relative to the two outermost photoionization bands, for which experimental data are available [499]. They illustrate that even in a transition metal complex rather good agreement can be obtained even at the LDA level, certainly sufficient to discriminate between the two orbitals $8a_{1g}$ (band A) and $4e_{2g}$ (band B) involved, both of which are mainly composed of Cr 3d orbitals, $3d_{z^2}$ in $8a_{1g}$ and $3d_{xy}$, $3d_{x^2-y^2}$ in $4e_{2g}$. The

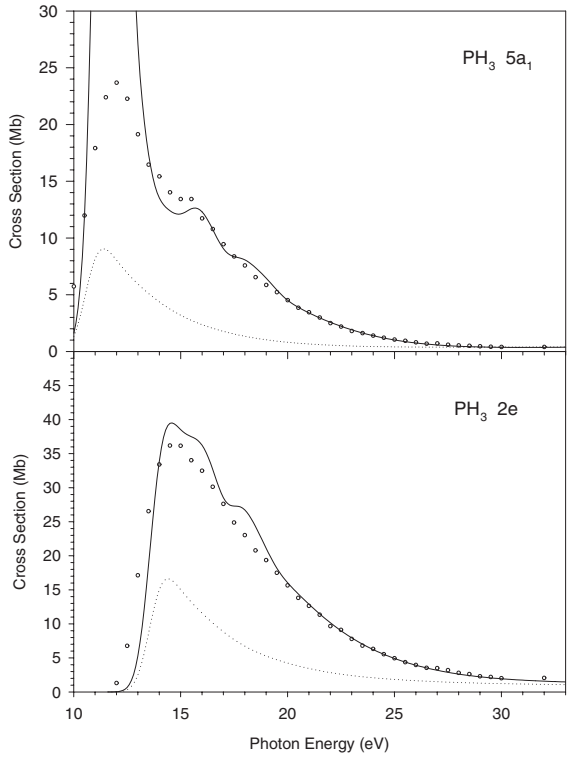


Figure 34. LDA (· · · · · ·) and TDLDA (——) photoionization cross sections relative to $5a_1$ and $2e$ orbitals in PH_3. Experimental data (○) from [503] (taken from [306]).

latter have a backbonding interaction with the LUMO (lowest unoccupied orbital) π^* orbital of the benzene moiety, which lowers its energy, and gives rise to a significantly different cross section profile. Although poorly resolved in the experiment, a very pronounced feature around 45 eV is clearly visible, and is completely missing from the calculated profiles. This is the 'giant resonance' associated with 3p $\rightarrow$ 3d excitation in unfilled 3d orbitals, typical of transition metal compounds, a strong characteristic signature of d-metal character. Of course such autoionization processes cannot be described at the LDA level, but are taken into account by the TDLDA scheme, which should find a very interesting application in this field.

C_{60} and M@C_{60}. Due to the fascinating properties of C_{60}, and its role as grandfather of much of the current research in carbon nanostructures, we shall consider a few interesting results on this molecule and its endohedral compounds, where a guest atom is trapped inside the cavity. I_h symmetry makes C_{60} ideal for an OCE approach, and it is possible to obtain completely convergent results despite the large radius of the system. The advantage of using the single LB94 potential for all ionizations is that it makes it possible to describe the full spectrum, valence and core, of this 360-electron system [301, 310]. The total absorption spectrum [310] compares quite well with the experimental data [500], despite the fact that strong response effects have been found in previous model calculations [485–487]. Indeed the rise of the cross section at the low-energy side is probably too steep, and it is expected that the TDLDA approach would redistribute oscillator strength towards higher energy, improving the shape. It is difficult to estimate absolute values experimentally, and a recent careful reexamination of available experimental data [501] gives a maximum at 20 eV, which is somewhat higher than currently calculated, but is subject to considerable uncertainty.

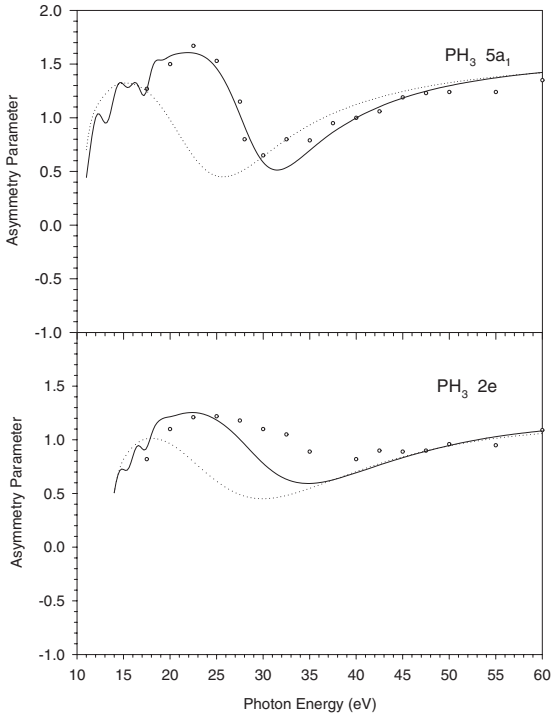


Figure 35. LDA (· · · · · ·) and TDLDA (——) photoionization asymmetry parameter relative to $5a_1$ and $2e$ orbitals in PH_3. Experimental data (○) from [502].

It is much easier to measure accurately intensity ratios, and much interest has focused on oscillations in the cross sections [488–490], which are best evidenced in the cross section ratio between the two lowest ionization bands (HOMO/HOMO -1 ratio). Accurate experimental results, together with a previous theoretical analysis [488] and LDA results, are reported in figure 39. As seen the calculated profile matches the experimental one pretty closely, including the damping at high energy, although a uniform shift of about 4 eV to higher energy has been added to the calculated profile to match the experiment, signalling that the potential employed is a little too attractive, an effect often observed.

Calculations of 1s core photoionization of the central atom in endohedral M@C_{60} is reported in figure 40. In comparison with the free metal atom profile one can note the appearance of sharp resonances induced by the cage. Indeed the presence of such structure has been predicted with a model δ-like spherical potential [491], with suitably adjusted strength. Recent results [492] suggest a more complicated picture. In fact, considering only the $l = 0$ potential component in our calculation, only weak oscillations are obtained. Moreover, the asymmetry parameter, which should stay fixed at the atomic $\beta = 2$ value for 1s ionization in spherical symmetry, shows considerable deviations. An analysis of the continuum wavefunction, which is quite complicated, indicates an almost pure p-wave propagating from the origin, until it reaches the C_{60} boundary, where heavy mixing with other angular components occurs, and then a slow decrease in the exterior region.

In conclusion the results reported indicate the variety of systems already within the scope of the OCE approach, the quality of the LDA approach to molecular photoionization and the

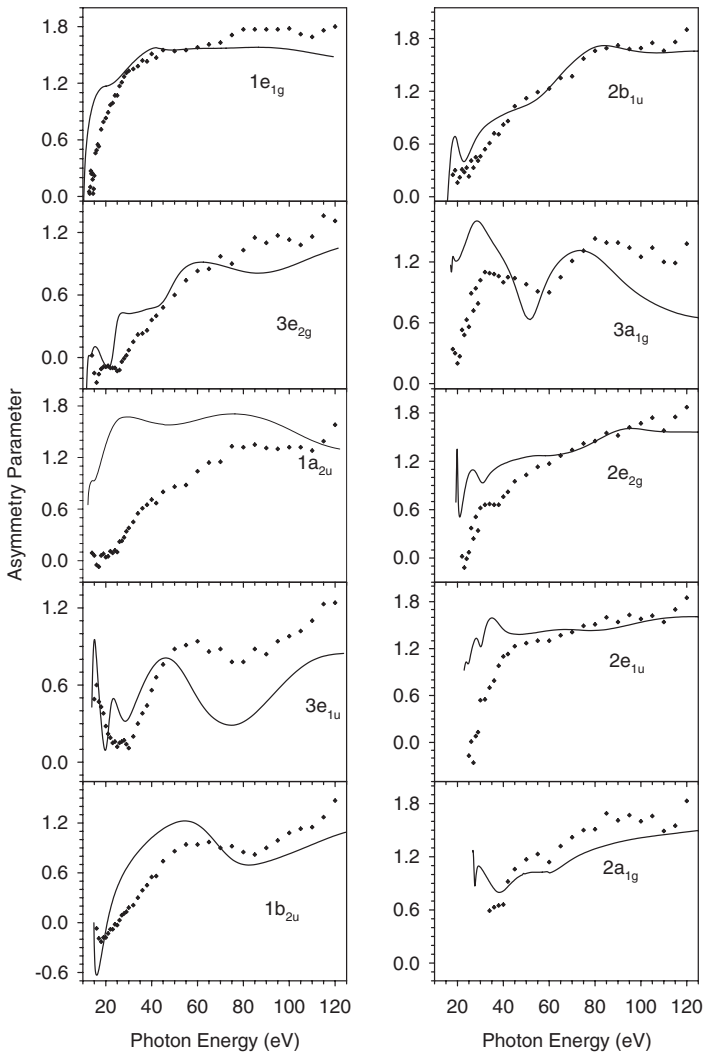


Figure 36. LDA photoionization asymmetry parameter profiles relative to all valence orbitals in C_6H_6. (——) calculated values; full diamonds, experimental data from [496] (taken from [309]).

potential of the *B*-spline LCAO approach as well as of the TDLDA scheme, which are still to be fully developed.

8. Conclusion and perspectives

In this review we have tried to show the reader that one of the most significant developments in computational atomic and molecular physics in recent years has been the introduction of *B*-spline basis sets in calculations of atomic and molecular structure and dynamics. *B*-splines were introduced in applied mathematics more than 50 years ago, but it has been in the 1990s, with the advent of powerful computers, that the number of applications has grown exponentially. This is best exemplified by inspection of the long list of papers given in the bibliography (see also figure 41). We have tried to make this list as extensive as possible.

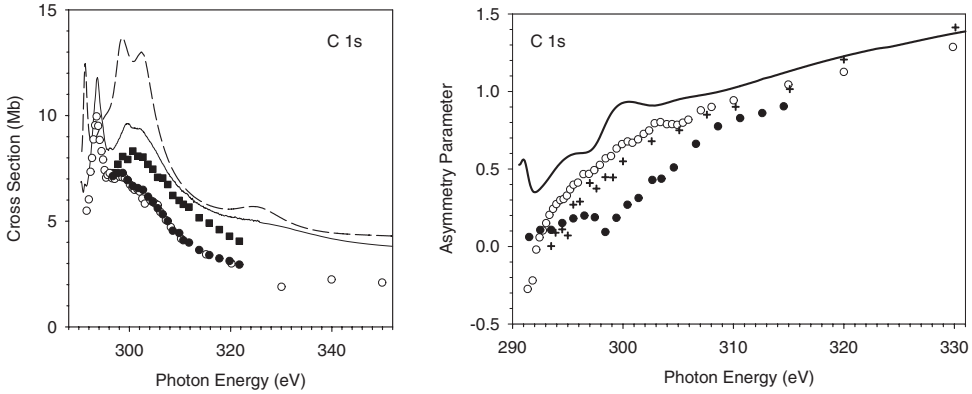


Figure 37. Left, total C 1s photoionization cross section in C_6H_6: (– – –) LDA calculated profile, (——) absorption experimental data, (○) and (●) main line photoemission from two different experiments, (■) main line + satellite photoemission experimental data from [498]. Right, C 1s asymmetry parameter in C_6H_6: (——) LDA calculated profile, (+) main line experimental data from [497], (○) main line and (●) satellite experimental data from [498] (taken from [309]).

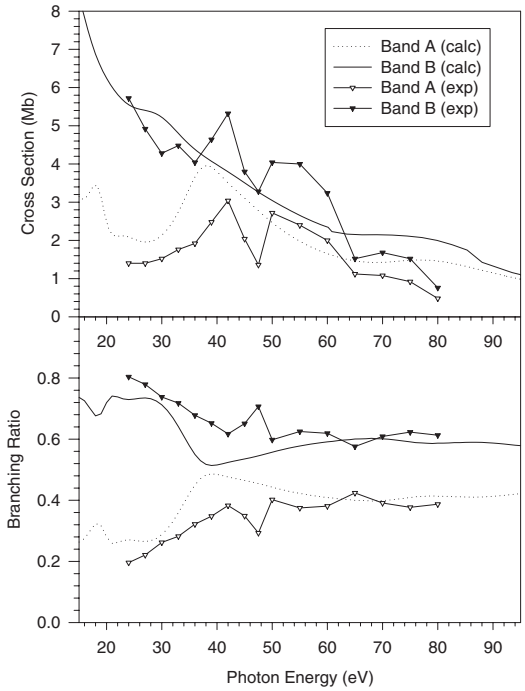


Figure 38. LDA photoionization cross sections (upper panel) and branching ratios (lower panel) relative to $8a_{1g}$ (band A) and $4e_{2g}$ (band B) orbitals in $Cr(C_6H_6)_2$. Experimental data from [499] (taken from [308]).

In this review we have discussed the main properties of *B*-splines and how they can be implemented to solve different problems in atomic and molecular physics. We have focused on those applications that have led to the discovery of new interesting phenomena and pointed out the reasons behind the success of the approach.

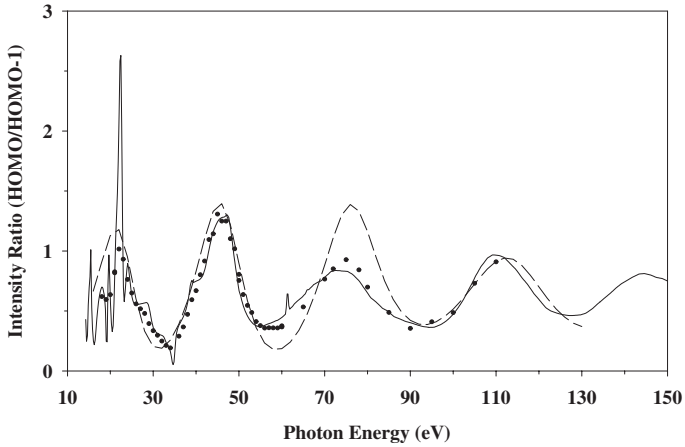


Figure 39. HOMO/HOMO −1 photoionization cross section ratio in C_{60}. Experimental data (●) and theoretical calculation (– – –) from [488], *B*-spline LDA results (——) (taken from [310]).

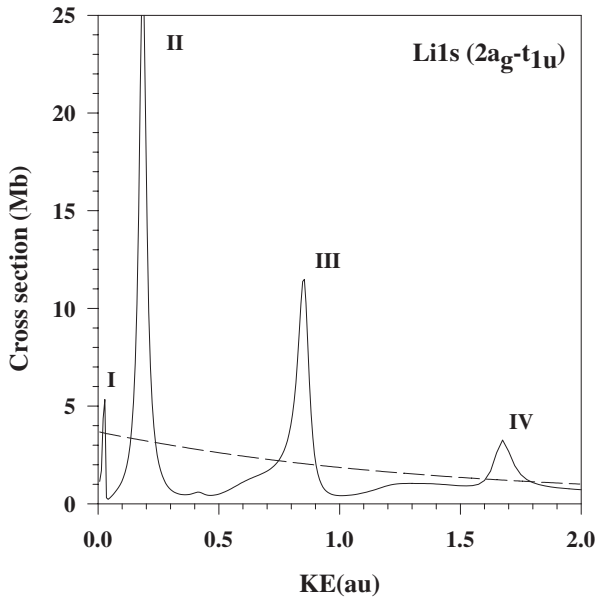


Figure 40. LDA photoionization cross section profile relative to Li 1s ionization in Li@C_{60}. (——) Li@C_{60} profiles, (– – –) free atom profile (taken from [303]).

As has been amply illustrated by the examples given in this review, a basis of *B*-spline functions is effectively complete. This has obvious advantages for MBPT calculations for example, but, in addition, *B*-splines are able to provide an accurate representation of continuum states, which makes them superior to more conventional L^2 basis sets. In particular, the asymptotic behaviour of these states can be accurately described by placing grid points at very large distances from the nuclei. Also the density of continuum states can be efficiently controlled. This has opened up new fields of application where traditional L^2 basis sets were not appropriate. Significant examples are the study of Rydberg and multiply excited states in

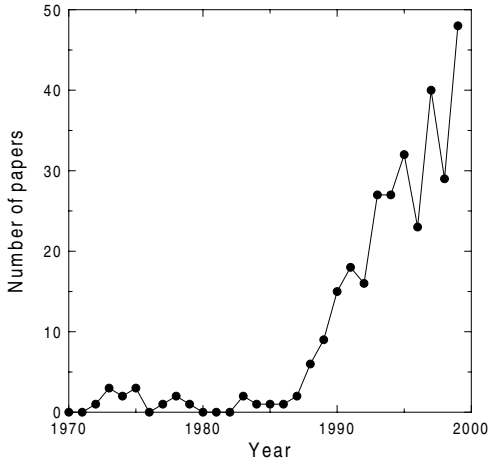


Figure 41. Plot of the number of papers that have made use of B-spline basis sets in atomic and molecular physics as a function of time. We are not aware of any paper before 1970.

atomic structure calculations, and ATI in laser–atom interactions. The effective completeness of B-splines basis sets is also an ideal property to accurately evaluate Green functions. This is advantageous, for instance, in calculations involving multichannel continua or nonlinear processes such as MPI. Furthermore, the use of B-splines has led to a major breakthrough in the description of continuum states of molecular systems, including the interference between the electronic and vibrational continua.

The future of B-splines is also promising. Existing atomic structure codes could be converted to a B-spline basis, with gains in numerical stability, accuracy and range of addressable problems. Moreover, the powerful approaches discussed in this review for the treatment of continuum states could extend the capabilities of these codes to treat continuum states in many-electron systems and multiply excited states such as those observed in recent syncrotron radiation experiments. Typical many-body approaches common in quantum chemistry (large-scale CI, coupled cluster, propagator methods for excitation and response properties, large-scale CC) could also be implemented with the same benefits.

A large number of applications are also foreseen in the study of the interaction of atomic systems with strong laser fields over a wide range of wavelengths and intensities. Indeed, although B-splines have been successfully applied to study He in strong XUV laser fields, the behaviour of this simple atom when it is exposed to longer-wavelength fields still remains a challenging problem for theoreticians. Multiphoton two-electron ionization is a fascinating aspect of the three-body problem that might also be addressed. All these problems require the handling of very large and dense basis sets, but the rapid increase in computational power will make these calculations possible in the near future.

Finally, many future applications will be devoted to the study of ionization and dissociation in large molecular systems. For instance, extension of the existing *ab initio* OCE and LCAO codes will allow the treatment of highly correlated molecules and clusters. Also, relativistic extensions of the TDLDA approach should make available accurate evaluation of cross sections and angular distributions for medium-size molecules containing heavy atoms. In addition, explicit expansions in interparticle coordinates could give extremely accurate results for three-body problems: H_2^+ and exotic systems containing e^+, μ etc. From a formal point of view, there is also no difficulty in extending the existing B-spline codes to study molecules in strong fields, in both the perturbative and nonperturbative regimes. Certainly, the loss of spherical symmetry and the inclusion of the nuclear degrees of freedom will lead to very demanding calculations, but the first *ab initio* calculations will be accessible in the near future.

Acknowledgments

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Appendix

A.1. Breakpoint sequences

We give a list of breakpoint sequences commonly used in B-spline calculations in atomic and molecular physics. In all cases, the radial coordinate is discretized in the interval $[r_{\min}, r_{\max}]$. In most cases, $r_{\min}$ is set to zero. This interval is divided into $(n-1)$ segments by arranging the n bps $[\xi_i]_{i=1}^{n}$ according to the given formula. An example of all sequences is displayed in figure A.1 for comparison.

A.1.1. Linear sequence. The most trivial sequence divides $[r_{\min}, r_{\max}]$ into $(n-1)$ segments of equal length.

$$\xi_i = r_{\min} + \frac{r_{\max} - r_{\min}}{n-1}(\mathrm{i}-1) \qquad i = 1, \ldots, n. \tag{203}$$

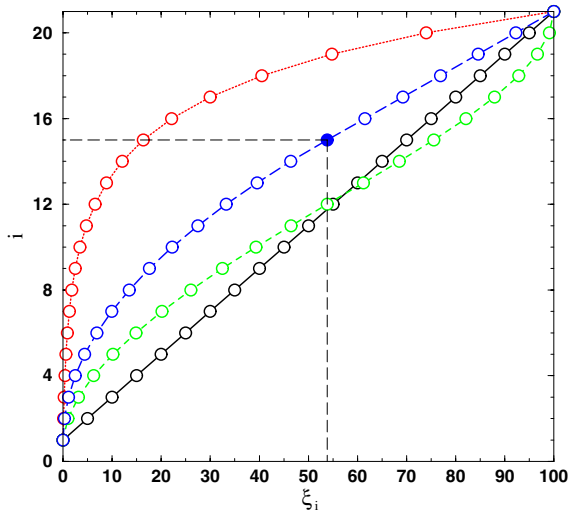


Figure A.1. Comparison of different types of breakpoint sequence: $n = 21, r_{\min} = 0, r_{\max} = 100$. Solid curve, linear sequence. Dotted curve, exponential sequence with $\gamma = 6$. Dashed curve, sinelike with $\alpha = 2$. Long-dashed curve, linear–parabolic with $i_0 = 15$ (the connection between the parabolic and the linear sequences occurs close to $r_0 = 53.8$).

(This figure is in colour only in the electronic version)

A.1.2. Exponential sequence.

$$\xi_i = r_{\min} + (r_{\max} - r_{\min})\frac{\mathrm{e}^{\gamma(\frac{\mathrm{i}-1}{n-1})} - 1}{\mathrm{e}^{\gamma} - 1} \qquad i = 1, \ldots, n. \tag{204}$$

This sequence tends to a linear sequence as $\gamma \to 0$ while all points exponentially accumulate close to $r_{\min}$ as $\gamma \to \infty$.

A.1.3. Sinelike sequence.

$$\xi_i = r_{\min} + r_{\max} \sin\left(\frac{\pi}{2}\left[\frac{\mathrm{i}-1}{n-1}\right]^{\alpha}\right). \tag{205}$$

The points accumulate close to $r_{\min}$ when $\alpha \to 0$ and close to $r_{\max}$ when $\alpha \to \infty$. For intermediate values ($\alpha \approx 2$) accumulation occurs at both ends.

A.1.4. Linear–parabolic sequence. A useful sequence adapted to a good description of both the bound and the continuum states associate a linear spreading at large distances with a quadratic sequence close to the origin. The connection is such that the function ξ_i and its first derivative are continuous. The parameter i_0 defines the value of i at which the connection is made.

$$\begin{aligned} r_0 &= \xi_{i_0} = \frac{r_{\max}(i_0 - 1) + r_{\min}(n - i_0)}{2n - i_0 - 1} \\ \alpha &= \frac{r_0 - r_{\min}}{(i_0 - 1)^2} \\ \beta &= \frac{r_{\max} - r_0}{n - i_0} \\ \xi_i &= \begin{cases} r_{\min} + \alpha(i-1)^2 & 1 \leqslant i < i_0 \\ r_0 + \beta(i - i_0) & i_0 \leqslant i \leqslant n. \end{cases} \end{aligned} \tag{206}$$

Other grids have been employed in the literature (see, for instance, [20, 23, 27]).

A.2. Density of states

The normalization problem discussed in section 3.3 requires an evaluation of $\left.\frac{\partial E(n')}{\partial n'}\right|_{n'=n}$. The method presented here is based on the polynomial interpolation of E_n. There is a unique polynomial $P_3(n)$ of order three interpolating the set of three discretized energies E_{n-1}, E_n, E_{n+1}. Once the polynomial is known, the density reads

$$\rho^{(3)}(E_n) = \left(\left.\frac{\partial E(n')}{\partial n'}\right|_{n'=n}\right)^{-1} = \left(\left.\frac{\partial P_3(n')}{\partial n'}\right|_{n'=n}\right)^{-1} = \frac{2}{E_{n+1} - E_{n-1}} \tag{207}$$

which is the result found in expression (21). By extending the set to five, seven and nine energies, the density is extracted using the corresponding polynomial, which leads to

$$\begin{aligned} \rho^{(5)}(E_n) &= \left(\frac{8(E_{n+1} - E_{n-1}) - (E_{n+2} - E_{n-2})}{12}\right)^{-1} \\ \rho^{(7)}(E_n) &= \left(\frac{45(E_{n+1} - E_{n-1}) - 9(E_{n+2} - E_{n-2}) + (E_{n+3} - E_{n-3})}{60}\right)^{-1} \\ \rho^{(9)}(E_n) &= \left(\frac{672(E_{n+1} - E_{n-1}) - 168(E_{n+2} - E_{n-2})}{840}\right. \\ &\quad \left. + \frac{32(E_{n+3} - E_{n-3}) - (E_{n+4} - E_{n-4})}{840}\right)^{-1}. \end{aligned} \tag{208}$$

A.3. Two-electron integrals

The calculation of two-electron integrals is a central problem in atomic structure calculations as soon as more than one electron is considered. The speed and accuracy with which such integrals can be calculated is crucial for how large problems can be attempted. The accuracy of the B-spline approach is superior to most other approaches that have been tried while the question of speed, which is coupled to the amount of memory available because integrals can be stored, is more complex and we recall that Altenberger-Siczek and Gilbert [42] concluded that B-splines could not be used for SCF calculations precisely because of the time and amount of memory required for the calculation of the two-electron integrals. Although time has helped to solve these two computer hardware problems, more progress is required. This applies also to software developments, for example to make full use of symmetry properties.

The traditional approach in HF calculations for atoms has been to use finite-difference techniques to obtain values of radial orbitals at a number of points between the nucleus and infinity. The use of B-splines means that the radial orbital is represented by an expansion in regular polynomials instead. We shall later discuss the situation when B-splines are used directly without first constructing single-particle orbitals. Integrals over polynomials can be carried out to machine accuracy using GL quadrature if the integration is over a closed interval. This requires the knowledge of the orbital at a number of so-called GL points. If n GL points are used, polynomials of order $2n$ or less can be integrated exactly. The GL points are zeros of Legendre polynomials, which can be obtained from mathematical tables (for example [533]) or from a library routine [534]. All integrals involve characteristically two B-splines and a low power of the integration variable. Landman and Hansen [56], who used B-splines of order seven (polynomials of degree six) found that the use of eight GL points gave very accurate results and that inclusion of further points did not lead to a significant improvement. However, this result does depend on the power of the integration variable r involved in a particular application.

The knot sequence $\{t_i\}$ defines how the integration region $[0, r_{\max}]$ is partitioned into sub-intervals. The total integral of a function is obtained by adding the contributions corresponding to each sub-interval. The length of the sub-intervals $[t_j, t_{j+1}]$ depends on j. Consequently, for each sub-interval the GL points are different and they can be obtained via a linear transformation of the Gaussian points in the interval $[-1, 1]$.

The integration of one-dimensional functions is straightforward with this approach. However, the two-electron integrals correspond to two-dimensional integrals. The general definition of the two-body (Slater) integrals is

$$R^k(l, m, n, o) = \int_0^\infty \int_0^\infty \frac{r_<^k}{r_>^{k+1}} \, \mathrm{d}r_1 \, \mathrm{d}r_2 \, P_l(r_1) P_m(r_2) P_n(r_1) P_o(r_2) \tag{209}$$

where $r_<$ and $r_>$ are the smaller and the larger of r_1 and r_2, respectively, and both integrations are over the whole region $[0, \infty]$. For B-splines the region will be $[0, r_{\max}]$. $P_l(r)$ is the radial orbital for electron l but could also be a B-spline basis function as discussed later.

The R-integral can be rewritten in the following way:

$$R^k(l, m, n, o) = \int_0^{r_{\max}} \frac{1}{r_1} Y^k(P_m, P_o; r_1) P_l(r_1) P_n(r_1) \, \mathrm{d}r_1 \tag{210}$$

where

$$Y^k(P_m, P_o; r_1) = \frac{1}{r_1^k} \int_0^{r_1} r_2^k P_m(r_2) P_o(r_2) \, \mathrm{d}r_2 + r_1^{k+1} \int_{r_1}^{r_{\max}} \frac{1}{r_2^{k+1}} P_m(r_2) P_o(r_2) \, \mathrm{d}r_2. \tag{211}$$

The Y^k-function is calculated at the Gaussian points needed for the integration of equation (209). The latter integration is straightforward and the only difficulty lies in

the calculation of the Y^k-function, equation (211). Two methods have been used for this calculation. One relies on transforming the integrals into a differential equation and was introduced by Hartree. The other is straight GL quadrature. We shall first discuss the latter.

The difficulty in this approach is due to the cusp at the point r_1 in the definition of the Y^k-function. An accurate determination of the contribution to the Y^k-function from this sub-interval is essential since the weighting function $r_<^k/r_>^{k+1}$ has its maximum here. Assume that r_1 lies in the sub-interval $[t_i, t_{i+1}]$. In this interval two different functions must be integrated, one between t_i and r_1 and another between r_1 and t_{i+1}. For each of these two parts a set of Gaussian points is defined and the integration of each function over its own region is carried out. That the existence of the cusp does not lead to an inaccuracy in the integration procedure is because only the function values in the Gaussian points are needed and not the function value in the cusp itself.

The second method involves transforming the Y^k-function into a differential equation and avoids the cusp problem. Traditionally a set of first-order differential equations was solved in which an additional function Z^k was introduced [531, 532]. For use with B-splines Froese Fischer and co-workers [21, 22, 24, 25] have combined the two first-order equations into a single second-order Poisson equation

$$\frac{\mathrm{d}^2}{\mathrm{d}r^2}Y^k(r) = \frac{k(k+1)}{r^2}Y^k(r) - \frac{2k+1}{r}P_m(r)P_o(r) \tag{212}$$

with boundary conditions

$$Y^k(0) = 0 \tag{213}$$

and

$$\frac{\mathrm{d}}{\mathrm{d}r}Y^k(r) = \frac{-k}{r}Y^k(r) \qquad \text{as} \quad r \to \infty. \tag{214}$$

Froese Fischer and Guo [21] have solved the differential equation (212) using the spline–Galerkin method similarly to solving a single-particle equation in a B-spline basis. This leads to a banded linear system of equations for the expansion of $Y^k(r)$, where the last equation arises from applying the boundary condition, equation (214), at $r = r_{\max}$.

Two-electron integrals for hydrogenic orbitals can be expressed exactly in terms of rational numbers [535]. Thus a good test of the numerical methods is to calculate such integrals and Froese Fischer *et al* [22] and Landtman and Hansen [56] have used this test to demonstrate that the use of B-splines leads to a very considerable improvement in the accuracy of two-electron integrals. Froese Fischer *et al* [22] used the differential equation technique equation (212) to determine Y^k while Landtman and Hansen [56] used the GL quadrature equation (211). In table A.1 we compare the two sets of results. Both sets of calculations are carried out on SUN workstations in double precision using B-splines of order seven. Froese Fischer *et al* used a linear grid from 0 to 1 au with eight intervals in this region and a exponential grid thereafter. The box size was 160 au and the total number of splines was 57. Landtman and Hansen used a exponential grid between 0 and 160 au, which leads to 11 intervals in the region 0–1 au. Since the total number of splines is 60, the two basis sets are rather similar. Also the accuracy of the two sets of results is similar despite the large difference in numerical approach.

In addition, earlier results obtained using finite-difference methods [536] are shown. The accuracy of the latter results is about four orders of magnitude worse than the B-spline values. The finite-difference results were obtained using exact hydrogenic wavefunctions, while both B-spline calculations use a spline approximation to the hydrogenic wavefunctions. Thus the B-spline results test the entire process while the finite-difference results tested only the integration procedures.

Table A.1. Comparison of analytic and numerical values for some F^k and G^k integrals using hydrogenic wavefunctions. The integrations using B-spline basis functions of order seven (columns 3 and 4) are carried out in two different ways as described in this appendix, while the values in column 5 are obtained using finite-difference methods. The notation $a\mathrm{E}-b$ means $a \times 10^{-b}$. From [56].

Integral	Exact value[a] [535]	Error [56]	Error [22]	Error [531, 536]
F^0(1s, 1s)	5/8	−2.59E−14	−3.00E−14	−5.09E−10
F^0(2s, 1s)	17/81	−1.53E−14	−2.00E−14	7.50E−10
F^0(2s, 2s)	77/512	−2.43E−14	−3.00E−14	1.56E−9
F^0(2p, 1s)	59/243	−2.60E−15	0.	2.77E−10
F^0(2p, 2s)	83/512	−1.79E−14	−2.00E−14	1.12E−9
F^0(2p, 2p)	93/512	−8.04E−15	−9.99E−15	6.53E−10
G^0(2s, 1s)	16/729	3.15E−16	9.99E−16	−1.36E−10
G^1(2p, 1s)	112/2187	6.46E−15	8.40E−14	−2.48E−10
G^1(2p, 2s)	45/512	−1.95E−14	1.18E−12	1.70E−10
G^0(2p, 3p)	96 768/9765 625	1.22E−15	1.80E−15	
G^2(2p, 3p)	110 592/9765 625	4.64E−15	9.00E−15	
G^1(2p, 3d)	1824 768/48 828 125	7.80E−15	1.02E−12	
G^3(2p, 3d)	1064 448/48 828 125	4.95E−15	5.00E−15	
G^0(2p, 4p)	560/177 147	1.00E−14	1.49E−14	
G^2(2p, 4p)	2128/531 441	1.47E−14	1.95E−14	
G^1(2p, 4d)	19 120/1594 323	2.49E−14	1.65E−13	
G^3(2p, 4d)	3920/531 441	1.48E−14	1.90E−14	
F^2(4f, 4f)	103 275/3670 016	−3.45E−14	1.10E−12	
F^4(4f, 4f)	69 003/3670 016	−3.03E−14	−6.00E−14	
F^6(4f, 4f)	7293/524 288	−2.51E−14	−5.50E−14	

[a] The actual values are taken from Froese Fischer *et al* [22].

It is interesting that the accuracy of the results does not depend on the k-value of the integral. Since the weighting factor $r_<^k/r_>^{k+1}$ becomes more and more peaked as k increases, the importance of handling the cusp correctly could be expected to increase with k. In fact for $F^2(4f, 4f)$ Froese Fischer *et al* [22] obtained a somewhat worse accuracy than for $k = 4$ and 6, while in the GL quadrature results by Landtman and Hansen, the error for $k = 2$ is similar to that for the larger k values. The 'large' errors Froese Fischer *et al* reported for the G^1(2p, 2s) and G^1(2p, 3d) integrals are not reproduced by the direct quadrature results either. In a more recent paper Qui and Froese Fischer [25] have also found a more uniform error when quadrature is used compared to the differential equation approach (equation (212). By changing the parameters involved, Landtman and Hansen [56] found that the accuracy is reduced to about 10^{-11} if either the number of splines is halved or the box size is reduced to 50 au.

Another approach is to work more directly in terms of the B-splines. If the radial function for orbital a is approximated by

$$P_a(r) = \sum_i a_i B_i(r) \tag{215}$$

then the two-electron integral becomes

$$R^k(a, b, c, d) = \sum_i \sum_j \sum_{i'} \sum_{j'} a_i b_j c_{i'} d_{j'} R^k(i, j, i', j') \tag{216}$$

where $R^k(i, j, i', j')$, calculated between the splines i, j, i' and j', is called a Slater matrix element [25]. If o is the order of the B-splines , then $R^k(i, j, i', j') = 0$ if either $|i - i'| \geqslant o$

or $|j - j'| \geqslant o$ due to the piecewise nature of the B-splines. Qui and Froese Fischer [25] have discussed a procedure to calculate the Slater matrix elements by GL quadrature which takes into account this and other simplifications due to the particular properties of the B-splines. In this procedure, the Slater matrix elements (216) are broken up into bits associated with a 'cell' in r_1 and r_2 defined by the knot sequence; recall that each B-spline is defined over o knot points. Two types of 'cell' integral exists, one off-diagonal in which r_1 is different from r_2 everywhere. This integral can be calculated as a product of two one-dimensional integrals, which are calculated and stored separately. The diagonal cells, where the cusp occurs, have to be considered differently and a new set of GL points defined, as discussed above. Although there are few of these integrals, Qui and Froese Fischer [25] state that most of the computer time is spent on calculating these. Qui and Froese Fischer also discuss the symmetry rules that can be used to reduce the number of individual Slater matrix elements that must be calculated. These authors use a composite grid which is logarithmic from a certain point and they show some translation relations which can be used to reduce the number of calculations in the logarithmic part of the grid. Qui and Froese Fischer use k GL points for B-splines of order k and have obtained an accuracy with B-splines of order eight a little better in general than shown in table A.1, where the order of the B-splines used was seven. With B-splines of order four the error has increased to roughly 10^{-11}.

If the Slater matrix elements are calculated once and stored in memory then the calculation of the Slater integrals is in principle reduced to carrying out the multiplications within the nested do-loop. Since there are fewer distinct cell integrals than Slater-matrix elements, it is advantageous to store the former and assemble the latter when need arises and Qui and Froese Fischer [25] reported that the summation over the cell matrix elements to construct the Slater matrix elements is so fast that this approach is feasible. To assemble the Slater integrals from the Slater matrix elements is also reported to be fast compared with the calculation of the Slater matrix elements.

Another approach is to use the B-splines directly as in the spline-Galerkin method. The calculation of particular integrals over splines proceeds as above in this case with the same advantages, including, for example, that many integrals are identically zero as discussed above. However, to assemble a large matrix by use of equation (216) requires care because a straightforward application of the equation scales as N^8, where N is the number of basis states. With a basis size required for a realistic calculation, this is beyond the power of present computers. However, a careful analysis shows that the size of the problem can be reduced to N^5, which is still large. Further reductions are possible depending on the type of basis set, as discussed above for B-splines. Nevertheless, for large matrices, as opposed to constructing Slater integrals with a relatively small number of splines, assembling the matrix is still a time-consuming task.

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